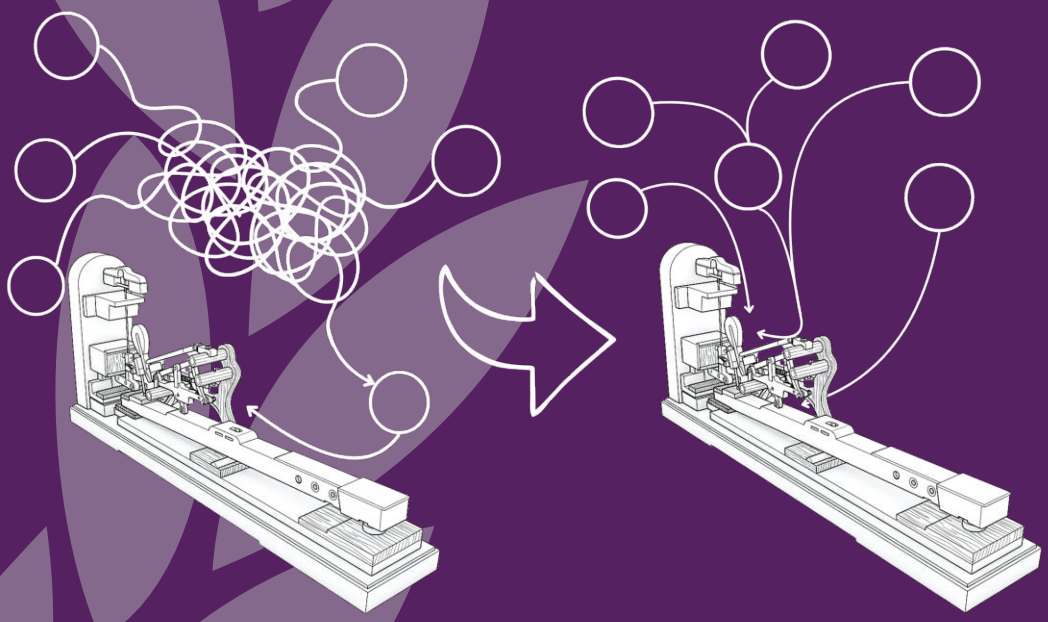
UNIVERSITY OF SKÖVDE



DOCTORAL DISSERTATION

COLLABORATIVE TECHNOLOGIES IN ENGINEERING DESIGN REVIEWS

FRANCISCO GARCIA RIVERA
Informatics

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Informatics



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ABSTRACT

Design reviews are central activities in product development, bringing together cross-functional teams to evaluate design proposals, identify issues, and decide on the progression of work. As product development becomes increasingly digital and distributed, these reviews are mediated by tools such as CAD software, videoconferencing platforms, and, more recently, extended reality (XR) systems. These technologies fundamentally shape how reviewers access shared product models, coordinate attention, and form a shared understanding. However, current industrial practice relies heavily on one person's screen sharing with the rest of the group, which limits direct interaction with product design representations and creates friction in communication and coordination.

This thesis examines how collaborative technologies can better support interaction and coordination in design reviews and develops practical guidance for their deployment in industrial settings. The research adopts a design-oriented mixed-method approach, combining field studies, experiments, and participatory workshops. Field observations of distributed design reviews identified four recurring friction situations: requesting specific viewpoints, indicating specific elements, expressing design change ideas, and evaluating ergonomics. Subsequent laboratory experiments and qualitative observations in shared 3D environments isolated how specific technological configurations, such as independent navigation and shared spatial scenes, influence how teams distribute attention and maintain common ground.

Synthesizing the results, the thesis identifies a set of core affordances that collaboration systems for design reviews need to support: shared spatial presence around the product model, in-scene deictic indication (pointing), and independent navigation with mechanisms for following a presenter. These affordances are translated into a framework and a set of practical guidelines that connect interaction capabilities with the preparation, conduct, and documentation of design reviews. These guidelines were developed and validated through workshops with two automotive companies, confirming the guidelines' clarity and applicability in real-world practice. Collectively, the thesis contributes to a deeper understanding of technology-mediated collaboration and provides actionable methods for integrating advanced digital tools into engineering workflows.

SAMMANFATTNING

Konstruktionsgenomgångar (design reviews) är centrala aktiviteter inom produktutveckling, där tvärfunktionella team samlas för att utvärdera konstruktionsförslag, identifiera problem och besluta om hur arbetet ska fortskrida. I takt med att produktutvecklingen blir alltmer digital och distribuerad medieras konstruktionsgenomgångar via verktyg såsom CAD-programvara, videokonferensplattformar och, på senare tid, XR-system (extended reality). Dessa tekniker påverkar i grunden deltagarnas tillgång till produktmodellerna, hur uppmärksamhet samordnas och hur deltagarna bildar en gemensam förståelse. Nuvarande industriell praxis är dock starkt beroende av presentatörscentrerad skärmdelning vilket begränsar deltagarnas interaktion med konstruktionsförslag och skapar friktion i kommunikation och samordning.

Denna avhandling undersöker hur samarbetsteknologier bättre kan stödja interaktion och samordning vid konstruktionsgenomgångar och utvecklar praktiska riktlinjer för deras användning i industriella sammanhang. Forskningen använder en designorienterad mixed-method-ansats, som kombinerar fältstudier, experiment och deltagande workshops. Fältobservationer av distribuerade konstruktions-genomgångar identifierade fyra återkommande friktionssituationer: begäran om specifika betraktningspunkter, indikering av specifika produktdetaljer, uttryckande av idéer om designförändringar, och utvärdering av ergonomi. Efterföljande laboratorie-experiment och kvalitativa observationer i delade 3D-miljöer isolerade hur specifika tekniska konfigurationer, såsom oberoende navigering och delade rumsliga scener, påverkar hur team fördelar sin uppmärksamhet och upprätthåller gemensam grund.

Resultaten sammanfattas i en uppsättning centrala funktioner som samarbetssystem för konstruktionsgenomgångar behöver stödja: delad rumslig närvaro kring produktmodellen, deiktisk indikation (pekning) i scenen, och oberoende navigering med mekanismer för att följa en presentatör. Dessa funktioner översätts till ett ramverk och en uppsättning praktiska riktlinjer som kopplar samman interaktionsmöjligheter med förberedelse, genomförande och dokumentation av konstruktionsgenomgångar. Dessa riktlinjer har utvecklats och validerats genom workshops med två fordonsföretag, vilket bekräftar riktlinjernas tydlighet och användbarhet i industriell verksamhet. Sammantaget bidrar avhandlingen till en djupare förståelse av teknikmedierad samverkan och ger praktiska metoder för att integrera avancerade digitala verktyg i ingenjörsmässiga arbetsflöden.

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I arrived in Skövde exactly 3,123 days before writing these words. As I mark the conclusion of my doctoral thesis, I cannot help but reflect on that journey. I was 22 years old when I first arrived, a vastly different person from the one here today. Living abroad is a lesson in evolving perspective; after two years, I thought I understood the experience, but by the third and fourth, that understanding was overshadowed by new realizations. A PhD follows a similar trajectory, where what you think you know at the beginning is stripped away and rebuilt by the end. I suspect my current perspective will be humbled by the wisdom of the years to come, but I know that this growth, while internal, is primarily defined and fueled by the individuals who have crossed my path and carried me through these years.

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PUBLICATIONS

The publications appended to this dissertation are listed below. Francisco Garcia Rivera's contributions, as the main author of all appended publications, are specified for each publication using the CRediT taxonomy (https://credit.niso.org). Co-authors include the supervision team: Professor Dan Högberg, Dr. Anna Brolin, Dr. Asreen Rostami, and Dr. Maurice Lamb. Additional collaborations throughout the thesis involved Dr. Beatrice Alenljung at the University of Skövde, and Professor Francesco Ferrise and Dr. Nicolò Dozio at Politecnico di Milano in Italy. The collective work across these publications reflects the multidisciplinary and collaborative nature of the research carried out during the doctoral studies.

APPENDED PUBLICATIONS

I. Rivera, F.G., Lamb, M., Högberg, D. and Brolin, A. (2022). The schematization of XR technologies in the context of collaborative design. In: SPS2022 - Proceedings of the 10th Swedish Production Symposium, April 26-29, 2022, Skövde, Sweden, Ng, A.H.C., Syberfeldt, A, Högberg, D., Holm. M. (Eds.), *Advances in Transdisciplinary Engineering*, 21, 520–529, IOS Press, Amsterdam. doi.org/10.3233/ATDE220170. Open Access.

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ADDITIONAL PUBLICATIONS

1. Cao, H., Rivera, F.G., Söderlund, H., Berlin, C., Stahre, J. and Johansson, B. (2025). Human centered design of VR interface features to support mental workload and spatial cognition during collaboration tasks in manufacturing. *Cognition, Technology and Work*, 27, 467–485. doi.org/10.1007/s10111-025-00809-6. Open Access.

   Journal paper. Published.

2. Rivera, F.G., Ferrise, F. and Panarotto, M. (2025). Leveraging Activity Theory and Functional Modelling for implementing extended reality in design reviews. *Proceedings of the 22nd Congress of the International Ergonomics Association, Volume 1*. doi.org/10.1007/978-981-95-0211-0_56.

   Conference paper. Published.

3. Cao, H., Söderlund, H., Despeisse, M., Rivera, F.G. and Johansson, B. (2024). VR interaction for efficient virtual manufacturing: Mini map for multi-user VR navigation platform. *Proceedings of the 11th Swedish Production Symposium*. doi.org/10.3233/ATDE240178. Open Access.

   Conference paper. Published.

4. Rivera, F.G., Rostami, A., Mattsson, S. and Söderlund, H. (2024). How can XR enhance collaboration with CAD/CAE tools in remote design reviews? *Proceedings of the 11th Swedish Production Symposium*. doi.org/10.3233/ATDE240182. Open Access.

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5. Rivera, F.G. and Lamb, M. (2024). Examining the impact of camera control on collaborative problem solving. *Proceedings of the 19th SweCog Conference*.

   Poster.

6. Almirón Santa-Bárbara, R., Rivera, F.G., Lamb, M., Víquez Da Silva, R. and Gutiérrez Bedmar, M. (2023). New technologies for the classification of proximal humeral fractures: Comparison between virtual reality and 3D printed models—a randomised controlled trial. *Virtual Reality*, 27, 1623–1634. doi.org/10.1007/s10055-023-00757-4. Open Access.

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7. Rivera, F.G., Högberg, D., Lamb, M. and Perez Luque, E. (2022). DHM supported assessment of the effects of using an exoskeleton during work. *International Journal of Human Factors Modelling and Simulation*, 7(3/4), 231–246. doi.org/10.1504/IJHFMS.2022.124308. Open Access.

   Journal paper. Published.

8. Rivera, F.G., Brolin, A., Perez Luque, E. and Högberg, D. (2021). A framework to model the use of exoskeletons in digital human modelling tools. *Advances in Simulation and Digital Human Modeling, AHFE 2021*. Lecture Notes in Networks and Systems, vol 264. Springer. doi.org/10.1007/978-3-030-79763-8_38. Open Access.

   Conference paper. Published.

9. Rivera, F.G., Brolin, E., Syberfeldt, A., Högberg, D., Iriondo Pascual, A. and Perez Luque, E. (2020). Using virtual reality and smart textiles to assess the design of workstations. *Proceedings of the 9th Swedish Production Symposium*. doi.org/10.3233/ATDE200152. Open Access.

   Conference paper. Published.

10. Igelmo, V., Syberfeldt, A., Högberg, D., Rivera, F.G. and Perez Luque, E. (2020). Aiding observational ergonomic evaluation methods using MOCAP systems supported by AI-based posture recognition. *Proceedings of the 6th International Digital Human Modeling Symposium*. doi.org/10.3233/ATDE200050. Open Access.

    Conference paper. Published.

11. Iriondo Pascual, A., Högberg, D., Syberfeldt, A., Rivera, F.G., Perez Luque, E. and Hanson, L. (2020). Implementation of ergonomics evaluation methods in a multi objective optimization framework. *Proceedings of the 6th International Digital Human Modeling Symposium*. doi.org/10.3233/ATDE200044. Open Access.

    Conference paper. Published.

12. Perez Luque, E., Högberg, D., Iriondo Pascual, A., Lämkull, D. and Rivera, F.G. (2020). Motion behaviour and range of motion when using exoskeletons in manual assembly tasks. *Proceedings of the 9th Swedish Production Symposium*. doi.org/10.3233/ATDE200159. Open Access.

    Conference paper. Published.

13. Reinhard, R., Mårdberg, P., Rivera, F.G., Forsberg, T., Berce, A., Mingji, F. and Högberg, D. (2020). The use and usage of virtual reality technologies in planning and implementing new workstations. *Proceedings of the 6th International Digital Human Modeling Symposium*. doi.org/10.3233/ATDE200047. Open Access.

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INTRODUCTION

1 INTRODUCTION

Developing new products is a complex process where collaboration among stakeholders from different backgrounds is critical to success (Permatasari et al., 2021). The product development process (PDP) is a multifaceted sequence of activities an enterprise employs to conceive, design, and commercialize a product (Ulrich et al., 2019). The PDP guides product designers through stages, typically beginning with establishing design specifications and ending with a detailed and fully defined design ready for production. Figure 1 illustrates Ulrich and Eppinger's model of a PDP, which shows typical stages of the process (Ulrich et al., 2019).

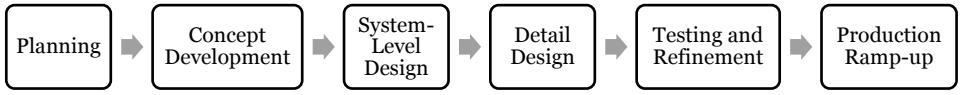


Figure 1. Ulrich and Eppinger's model of a PDP. Adaptation from Ulrich et al. (2019).

There are several models to describe activities in the PDP. Nigel Cross (2021) defined a basic descriptive model of the PDP consisting of four key stages (Figure 2), which can also be understood as labels of fundamental design activities: exploration of the problem space to understand the problem and the requirements, generation of solutions that build up the solution space, evaluation of the generated solutions against the criteria of the requirement specification, and communication of the final design of the product. In reality, there are typically many iterations and integrations of these activities (Cross, 2021; Dorst & Cross, 2001; Gericke & Blessing, 2012; Lawson, 2006). Still, Cross's model illustrates the fundamental stages from start to finish of a PDP and can be used as a labelling system for explaining and classifying key design activities within the PDP at various stages.

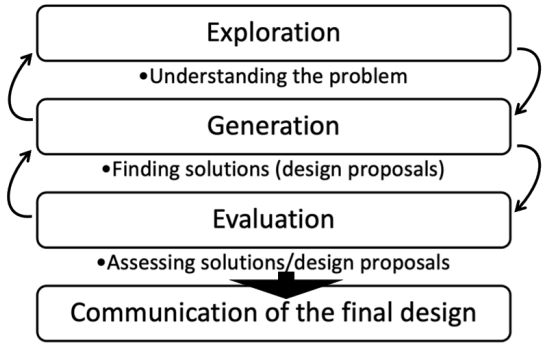


Figure 2. Four key stages of the PDP. Adaptation from Cross (2021).

A common evaluative activity found in many organizations' PDPs is design reviews (DRs), which are central to this thesis. DRs act as formal checkpoints within the PDP where cross-functional stakeholders examine a design proposal against requirements, risks, and project objectives to determine its readiness for progression to subsequent stages of the PDP (Jamnia, 2018). DRs involve structured evaluation, and integrating perspectives from different stakeholders, including engineering, manufacturing, ergonomics, quality, service, and management, to align understanding, surface issues early, with an aim of reducing downstream rework (Adams et al., 2016; Gräßler et al., 2022; Horvat et al., 2022). In this context, DRs serve as governance points that link technical evidence to decision-making and record the reasoning behind progressing, redirecting, or postponing work (Chen et al., 2019; Huet et al., 2007). While the term "design review" is also common in fields such as user experience (UX) research and human-computer interaction (HCI), in the context of this thesis and the appended papers, DRs refer specifically to these formal technical gates performed within an industrial engineering PDP. This is sometimes also referred to as 'engineering design reviews', hence the title of this thesis. For simplicity, the term DR will be used throughout the thesis.

DRs can occur at multiple stages of the PDP. Hence, adopting Ulrich and Eppinger's terminology in Figure 1, DRs occur both at the concept development, system-level design, and detail design stages (Clarkson & Eckert, 2005; Ulrich et al., 2019). In practice, DRs are organized as scheduled meetings aligned with the progression of the PDP (Bedingfield & Clarkson, 2020; Chen et al., 2019). In DRs, it is common for a designated presenter, or "driver," to walk participants through the relevant design representations while different stakeholders contribute domain-specific assessments (Détienne et al., 2005). The setup tends to follow the same pattern, whether the session takes place co-located around a physical display, in a hybrid format where on-site participants gather around a shared screen that is also streamed to distributed attendees, or fully distributed, with one participant (typically the presenter) sharing their screen while others follow along (either in co-located or distributed settings). The conduct of DRs is prepared around a defined scope and a set of artifacts, typically including models and drawings, requirement lists, bills of materials (BOM), simulations, test reports, mock-ups, or prototypes, selected to support the specific evaluation at hand, where Computer-Aided Design (CAD) models are the most commonly used support artifact during DRs (Chen et al., 2019).

During a typical DR, the participants rely on the presenter to navigate and display the design artifacts. This means that the participants' attention is sequentially concentrated on specific aspects of the product design. At the same time, the participants' contributions are mainly verbal or gestural. This centralization, primarily governed by the presenter, provides a clear and consistent structure for the DR and shapes how collaboration occurs. As mentioned, these coordination patterns are evident in co-located DRs, i.e., carried out in physical rooms where groups gather around a shared display or prototype, as well as in distributed or hybrid DRs (Hirschmann et al., 2025). This arrangement works effectively for coordination and documentation; however, it can limit participants' opportunities for direct engagement with the design representation, parallel exploration of issues, and can hamper the ability of specialists to contribute their expert input.

As the development of physical products is increasingly mediated by digital tools, in which designers interact primarily with digital models rather than the physical objects themselves, these centralized interaction patterns have a greater impact on collaboration. In a DR, even when all reviewers share the same room, most discussions still occur through a single visual channel that one person controls (Cheng et al., 2023). Whether the team is co-located, hybrid, or distributed, the collaborative logic of the DRs essentially remains the same: a shared focal point structures the conversation, exploration, evaluation, and decisions. What remains less understood is how this inherent centralization influences communication, attention, and understanding across different technological settings. Addressing this gap requires examining DRs not only as procedural checkpoints, but as collaborative activities where the team's ability to exchange information is strictly defined by the technical constraints of how the design is displayed and controlled.

1.1 AIM

The aim of this thesis is to understand how collaborative technologies support interaction and coordination in DRs and to translate this understanding into frameworks that inform both technology development and practical use in industry. This is pursued through three objectives: identifying the collaboration challenges that limit DR effectiveness, determining which technological capabilities address these challenges, and integrating these insights into validated guidelines for industrial application.

1.2 DELIMITATIONS

This thesis is limited to physical product development, where most design work and collaboration take place around digital artifacts and digital representations, including CAD models, CAD viewers, 3D simulations, and digital prototypes. The focus is on DRs and the interaction and coordination problems that occur when teams use digital tools to explore, discuss, and evaluate design proposals, and decide and document design decisions and follow up work. The thesis does not address software development, service design, or other domains where design reviews follow different practices.

The research seeks to maintain a general perspective on engineering DRs, but the empirical studies and industrial collaborations are primarily situated in the Swedish automotive industry. As a result, the findings reflect practices and constraints that are common in this context. Broader organizational factors such as long-term process maturity, large-scale tool deployment, or extended enterprise collaboration are outside the scope of this work.

1.3 THESIS OUTLINE

The thesis is structured into six chapters that introduce the research problem, present the theoretical background, describe the research approach, summarize the results, and discuss their implications.

- Chapter 2: presents the background needed to understand the context of the research. It introduces DR in product development, human-centric perspectives, technology-mediated collaboration, asymmetric collaboration, and the theoretical lenses used in the thesis.
- Chapter 3: describes the research method. It introduces the research questions, explains the overall research approach and its connection to design research methodology, and outlines the empirical and analytical methods used in the studies.
- Chapter 4: summarizes the main results. It presents the findings of the empirical and conceptual studies and shows how they answer the research questions.
- Chapter 5: discusses the implications of the results. It reflects on the collaboration challenges observed in design reviews, the capabilities that shape coordination, and the contribution of the proposed framework and guidelines. It also outlines methodological reflections and limitations.
- Chapter 6: concludes the thesis by answering the overarching research question and proposing directions for future work.

BACKGROUND

2 BACKGROUND

This chapter establishes the theoretical foundation for the research by reviewing literature across the disciplines that intersect in DRs. To map this multidisciplinary space, an Area of Relevance and Contribution (ARC) diagram is employed (Blessing & Chakrabarti, 2009), shown in Figure 3. The diagram organizes the established fields informing the study into three categories: *Context* (PDP and Distributed work), *Enablers* (XR, Collaborative Virtual Environments, and Asymmetric Collaboration), and *Theoretical lenses* (Presence, Activity Theory, and Affordances). The contribution area at the centre of the diagram represents the focus of this thesis: technological and procedural support for DRs.

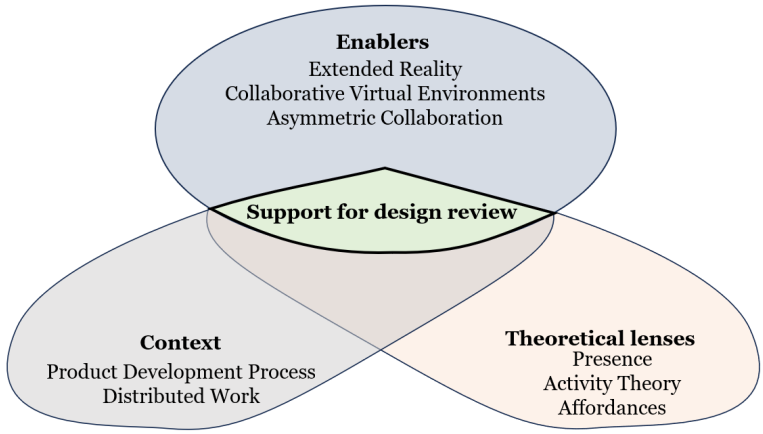


Figure 3. Area of relevance and contribution for the thesis.

Foundational literature in the area of relevance and contribution was identified through two structured searches conducted at different stages of the research, supplemented by ongoing discovery throughout the project. An initial search was conducted in 2021 using Scopus with the search strings illustrated in Table 1. This search yielded approximately 700 candidate papers, which were screened by title and abstract, reducing the set to around 150 papers for full-text review. A second structured search was conducted toward the end of the research (2025), following PRISMA guidelines for scoping reviews, using a refined search string. This search identified 42 papers after screening.

Table 1. Literature search methodology, search strings, and screening results across the two research phases.

Search Phase	Database & Year	Search String / Criteria	Results
Initial Scoping Search	Scopus (August 2021)	TITLE-ABS-KEY(("virtual reality" OR "augmented reality" OR "mixed reality" OR "immersive") AND ("collaborative" OR "collaboration" OR "co-design" OR "distributed") AND ("design" OR "design review" OR "prototyping"))	~700 candidate papers identified. Screened down to ~150 papers for full-text review.
Final Scoping Review (PRISMA)	Scopus (April 2025)	TITLE-ABS-KEY("design review*" OR "engineering design review*" OR "product design review*" OR "technical review*" OR "engineering review*" OR "design evaluation*" OR "review meeting*" OR "gate review*" OR "design critique*") AND TITLE-ABS-KEY("product*" OR "engineering design") AND TITLE-ABS-KEY("collaborat*" OR "stakeholder*" OR "cross-functional" OR "co-design" OR "teamwork" OR "participatory") AND TITLE-ABS-KEY("CAD" OR "computer-aided design" OR "virtual reality" OR "VR" OR "augmented reality" OR "AR" OR "mixed reality" OR "MR" OR "digital twin*" OR "AI" OR "cloud CAD" OR "remote collaboration" OR "distributed team*" OR "immersive technolog*" OR "XR" OR "collaboration platform*") AND PUBYEAR > 2009	42 relevant papers identified after screening.

The initial literature review helped establish the theoretical directions and scope of the research, while naturally leading to additional relevant work through citation tracking and recommendations from collaborators. The final scoping review served to confirm the continued relevance of the research contributions and to situate the findings within the current state of the field. Although this final review followed formal PRISMA guidelines to ensure methodological rigor, it was not published as a standalone article, as comprehensive literature reviews already exist within this domain (Horvat et al., 2022). This last structured literature review was used internally to benchmark the thesis findings against the latest technological advancements.

Drawing upon the literature identified through this process, the remainder of this chapter synthesizes the current state of knowledge across the areas outlined in the ARC diagram (Figure 3). It begins by examining the structural role of design reviews in the product development process (Section 2.1), before exploring the human-centric needs of these evaluations (Section 2.2), the impact of technological mediation on team collaboration (Section 2.3), and the theoretical lenses used to analyze them (Section 2.4).

2.1 DESIGN REVIEWS IN PRODUCT DEVELOPMENT

The product development process (PDP) varies widely across organizations, products, and industry sectors, making it hard to describe or prescribe a single, uniform process that companies use or can adopt (Ulrich et al., 2019). To make sense of this variety,

researchers and practitioners have developed models of the PDP that represent how PDP is carried out in practice (descriptive models) or how it should be carried out ideally (prescriptive models). Prescriptive models, such as Ulrich and Eppinger's model (Figure 1), are used to guide the PDP and encourage product designers to adopt better practices (Cross, 2021). Descriptive models, such as Cross's basic four-stage model, shown in Figure 2, aim to characterize how PDPs are actually performed (Cross, 2021).

Beyond the specific formulation, descriptive and prescriptive models share the common idea that PDP alternates between divergent and convergent thinking phases. Divergent thinking refers to expanding the solution space. In contrast, convergent thinking focuses on narrowing the solution space and selecting the most suitable solutions (Adikari et al., 2013), as shown in Figure 4.

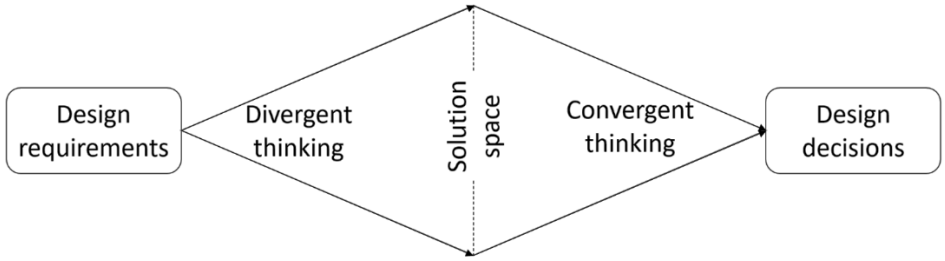


Figure 4. Divergent and convergent nature of PDP (Adikari et al., 2013).

Some PDPs have this alternation implicit in their formulation, for example, Ulrich and Eppinger's model (Ulrich et al., 2019) prescribes phases of concept generation and selection, meaning that it alternates between finding solutions (i.e., increasing the solution space through divergent thinking) and selecting solutions (i.e., decreasing the solution space through convergent thinking). Similarly, Cross's model, as explained, describes four types of activities: explanation, generation, evaluation, and communication of the final design, each phase implicitly containing both convergent and divergent thinking. The concept of divergent and convergent thinking has also served as a foundation for other PDP models. For example, the double diamond model in Figure 5 is an example of PDP models that explicitly utilize the concept of divergent and convergent phases to model the PDP (Design Council, 2024).

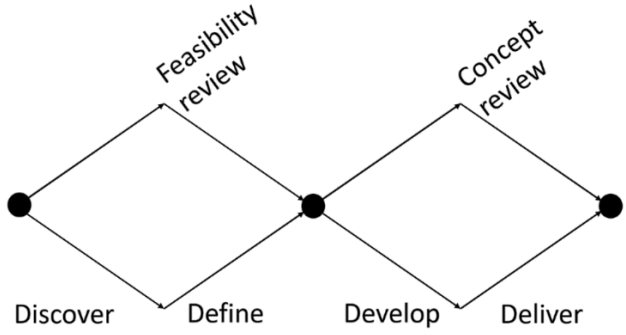


Figure 5. Double diamond model (Design Council, 2024).

Divergent and convergent thinking occur not only at the level of the PDP as a whole but also within the specific sub-processes that compose it. Consequently, the PDP can be described as a nested structure of interconnected processes, ranging from the overall project progression down to the fine-grained tasks performed by individual designers. To analyze work across these different levels of granularity, Activity theory provides a useful framework for describing this structure by decomposing work into three nested levels: *activities*, *tasks*, and *actions*, each with its own goal and temporal scope (Bedny & Harris, 2005; Qureshi et al., 2014). Actions are the discrete operations that fulfil immediate goals, such as taking a note or asking a question. These actions are logically combined and organized into tasks that pursue specific goals. For example, documenting a meeting to evaluate a design proposal may involve discussing issues with a colleague, highlighting design changes, and taking notes. At the highest level, activities represent broader goal-directed systems that include several related tasks; a detailed evaluation of a prototype can be seen as one such activity, integrating modelling, testing, and decision-making. Cash et al. (2015) extend this reasoning by defining *macro-*, *meso-*, and *micro-scale* processes as complementary descriptions of activities, tasks, and actions (Figure 6). Macro-scale processes comprise sequences of activities linked by a common focus, meso-scale processes include sequences of tasks motivated by a shared objective, and micro-scale processes consist of sequences of actions oriented toward an immediate goal.

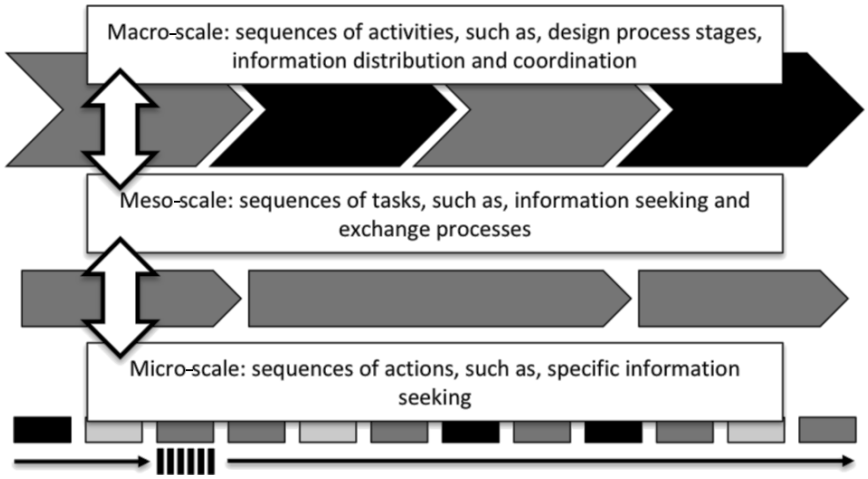


Figure 6. Macro-scale (activities), meso-scale (tasks), and micro-scale (actions) (Cash et al., 2015).

Within this multi-scale view of the PDP, DRs are mainly seen as convergent activities because their overall goal is to evaluate and identify issues with the current design proposal. Still, DRs may also contain elements of divergent activities in terms of suggesting and discussing potential solutions to resolve identified issues, to refine and bring the design closer to an acceptable solution. In this light, the main aim of DRs is to evaluate design proposals against requirements, reduce uncertainty, and determine subsequent actions (Adams et al., 2016). DRs can occur at multiple stages of the PDP, from early concept evaluations to late pre-production verifications (Chen et al., 2019; Clarkson & Eckert, 2005). This industrial variety is formalized in standards like the NASA NPR 7120.5 framework (Osborne, 2022), which defines specific technical

reviews based on project maturity. These include the Preliminary Design Review to ensure the system is ready for detailed design, the Critical Design Review to authorize fabrication and integration, and the Production Readiness Review to verify manufacturing controls. Each DR carries different procedural goals to ensure the project meets its designated criteria.

As a result, while DRs share the same overarching function of convergence, the content, structure, and participants of DRs vary not only depending on the stage in the PDP process but also due to other factors such as project complexity, organizational practices, regulatory constraints, and the specific objectives of the DR. Therefore, variations in DRs reflect both their timing within the PDP and broader project or organizational context. From a pragmatic perspective, these differences can be described in terms of the tasks and actions that compose each DR. For instance, an early-stage DR may involve tasks such as assessing feasibility, reviewing alternative concepts, or revising high-level requirements. In contrast, a late-stage DR may focus on validating tolerances, verifying assembly procedures, or approving tooling modifications.

In addition to consolidating design information and decreasing the solution space, DRs often serve an evaluation gate function within the PDP (Figure 7). At these gates, the outcome of a DR determines whether the development moves forward to the next stage of the PDP, whether the design needs further refinement, or whether the project should be paused or discontinued (Tzokas et al., 2004). This gate logic formalizes the role of DRs as decision checkpoints that control the pace and direction of PDP. Each DR thus contributes not only to the technical validation of design proposals but also to the governance of the overall PDP, linking local collaborative activity to strategic project progression.

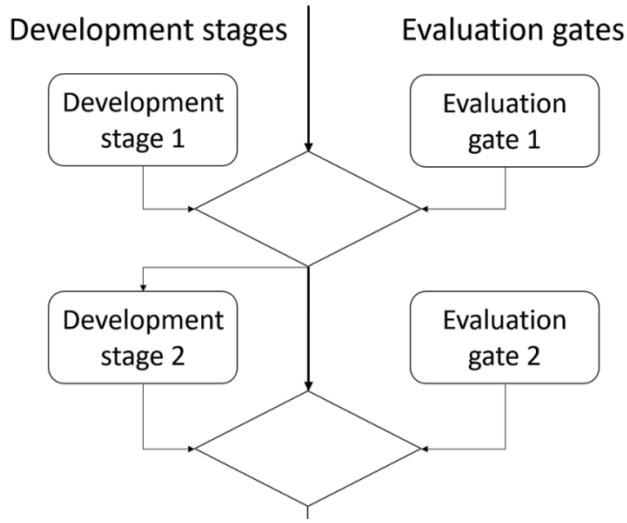


Figure 7. Evaluation gates. Adaptation from Tzokas et al. (2004).

In summary, DRs occupy a central role in the PDP as structured, convergent activities that integrate multiple perspectives and govern project progression. Analyzing DRs at the level of the tasks and actions that compose them reveals how differences in artifacts, participant roles, and technological contexts influence distinct patterns of collaboration. The following sections apply this analytical approach to examine the specific communication and coordination factors that shape these exchanges. This analysis establishes the foundation for the proposed improvements to how DRs are conducted and supported.

2.2 HUMAN-CENTRICITY IN DESIGN REVIEWS

Human-centred product development recognizes that a successful design requires understanding both the people involved in creating the product as well as the end users who will interact with it, since the decisions and interpretations of designers shape the final outcome (Norman, 1986). Human-centred PDP is characterized by a strong focus on the processes, insights, and activities for which the product, services, or systems are intended, instead of the designers' creativity (Giacomin, 2015). Figure 8 and Figure 9 show the shift in thinking from a *machine-centred* to a *human-centred* view (Chapanis, 1996; Wasserman, 1989). In the machine-centred view (Figure 8), product designers conceive the product (in this case, a photocopier) as a set of mechanisms that work together, with the only external element being pieces of paper. In this conception, the human is a generic element that only "brings the job to the machine." In this approach, the user is expected to compensate for system deficiencies rather than being treated as a specific component with defined performance limits (Chapanis, 1996). In the human-centred view (Figure 9), the user is at the centre of the system, and it is no longer a generic human but a person with specific abilities, needs, and attributes. Human-centred design acknowledges that the product's success is linked to its ability to accommodate its users' requirements (Wasserman, 1989).

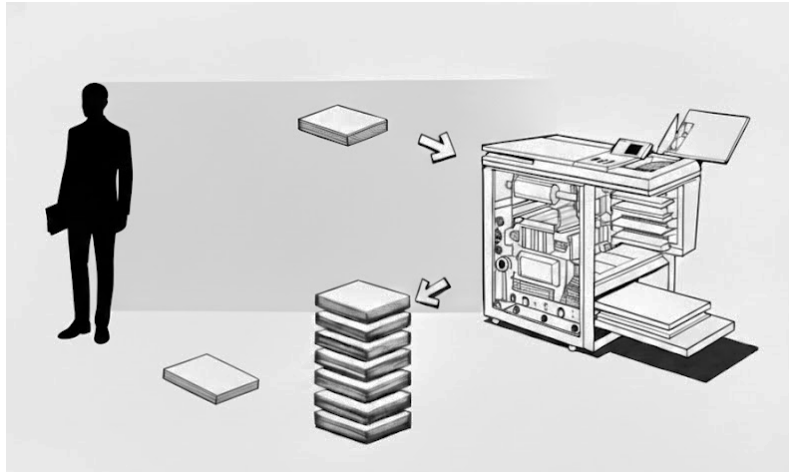


Figure 8. Machine-centred view. Adaptation from Wassserman (1989).



Figure 9. Human-centred view. Adaptation from Wasserman (1989).

Within DRs, the notion of human-centeredness applies at two complementary levels: the content of evaluation and the process of evaluation. At the content level, DRs frequently address questions concerning how people will interact with the designed product across its lifecycle, from use and maintenance to assembly and recycling (Khalid & Helander, 2006; Maguire, 2001). Hence, evaluations of ergonomics, usability, and safety are central to these DRs (Ahmed et al., 2018; Crilly et al., 2008). To support such analyses, organizations rely on tools that simulate or measure human-product interaction. For example, Digital Human Modelling (DHM) enables designers to examine posture, fit, reach, and visibility for users with different anthropometries (Högberg et al., 2019). Motion-capture systems allow the collection of empirical movement data for assessing workload and efficiency (Menolotto et al., 2020). Extended reality (XR) technologies extend these capabilities by enabling immersive

inspection of virtual prototypes, allowing participants to perceive scale, spatial relationships, and ergonomics in context (De Crescenzio et al., 2019; Grandi et al., 2018; Shi et al., 2019). These technologies support a human-centred approach by providing early, evidence-based insights into how designs accommodate human capabilities.

At the process level, the organization of the DR itself must be human-centred. This is because collaboration depends on participants' perceptual, cognitive, and communicative capacities (Cross, 2021). DRs require participants from multiple disciplines to interpret shared visualizations, coordinate viewpoints, and collectively negotiate meaning. When meeting structures or tools do not support these needs, such as when control of the shared model is overcentralized or when the design representations overload attention, participants experience increased cognitive effort and reduced engagement (Gräßler et al., 2022; Horvat et al., 2022). A human-centred DRs process, therefore, seeks to align the representational and procedural environment with the capacities of its participants, ensuring that all contributors can effectively perceive, reason, and communicate.

In practice, these two dimensions reinforce one another. Tools that visualize human-product interaction also change the social dynamics of the DR by enabling participants to explore, point, and discuss together (Berg & Vance, 2017). As a result, the pursuit of human-centred products depends on human-centred DR processes. In this thesis, this dual view of human-centredness is central. The designed products should support the people who use, assemble, maintain, or interact with them, and the DR process itself, and associated use of technologies, must support the people who evaluate those aspects. This means that DRs need tools and procedures that respect human perceptual, cognitive, and communicative limits, since the ability to evaluate human-centred qualities in a design depends directly on whether the DR environment is itself human-centred.

2.3 COLLABORATION IN DESIGN REVIEWS MEDIATED BY TECHNOLOGY

In this thesis, technology-mediated collaboration refers to DRs where the activity is conducted through digital tools. While in many instances technology mediates the social collaboration between participants (such as in distributed DRs using videoconferencing), its broader role is mediating the DR as a whole. Because the DRs in focus throughout this thesis rely on digital representations, the technology serves as the primary channel for the entire DR session, defining how the team accesses information, coordinates attention, and reaches decisions. This mediation includes co-located settings where the use of digital tools to explore a virtual object fundamentally shapes the collaborative dynamics of the group.

In modern PDP, most of the activities are carried out using computers and digital representations of the product. While DRs can use physical prototypes, this research focuses on DRs in which the visual representation is exclusively a virtual object, which is the most common approach (Chen et al., 2019). Collaboration during DRs is shaped by the tools through which participants can access and interact with the virtual environments used to display the design representations and each other. While DRs

have always relied on visual artifacts to facilitate communication and decision-making, the digitalization of PDP has transformed how such artifacts are created, shared, and discussed. Figure 10 illustrates the transformation of engineering design work from the analogue drafting era to the digitally mediated environments that dominate today. The integration of CAD and Computer-Aided Engineering (CAE) environments has made the 3D model the central representation around which collaboration occurs (Chen et al., 2019). As a result, the interface between humans and digital artifacts has become a key determinant of how effectively knowledge is exchanged and consensus is achieved (Lee & Ostwald, 2022).

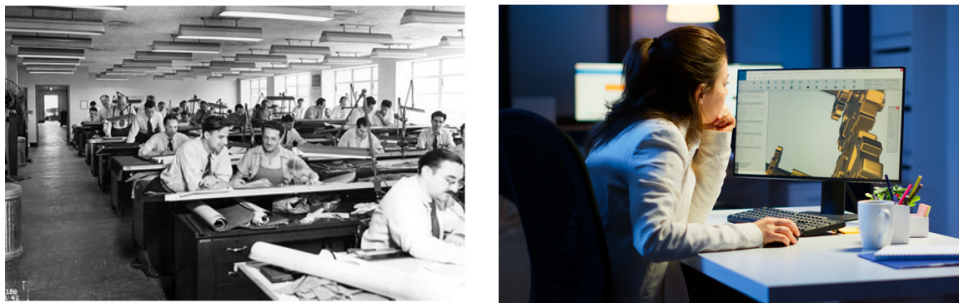


Figure 10. Transition from physical drafting rooms (left) to computer-aided design environment (right). Left: Image by NASA (Public Domain). Right: Image by DC studio on Freepik.

In the current state of practice, CAD and CAE tools are generally targeted toward different phases of the design process. Some platforms are authoring-centric, such as cloud-native tools like Onshape (PTC, Massachusetts, USA) or Fusion 360 (Autodesk, California, USA), where the focus is on the creation of models and the DR is a secondary extension of the modeling process. Other traditional authoring suites, such as Siemens NX (Munich, Germany), SolidWorks (Vélizy-Villacoublay, France), or Inventor (California, USA), utilize integrated hybrid setups where XR or online collaboration is added as a plug-in to the offline tool, allowing teams to collaborate without relying on external videoconferencing. Finally, there are dedicated DR environments like VRED (Autodesk California, USA), CoLab (CoLab, Newfoundland, Canada), or Campfire (Campfire, California, USA), which are made specifically for evaluation; these embed all necessary inspection and communication tools but typically lack the ability to modify the underlying engineering geometry.

As discussed, in traditional co-located DRs, teams gather around a large display providing the visual representation, with one participant, the presenter, controlling the visual representation. This centralized control structure simplifies coordination but also constrains interaction, as other participants must verbalize requests or rely on the presenter interpretation to access alternative viewpoints (Bedingfield & Clarkson, 2020; Horvat et al., 2024). The growing prevalence of distributed and hybrid work has accentuated these limitations. To conceptualize the different ways DRs are conducted, they can be categorized along two intersecting axes: Spatial Distribution (the rows: where participants are) and Symmetry (the columns: the types of tools they use). Table 2 illustrates how these terms relate to one another.

Table 2. Matrix of definitions dependent on tool use and geographical location.

	Symmetric (Same medium)	**Asymmetric (Different interfaces)**
Co-located	Everyone gathers around a single shared display or medium (e.g., laptops, VR headsets).	Participants in the same room use different mediums and levels of immersion (e.g., one in VR, others on PC).
Distributed	Geographically separated participants all use the same medium and level of immersion	Remote participants use different mediums and levels of immersion (e.g., VR and desktop as in Papers II and V).
Hybrid	A co-located group and distributed individuals all use a similar medium and levels of immersion.	A mix of co-located and remote participants using different mediums and levels of immersion (e.g., the workshops in Paper VI).

While early research by Törlind (2003) demonstrated that effective distributed engineering collaboration was achievable using sophisticated, dedicated hardware systems, modern industry practice has largely shifted toward accessible commodity tools. A common practice today when organizing DRs is to use videoconferencing applications such as Zoom (Zoom Video Communications, Inc., California, USA) and Teams (Microsoft, Washington, USA), combined with screen-shared CAD models, enabling distributed DRs but further concentrating control and reducing embodied cues. Current tools remove critical social signals, making it impossible for participants to use gaze to establish joint attention or use gestures to intuitively reference model details (Lapointe et al., 2021; Tea et al., 2021). Participants must compensate for the loss of social cues, such as pointing or using gaze to direct attention, by engaging in more explicit verbal negotiation, which can slow discussions and increase cognitive effort (Horvat et al., 2022).

These dependencies on verbal communication to replace spatial cues aligns with research on distributed cognition, which emphasizes that team understanding depends on shared representations and visible cues that allow participants to align their interpretations (Hutchins, 1995; Lee & Ostwald, 2022). When the shared representation is mediated through a single user's viewpoint, opportunities for parallel exploration and rapid grounding diminish. Participants must reconstruct spatial context from screen movements and verbal descriptions, which increases the effort required to maintain common ground (Tserenchimed & Kim, 2024). This reduced visibility of others' actions also weakens feedback loops that support coordination, such as noticing where someone is looking or what element they are referring to (Piumsomboon et al., 2019).

At the same time, digital tools provide analytical capabilities that physical artifacts cannot. CAD and CAE systems offer precise control of geometry and the ability to simulate structural, mechanical, and ergonomic properties with high fidelity (Ulrich et al., 2019). However, screen-based environments have limitations when tasks involve embodied or spatial reasoning. Research shows that 2D displays constrain depth perception, scale assessment, and natural spatial navigation, which can make certain evaluative tasks more difficult (Berg & Vance, 2017). These constraints influence collaboration by shaping how participants interpret the model, how they

coordinate their attention, and how they communicate their observations. When the environment does not provide intuitive spatial cues, verbal explanations often take precedence, which can increase mental workload.

For these reasons, researchers have explored immersive systems such extended reality (XR) to support DRs (Gräßler et al., 2022; Horvat et al., 2024). In this context, XR refers to an umbrella term that covers virtual reality (VR), augmented reality (AR), and mixed reality (MR) technologies. In most examples, XR refers to head-mounted display systems that allow users to experience and interact with 3D digital content in an immersive way, while still being part of a broader ecosystem of tools that includes conventional displays. XR provides additional perceptual and interaction capabilities. XR allows multi-user inspection of virtual prototypes at full scale and supports more embodied interaction, which can improve spatial understanding and facilitate certain forms of collaborative reasoning (Gugenheimer et al., 2017). However, immersive environments also introduce new constraints, such as reduced input precision, limited text entry, and weaker integration with enterprise tools (Burova et al., 2022; Pham & Stuerzlinger, 2019; Yigitbas et al., 2022).

2.3.1 XR SYSTEMS FOR COLLABORATIVE EVALUATION

XR systems expand the range of representational and interaction possibilities available during DRs by providing immersive environments where participants can experience design representations at full scale. The concepts underlying XR originated in early research on head-mounted displays and the Reality-Virtuality continuum (Milgram & Kishino, 1994; Sutherland, 1968). However, the technology has become increasingly relevant in DRs due to improved device reliability and the ability to integrate large and complex 3D models into interactive spatial environments.

A central contribution of XR to DRs is the ability to simulate realistic scale and spatial depth. By combining head-tracked perspective with stereoscopic cues, these systems allow participants to judge size, distance, and orientation in a manner that approximates real-world perception. This spatial fidelity enables users to intuitively grasp the physical constraints of a design (Çöltekin et al., 2020). Consequently, these perceptual advantages are especially relevant in DRs that involve ergonomics analysis, workspace layout, visibility studies, packaging checks, and the assessment of human scale interaction (De Crescenzio et al., 2019; Grandi et al., 2018; Shi et al., 2019). Studies have shown that immersive environments can help designers identify design issues that remain unnoticed in 2D or non-immersive 3D environments, particularly when tasks require posture evaluation, or inspection of spatial constraints (Berg & Vance, 2017).

XR further supports collaborative processes by reintroducing embodied communication cues that are often lost in traditional distributed settings. Through the use of avatars and motion tracking, the system renders gaze direction, posture, and pointing gestures, facilitating deictic indication and allowing participants to see where others are looking and what they are referencing (Gamelin et al., 2018, 2021; Lapointe et al., 2021; Oh et al., 2018). These non-verbal signals help participants coordinate their attention and intentions, thereby reducing communicative ambiguity during DRs.

Finally, XR bridges the gap between abstract input and virtual action by facilitating natural interaction. Unlike screen-based tools where the connection between mouse movement and cursor action is indirect, XR allows participants to point, walk, lean, or reach using natural body movements. This direct correspondence allows users to inspect the model using the same motor skills and spatial reasoning they employ in real-world evaluations (Gugenheimer et al., 2017; Thorp et al., 2024). Consequently, for assessments involving human-product interaction product designers can utilize their own bodily judgments for first-person assessments of reachability, visibility, and fit.

Despite these advantages, XR also presents challenges that limit its use as the only device for DRs. In the current state of XR, text input and precision selection are significantly less efficient than in keyboard and mouse-based systems due to limited tactile feedback and the precision of tracked controllers or hand tracking (Pham & Stuerzlinger, 2019; Yigitbas et al., 2022). Consequently, mandatory administrative tasks in formal DRs, such as logging action items, consulting test reports, or verifying information from suppliers, can become cumbersome when relying on gesture-based interactions with virtual interfaces. While connecting physical peripherals to the XR hardware like keyboards could mitigate this, doing so attaches the user to a desk and effectively replicates a desktop workflow inside a headset, negating the mobility and spatial benefits of XR. Additionally, XR can increase cognitive load when sensory fidelity is high but task demands require analytical reasoning or referencing external information (Thorp et al., 2024). This creates a trade-off where the attentional resources required to interpret the virtual environment compete with those needed for processing data, often resulting in performance decrements for documentation-heavy tasks (Makransky et al., 2019). Recent industrial developments have seen the emergence of high-fidelity collaborative platforms, such as NVIDIA Omniverse (California, USA), which utilizes frameworks such as Universal Scene Description to enable integration between different CAD and CAE tools (Ahmed et al., 2024). These platforms promise real-time synchronization between design changes and immersive visualization, potentially bridging the gap between XR sessions and the documentation needed in formal PDP.

However, achieving a high level of integration remains a barrier in practice. While many XR tools support importing geometry but do not yet provide robust support for product lifecycle management (PLM) integration, persistent annotations, or version control, which are central to industrial DR workflows (Howie et al., 2023; Wolfartsberger, 2019). As a result (and consistent with the observations in this research), XR sessions often remain parallel to, rather than embedded within, established PDP processes, and outputs from XR evaluations usually need to be manually transferred into documentation or decision-tracking systems.

Accordingly, XR systems offer substantial benefits for spatial perception, ergonomics analysis, and embodied communication, while standard desktop workstations offer stronger support for precision, data management, and documentation. XR is therefore best understood not as a replacement for established DR tools but as a complementary set of tools that can enhance specific phases or aspects of evaluation. This complementarity provides the foundation for emerging approaches that combine immersive and non-immersive tools in shared collaborative digital environments, as discussed in the next section.

2.3.2 ASYMMETRIC XR COLLABORATION

As PDP processes increasingly depend on distributed teams and diverse technological setups, DRs often involve participants using different types of devices, each offering unique capabilities and limitations. *Asymmetric collaboration* describes collaborative environments where participants engage in a shared virtual environment through different types of devices, such as XR headsets and flat screen devices (Figure 11). Instead of assuming all collaborators should use the same type of device, asymmetric settings recognize that different roles and tasks may benefit from various forms of interaction (Gugenheimer et al., 2017; Ouverson et al., 2021).

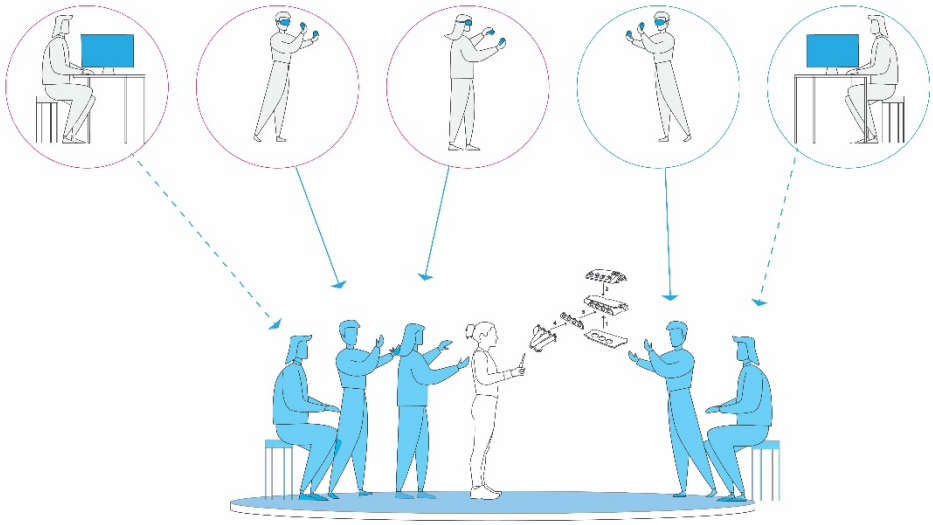


Figure 11. Conceptual illustration of asymmetric collaboration, where users with different devices (PC, VR, AR) contribute to a shared object manipulation and review task. From Paper II.

Interest in asymmetric collaboration has grown alongside the broader shift toward hybrid work and the increasing availability, functionality, and usability of XR technologies. Researchers have shown that immersive and non-immersive devices afford complementary interaction styles (Pham & Stuerzlinger, 2019; Zheng et al., 2025). These differences mean that requiring all participants to adopt the same devices can limit the effectiveness of the DR, especially in multidisciplinary settings where tasks vary from geometric inspection to documentation and decision tracking.

Asymmetric configurations have been explored in several collaborative environments. including industrial maintenance (Burova et al., 2022), architectural design (Ibayashi et al., 2015), and immersive data analysis (Reski et al., 2022). Studies on co-located asymmetric XR show that when combining immersive and non-immersive devices, users in the same shared space can facilitate joint exploration by allowing immersive users to act as embodied inspectors while 2D users provide analytical guidance or handle detailed controls (Burova et al., 2022; Gugenheimer et al., 2017). Other systems enable 2D users to observe or direct actions performed by XR users in real time, supporting workflows where only some participants require full spatial immersion (Burova et al., 2022; Ibayashi et al., 2015; Reski et al., 2022; Thoravi

Kumaravel et al., 2020). These studies highlight how asymmetric setups can preserve the advantages of immersive interaction without imposing its constraints on every participant.

However, asymmetric collaboration introduces challenges related to integration and communication. The system must maintain a consistent shared environment that updates reliably across platforms, ensuring that actions performed by immersive and 2D users remain synchronized (Ouverson et al., 2021; Reski et al., 2022). Differences in field of view, interaction techniques, and navigation methods can also create mismatches in what participants perceive or can access, which may require additional cues or visualization tools to maintain common ground. Achieving fluid collaboration, therefore, depends on designing mechanisms that make participants' actions visible and interpretable across device types.

The literature suggests that asymmetric collaboration is a promising approach for DRs because it reflects how multidisciplinary teams naturally distribute tasks and responsibilities. Immersive tools support embodied understanding, while screen-based tools support precision and documentation. When coordinated effectively, these heterogeneous capabilities can create a collaborative environment that leverages the strengths of each type of device while mitigating its individual limitations. This perspective positions asymmetric collaboration as an important area for advancing technological support for DRs.

2.4 THEORETICAL LENSES

2.4.1 PRESENCE

Marvin Minsky (1980) introduced the concept of *presence* in discussing telepresence technologies and the manipulation of physical objects remotely. Since Minsky's seminal work, various theories and definitions of presence have been developed; however, there is still a need for more consensus about what presence is. As XR devices matured, the definition became increasingly debated in the 1990s, and in the early 2000s, where considerable effort was put towards developing a consensus definition of presence. Several researchers identified factors that influenced the sense of presence during this time. Witmer and Singer (1998) defined presence as the subjective experience of being in one place or environment, even when one is physically situated in another. Slater (1996) defined presence as a function of two orthogonal variables: the extent to which the displayed sensory data matches the internal representation systems and subjective world models of the participant, and the extent to which proprioceptive feedback matches the sensory data generated by the system.

While the definition of presence, or an accurate measure of presence, is not a central objective of this research, it is important to understand how different hardware and software characteristics can influence users' experience. To that end, one definition that can be particularly useful when discussing collaborative virtual environments is the definition of Kwan Min Lee (2004), who defined presence as the "psychological state in which virtual objects are experienced as actual objects in sensory or non-sensory ways."

Relevant to this thesis is Kwan Min Lee's division of the concept of presence into three dimensions (Lee, 2004), as this is a useful notion for considering how XR devices can support different tasks in PDP.

- *Physical presence*: psychological state in which virtual physical objects are experienced as actual physical objects in sensory or non-sensory ways.
- *Social presence*: psychological state in which virtual social actors are experienced as actual social actors in sensory or non-sensory ways.
- *Self-presence* is a psychological state in which the virtual self is experienced as the actual self in sensory or non-sensory ways.

In the context of DRs, presence has implications for both perception and coordination. *Physical presence* enhances spatial understanding of the product by allowing participants to experience its geometry and scale directly, supporting tasks such as ergonomics assessment or spatial layout evaluation (De Crescenzio et al., 2019; Grandi et al., 2018). *Social presence* contributes to mutual awareness and engagement, helping participants to interpret gestures, attention, and intent; key elements for maintaining shared understanding during collective evaluation (Oh et al., 2018). *Self-presence*, in turn, influences how naturally participants act within the environment and how confident they feel contributing to the discussion (Li et al., 2021).

However, higher levels of immersion or presence do not automatically lead to more effective collaboration. Professional DRs often require precise, information-dense communication and simultaneous attention to multiple data sources (Pham & Stuerzlinger, 2019). Excessive sensory fidelity can increase cognitive demand or distract from analytical reasoning (Thorp et al., 2024). The challenge for collaborative technologies is therefore to balance the degree of presence with the functional requirements of the task, supporting a sense of co-presence, the awareness of others' actions, and the ability to coordinate with others, without overwhelming the participants.

For DRs, this perspective implies that presence should be considered not as an end in itself but as a resource that supports understanding and coordination. The design of collaborative environments should aim to cause the types of presence most relevant to the DR's objectives, such as strong physical and social presence to facilitate spatial reasoning and group discussion, while maintaining interfaces that support precision and cognitive control.

2.4.2 COMMON GROUND AND DISTRIBUTED COGNITION

Collaboration in design depends on participants' ability to develop and sustain a shared understanding of the problem, the artifacts being discussed, and the reasoning behind proposed decisions. This shared understanding, often referred to as *common ground*, enables individuals with diverse expertise, perspectives, and priorities to coordinate their actions effectively (Clark & Brennan, 1991). In DRs, common ground is created and maintained through ongoing exchanges that align interpretations of visual and verbal information.

The development of common ground in teamwork is not limited to spoken communication. Much of the understanding among team members comes from interacting with shared representations and environmental cues (Détienne et al., 2005). A similar perspective is developed in the theory of *distributed cognition*, which

sees cognitive processes as spread across people, tools, and representations instead of being contained within individual minds (Hollan et al., 2000; Hutchins, 1995). From the perspective of distributed cognition, tools and representations, such as sketches, CAD models, or digital notes, serve as cognitive support that distributes cognitive load, aids memory, and facilitates coordination (Cross, 2006).

During DRs, shared representations act as the primary medium through which distributed cognition develops. Participants depend on the visual model to set reference points, identify specific features, or verify alignment between verbal descriptions and physical geometry (Eckert & Stacey, 2003). When these representations are accessible to everyone, they act as joint perceptual spaces that support conversation and reasoning (Ball & Christensen, 2019). Conversely, when access is mediated, and participants do not have the opportunity to interact with the design representation, shared understanding becomes more difficult to maintain, as others must rely on verbal negotiation to infer context or intention (Bedingfield & Clarkson, 2020).

Participants continually monitor one another's verbal and nonverbal cues, adjusting their contributions to ensure mutual comprehension (Détienne et al., 2012). Maintaining common ground during DRs is therefore a dynamic process of alignment and repair. In digital or hybrid settings, this monitoring becomes more difficult, as gaze direction, gestures, and spatial positioning are partially lost or distorted (Lapointe et al., 2021). Technologies that enhance the visibility of others' actions, such as shared cursors, pointing tools, or embodied avatars, can partially compensate by restoring mutual awareness and grounding communication (Pham & Stuerzlinger, 2019).

From a theoretical standpoint, distributed cognition offers a valuable lens for examining how collaborative technologies impact DRs. Collaborative tools, such as shared 3D viewers, annotation systems, or pointing mechanisms, not only transmit information but also structure how knowledge is distributed and reconstructed across the group. When visual representations and interaction mechanisms make reasoning transparent, allowing participants to see what others see and understand why, they support the establishment of common ground.

In this thesis, the concepts of common ground and distributed cognition serve as a foundation for examining how interaction features in collaborative environments, such as viewpoint control, shared annotations, or deictic gestures, mediate understanding and coordination during DRs. Recognizing that cognition in DRs is inherently distributed underscores the need for technologies that accurately represent the product while also making the collaborative reasoning process itself visible and accessible to the participants.

2.4.3 AFFORDANCES AND MEDIATED ACTIONS IN DESIGN REVIEWS

As discussed in Section 2.1, from the viewpoint of Activity theory, design work can be described as a hierarchy of goal-directed levels, activities, tasks, and actions, each oriented toward a specific objective (Bedny & Harris, 2005; Qureshi et al., 2014). At the action level, designers modify or add information to design representations, such as rotating a 3D model, adding annotations, or highlighting a geometry issue, which

in essence translate to actions such as pressing buttons and clicking and dragging the mouse. The tools and representations used in a DR mediate the relationship between the participant and the design object, determining what can be seen, what can be manipulated, and how meaning is shared across the group (Kaptelinin & Nardi, 2006). This notion of mediation implies that the characteristics of the tools available in a DR, interfaces, control structures, and visualization modes, directly shape the types of actions that can be performed and, consequently, the trajectory of the overall design activity.

Affordances, introduced by Gibson (1979), refer to the action possibilities that an environment offers to an individual relative to their action capabilities. In Gibson's view, affordances point both ways: they are facts of the animal embedded in the environment. Later work in design and human-computer interaction built on this idea by distinguishing between the physical capability of an object and the perceived properties that suggest how it should be used (Gaver, 1991; Norman, 2013). In collaborative contexts, affordances extend beyond the individual to encompass social and procedural dimensions, describing what people can do together within a shared workspace (Leonardi, 2011).

Within the context of DRs, affordances serve as a functional framework for analyzing how tools shape collective activity. They define the range of possible interactions that participants can perform with both the design representation and each other. The presence or absence of specific affordances such as shared pointers, synchronized views, or independent navigation controls determines whether participants can jointly explore, indicate, or modify aspects of the design, and how easily they can coordinate their contributions. In this sense, affordances bridge the perceptual and procedural levels of collaboration, linking what participants can see and do with how shared meaning and coordinated action emerge.

Connecting this view with Activity theory clarifies how tool capabilities shape the structure of work. Activity theory explains the hierarchy of goals that guide DR work, from broader activities to tasks and the actions used to carry them out. Affordances explain which of those actions are actually possible in a given environment. This means that the way a task occurs in a DR depends on the affordances of the tools in use. For example, if a tool affords independent viewpoint control, participants can act on a task by inspecting a feature or issue themselves. If the tool does not afford this, participants must act through verbal requests. In both cases, the task is the same, but the available actions differ because the tool offers different possibilities. Building on this theoretical foundation, this thesis operationalizes the concept of affordances to define the essential capabilities of advanced collaboration tools. While some research in this domain provides granular, micro-level taxonomies of system functionalities (Horvat et al., 2022), this thesis intentionally focuses on a concise, high-level set of affordances as illustrated in Table 3.

Table 3. Example of identified affordances.

Affordances	Description of Action Possibilities
Shared Spatial Presence	Allows participants to perceive the 3D product model at 1:1 scale and maintain mutual awareness of each other's physical proximity and posture within the virtual environment.
In-scene Deictic Indication	Enables participants to use 3D pointers, avatars, and sketching tools to anchor their verbal feedback directly to the shared geometry.
Independent Navigation	Provides users with the autonomous capability to move their viewpoint around the model to inspect different areas, breaking reliance on a single presenter.

This theoretical framing serves as the foundation for Paper VI developed later in this research (fully detailed in Table 1 of Paper VI).

2.5 SYNTHESIS AND RESEARCH GAP

In conclusion, DRs act as critical convergence points within the PDP, governed by both procedural requirements and the human need for shared understanding. While digital tools have revolutionized the analytical depth of DRs, current screen-based practices often constrain the embodied and spatial interactions necessary for effective collaboration. Conversely, while immersive XR technologies restore presence and intuitive manipulation, they currently lack the precision and integration required for formal documentation and decision-making. The literature thus reveals a gap between the potential of advanced visualization technologies and their practical application in industrial workflows.

This thesis addresses this gap by investigating how collaborative technologies can be designed to support the specific communicative and cognitive demands of DRs. By framing the problem through the lens of asymmetric collaboration, this research moves beyond a comparison of "2D versus XR" to explore how heterogeneous systems can leverage the strengths of both. This approach acknowledges that effective support for DRs requires not just new visualization tools, but a holistic consideration of how these tools mediate interaction, coordinate attention, and integrate with the established social and procedural structures of engineering design.

RESEARCH METHOD

3 RESEARCH METHOD

This chapter presents the methodological foundation of the thesis. It begins by stating the research questions (Section 3.1) and objectives that guided the work. It then describes the overall research approach (Section 3.2) and the iterative progression across studies. Following this, the empirical and analytical methods are presented (Section 3.3), covering field observations, laboratory experiments, qualitative observations, and industrial workshops. The chapter concludes with a discussion of the data analysis strategies employed (Section 3.4) and the ethical considerations that governed the research (Section 3.5).

3.1 RESEARCH QUESTIONS

The aim of this thesis is to understand how collaborative technologies support interaction and coordination in DRs and to translate this understanding into frameworks that inform both technology development and practical use in industry.

To achieve this, the research has three objectives: (1) to identify and characterize the collaboration challenges that limit the effectiveness of DRs; (2) to determine and explain which technological capabilities address these problems and how they shape collaboration dynamics; and (3) to integrate these insights into a framework and guideline system and evaluate their clarity and applicability in industrial contexts. These objectives are captured in the overarching research question (RQ):

How can collaborative technologies support more effective and traceable design reviews?

The overarching RQ is further specified through three focused RQs:

RQ1 *What collaboration challenges arise in design reviews?*

RQ2 *What technological capabilities help overcome collaboration challenges in design reviews?*

RQ3 *How can these insights guide better use and development of digital tools for design reviews?*

Each research question is associated with a corresponding objective. Table 4 summarizes the relationship between the RQs and their corresponding objectives.

Table 4. Research questions, objectives, and descriptions.

Research question	Objective	Description
RQ1: What collaboration challenges arise in DRs?	O1: Identify and characterize the collaboration challenges that limit the effectiveness of DRs.	Understanding of recurring problems in DRs that define the problem space for improvement.
RQ2: What technological capabilities help overcome collaboration challenges in DRs?	O2: Determine and explain which technological capabilities address these problems and how they shape collaboration dynamics.	Evidence of how specific technological capabilities affect coordination, attention, and shared understanding.
RQ3: How can these insights guide better use and development of digital tools for DRs?	O3: Integrate insights into a framework and guideline system and evaluate their clarity and applicability in industrial contexts.	A validated framework and guideline set that connects collaboration capabilities with structured DR procedures.

3.2 RESEARCH APPROACH

This research follows a design-oriented mixed-method approach that combines conceptual, empirical, and participatory methods. The overall ambition is to understand how collaboration occurs during DRs, and to produce practical guidance on how DR support technologies can be developed and deployed in real-world settings. To ensure this guidance is grounded in actual operational constraints, the research is conducted in close collaboration with industrial partners.

The research draws from the Design Research Methodology (DRM), a methodological framework presented by Blessing and Chakrabarti (2009). DRM serves two primary purposes: first, it helps formulate and validate models and theories related to design phenomena, and second, it aids in developing and validating support systems based on these models to enhance design practice and outcomes, including education (Blessing & Chakrabarti, 2009). The value of DRM is its role as a framework that provides structure and direction for researchers to systematically explore and improve design practices, rather than simply serving as a collection of methods (Blessing & Chakrabarti, 2009).

Researchers who adopt DRM usually structure their work around four main activities. *Research Clarification* aims to provide the researcher with an understanding of the problem space by reviewing existing evidence and stating the goals for the work. In an initial descriptive study (*Descriptive Study I*) the researcher examines how the current system works in practice, going beyond what can be learned from the literature. Insights from these two activities inform the *Prescriptive Study*, in which support is proposed by the researcher to move the system toward a desired state. In DRM, *support* refers to the concrete aid developed by the researcher, such as a tool, guideline, or procedure that helps users carry out a task more effectively. Then, in a

secondary descriptive study (*Descriptive Study II*), the researcher investigates how the support works in practice and the extent to which it helps achieve the intended outcome.

In DRM, these activities are not strict stages that must be followed in a fixed order or quantities; rather, they are fundamental activities that the researcher is advised to carry out for a successful research outcome. In many design-oriented research projects, including this thesis, the activities serve as a guiding structure for combining different studies. The results of DRM typically take the form of prescriptions that can be instantiated as artifacts, frameworks, guidelines, or best practices that support some kind of design activity. In this research, the proposed support is provided as a framework and guidelines, where the framework refers to a structured conceptual model that provides a systematic way to understand, organize, and address complex problems within PDP, with particular focus on DRs, aiming to help organize knowledge, guide action, and support decision-making. As illustrated in Figure 12, the research process of this thesis was iterative, with the findings and limitations of each study directly motivating the query of the subsequent phases.

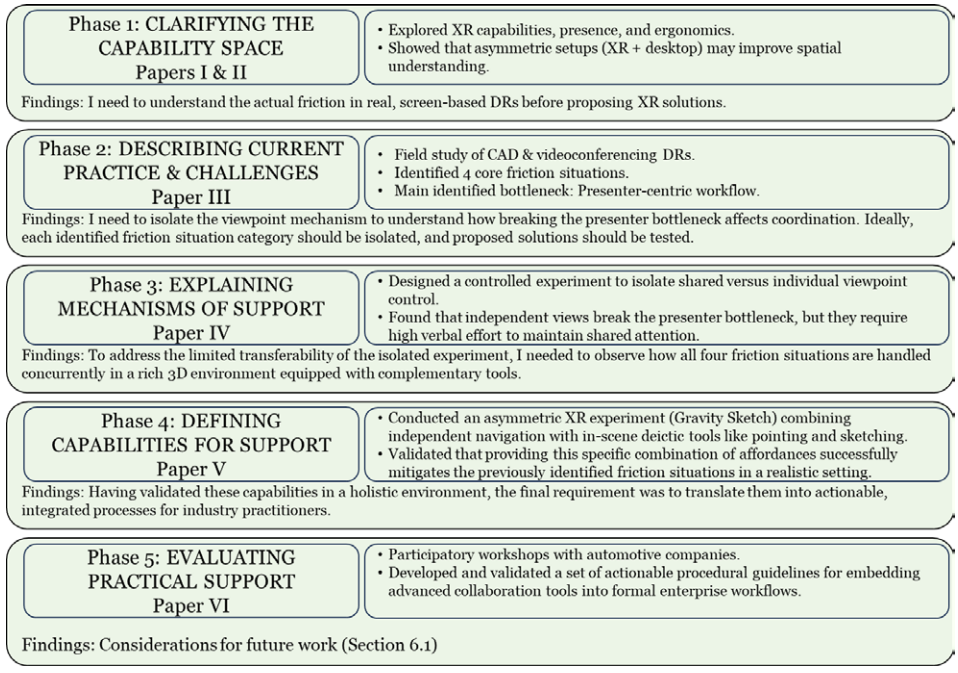


Figure 12. Visual representation of the research journey, illustrating how the findings and limitations of each study directly motivated the subsequent phases.

The research began with a Research Clarification phase triggered by the recognition that DRs face long-standing difficulties when conducted remotely ("Clarifying the capability space", Figure 12). The literature shows a growing interest in using XR for collaborative evaluation and, more specifically, in DRs, but most studies treated XR as a direct replacement for conventional 2D setups. In the initial stages of this thesis, instead, I explored how XR could complement existing workflows through asymmetric collaboration, where participants join from different devices and levels of immersion,

depending on their role. Early conceptual work mapped XR technologies and related their capabilities to DR tasks, using ideas of social, physical, and self-presence to describe how different configurations support interaction (Paper I). An experimental study then compared an asymmetric XR setup with a screen-sharing configuration and showed that allowing both XR and desktop participants to move freely in the scene can improve communication and mutual understanding (Paper II). Together, these studies clarified the initial capability space and suggested that viewpoint autonomy and asymmetric XR collaboration were promising directions to pursue.

Building on this foundation, the research moved into an initial descriptive study (Descriptive Study I) to capture the reality of DRs in industrial settings ("Describing current practice & challenges", Figure 12). A field study of distributed DRs at an automotive company defined recurring friction situations, namely: requesting specific viewpoints, indicating specific elements, expressing design change ideas, and evaluating ergonomics (Paper III). These observations, validated by a follow-up workshop with industrial partners, provided a grounded picture of the main collaboration challenges and defined the practical problem set addressed in the rest of the thesis. For example, the field observations revealed that relying on a single presenter to navigate the model caused recurring friction, as participants were forced to verbally describe the specific viewpoints they needed.

To address this issue, the research progressed to a set of Prescriptive Studies, focusing on developing and selecting technological support ("Explaining mechanisms of support" and "Defining capabilities for support", Figure 12). Although the methods used were experimental and observational, their specific role in this thesis was to inform the selection and design of the support capabilities. One experiment focused on viewpoint control and compared shared and individual navigation policies in a collaborative 3D task, studying how different viewpoint control configurations influenced communication patterns (Paper IV). A complementary study observed DR-like sessions in a shared 3D workspace where participants could use independent viewpoint control, pointing, sketching, and digital human models (Paper V). Together, the studies in Papers IV and V showed that viewpoint control and support for deictic reference actively shape how participants direct attention, refer to geometry, and coordinate their contributions in DRs. The studies also documented recurring ways in which participants combined these features to direct attention, refer to geometry, and express design changes, and they exposed typical problems such as viewpoint drift when navigation was not coordinated. Together, these studies identified the concrete interaction patterns that later work incorporated into core affordances and procedure elements in the framework and guidelines.

At this point, a framework outlining the requirements for advanced collaboration systems for DRs began to be defined. The framework focuses on three core affordances that recur across the work: (1) shared spatial presence around the same product model; (2) in-scene deictic indication to direct attention; and (3) independent navigation with presenter-follow to balance autonomy and coordination. Finally, the framework was instantiated into a set of practical guidelines (Paper VI). While the creation of these guidelines represented the culmination of the Prescriptive Study, the research then transitioned into a secondary descriptive study (Descriptive Study II) ("Evaluating practical support", Figure 12) to evaluate their impact. The goal of this study was to observe and describe how the proposed support actually functioned when introduced into a real workflow. To this end, real DR cases were reconstructed with

two automotive companies (Paper VI). These sessions were used to refine the wording, timing, and ownership of the proposed guidelines and to assess their clarity and feasibility in everyday work. The resulting guideline set connects interaction features directly to meeting structure and decision capture (Paper VI).

Throughout this progression, the activities of clarification, description, prescription, and evaluation were not followed in a strict sequence, but rather as overlapping functions that guided the query and design of each study. Table 5 shows a summary of the different activities carried out and the corresponding RQs they addressed. In conclusion, these research activities allowed the thesis to address RQ1 by identifying and characterizing the main collaboration challenges in DRs, RQ2 by determining and explaining which technological capabilities influence coordination and shared understanding, and RQ3 by integrating these insights into a framework and a validated set of guidelines for conducting DRs with advanced collaboration tools.

Table 5. Summary of research activities, objectives, and corresponding papers.

Research activity	**Role in the project**	**RQs / objectives**	**Papers**
Clarifying the problem and the capability space	Map collaborative technologies for DRs and identify promising capabilities and configurations.	Mainly RQ2, O2, with early input to RQ1.	I, II
Describing current DR practice and challenges	Show how distributed DRs are conducted in the industry and identify recurring collaboration problems.	RQ1, O1.	III
Explaining mechanisms of technological support	Analyze how configurations and features influence coordination, attention, and shared understanding.	RQ2, O2.	IV, V
Define the main ideas for supporting DRs	Formulate core affordances and relate them to existing DR technology research.	Bridge RQ2 and RQ3, O2 and O3.	V
Evaluating and refining practical support	Turn the framework into guidelines and test their clarity and applicability with the industry.	RQ3, O3.	VI

3.3 EMPIRICAL AND ANALYTICAL METHODS

This thesis employs a combination of empirical and analytical methods, each selected to address specific aspects of the RQs while respecting the constraints and requirements of the industrial context. The methodological design integrates field observations to capture the complexity of real-world practice, controlled experiments to isolate interaction mechanisms, and participatory workshops to explore, develop, and validate solutions with industrial partners. This mixed-method approach was a deliberate strategy to balance the need for experimental control with ecological validity, ensuring that the research could adapt to the specific environments where data was collected. By triangulating data across these different settings, the research aims to provide a holistic understanding of DRs that is both rigorously tested and practically relevant to the needs in industry. To provide transparency regarding the

scope and scale of these methods, Table 6 presents an overview of the empirical data collection across the included studies.

Table 6. Overview of empirical data collection throughout the research.

Paper	Study Type	Participants (N)	Sessions
Paper II	Laboratory Experiment	15 participants (Groups of 3)	5 sessions
Paper III	Field Observations	13-16 per session (mostly same participants)	15 sessions
Paper IV	Laboratory Experiment	60 participants (groups of 3)	20 sessions
Paper V	Qualitative Lab Observation	9 participants (groups of 3)	3 sessions
Paper VI	Industrial Workshops	11 participants (6 at Company 1; 5 at Company 2)	4 sessions

3.3.1 FIELD OBSERVATIONS

Field observations were conducted to understand how DRs are carried out in industrial contexts and to identify the recurring collaboration challenges that arise in actual work environments. Observation is a fundamental method in design and social research because it captures how activities occur in their natural context and reveals tacit practices that are often not accessible through self-report or controlled experimentation (Blomberg et al., 2017; Crabtree et al., 2012; Randall et al., 2007).

The motivation for using field observations was to ground the research in real industrial practice. While laboratory studies can reveal mechanisms of interaction, they cannot show how those mechanisms occur in practice when influenced by time pressures, organizational and technological constraints, and organizational dependencies. Observing DRs as they naturally occur allowed the research to capture both the formal and informal aspects of collaboration, including how participants shift attention, negotiate control of shared models, and respond to technical or communicative breakdowns.

Fieldwork was conducted at a Swedish automotive company over a three-month period. Fifteen distributed DRs were observed. The meetings were held using CAD software for visualization of component geometries, and videoconferencing for synchronous collaboration among distributed participants. Each session involved cross-functional participants, including engineers, managers, and specialists from different disciplines. I attended the meetings as a non-participating observer, taking detailed notes following a structured observation protocol derived from

ethnographically informed design studies (Blandford et al., 2016; Goodman et al., 2013). Notes included information about the sequence of actions, the participants involved, references to on-screen elements, requests for viewpoint changes, and moments of confusion or delay.

Immediately after each session, notes were expanded and structured in a standardized template that included context, duration, participants, observed interaction issues, and examples of verbal exchanges. The material was later coded using qualitative thematic analysis to identify recurring problems and categorize them by theme (Clarke & Braun, 2017). Detailed examples of this analytical progression (from raw meeting notes to refined notes and the final coding structure) can be found in Section 2 of Paper III. The analysis showed four major friction situations that repeatedly disrupted meeting flow: requesting specific viewpoints, indicating specific elements, expressing design change ideas, and evaluating ergonomics. These categories became the foundation for the subsequent stages of the research, directly addressing RQ1.

The advantage of field observation is ecological validity. It captures the real organizational setting, tool configurations, and power dynamics that shape how design teams collaborate. Observing real DRs also provided exposure to the subtle coordination work that often is difficult to capture in interviews, such as verbal shortcuts, gesture substitutes, or reliance on established team habits. This perspective allowed me to document not only the formal procedure, but the workarounds teams rely on to overcome the limitations in their existing tools for DRs.

The main limitation of field observation is the lack of experimental control. It is not possible to manipulate variables or to ensure that comparable situations arise across meetings. The role of the observer must also remain carefully balanced to minimize intrusion and bias (Hammersley & Atkinson, 2019). Additionally, confidentiality agreements limited the ability to record video or audio in some cases, making detailed behavior analysis dependent on comprehensive field notes.

Despite these limitations, the field observations provided the empirical basis and set the research directions. They supplied a realistic picture of how DRs are currently conducted, exposed the recurrent barriers to effective collaboration, and framed the problem set that guided subsequent experiments, workshops, and framework development.

3.3.2 LABORATORY EXPERIMENTS

Laboratory experiments were used to investigate how specific collaboration configurations influence communication, coordination, and task performance during DRs. Experiments are one of the most common research methods in design, cognitive science, and human-computer interaction when the objective is to isolate causal relationships and quantify behavioral effects (Goodman et al., 2011; Lazar et al., 2017; Oates, 2006). In this research, laboratory experiments were applied in two studies focusing on asymmetric collaboration (Paper II) and viewpoint control (Paper IV).

The motivation for using laboratory experiments was to obtain controlled evidence on how particular technological features shape collaborative behavior. Field data showed that certain recurring frictions, such as difficulties in referencing geometry or managing viewpoints, could be attributed to technical limitations in shared interfaces. The laboratory setting enabled reproducing these conditions in a simplified form and

manipulating a single variable, in this case, whether participants share or control their own viewpoint, while keeping all other factors constant. This allowed direct comparison of how different configurations affect the way participants communicate and coordinate.

Two main experimental setups were implemented. The first study compared distributed ergonomics evaluations conducted via a standard video call with an asymmetric configuration that combined VR and 2D screens (Paper II). The second study examined viewpoint control by assigning participants to one of two conditions: shared viewpoint control, where only one person could navigate the model, and independent viewpoint control, where all participants could move freely (Paper IV). In both cases, small groups of three participants collaborated on a structured DR-like task designed to reproduce the reasoning, interaction, and communication typical of DRs.

Quantitative data included task completion times, navigation and manipulation logs, viewpoint angle distribution, number of speaker changes, and distribution of talk time. These metrics were complemented by qualitative coding of speech transcripts to categorize words related to navigation, placement, and clarification. The combination of behavioral and conversational data enabled a detailed analysis of how different control policies influence collaboration patterns.

The main advantage of this approach was the ability to isolate dependent and independent variables and reduce confounding factors. Identical tasks were repeated across conditions, which increased reliability and made subtle behavioral differences more salient, such as shifts in turn-taking or the proportion of speech devoted to alignment. This controlled setting was essential because field studies alone could not reveal how specific capabilities, such as asymmetric device use (Paper II) or viewpoint control (Paper IV), directly influenced collaborative behavior. The experiments provided the causal evidence needed to interpret the friction observed in industrial practice and to link it to specific technological features. The main limitation was ecological validity. Real DRs involve complex decision processes, multiple roles, and longer timeframes than could be reproduced in a laboratory. Furthermore, industrial DRs are conducted by domain experts with deep professional knowledge, a level of expertise that is often difficult to recruit for controlled experiments. Consequently, the tasks used were simplified to ensure control and comparability, meaning that findings should be interpreted as evidence of mechanisms rather than direct reflections of industrial practice.

The experimental design was therefore guided by a trade-off between control and realism. Tasks were designed to remain abstract enough to generalize beyond a specific product, yet structured enough to require the same kind of spatial reasoning and verbal coordination seen for the navigation models part of the DRs. Roles were kept distinct to reproduce typical DR dynamics, such as one participant manipulating a model while others provide feedback. The resulting data revealed how key capabilities, such as viewpoint autonomy and asymmetric participation, influence communication structure, attention management, and collective understanding, providing evidence that later informed the interpretation of field data and the formulation of the framework.

3.3.3 QUALITATIVE OBSERVATIONS IN LABORATORY SETTINGS

Qualitative observations were conducted in laboratory environments to examine how participants use advanced collaboration tools during DR-like activities. This approach occupies a middle ground between controlled experimentation and open fieldwork. It allows researchers to study realistic interaction in a setting that remains structured enough for systematic observation and comparison (Hannes et al., 2022).

The motivation for using qualitative observation was to understand how participants naturally employ different collaborative features when working together in 3D environments. While field studies reveal existing problems and experiments test isolated variables, this approach captures how people collaborate when simultaneously using multiple features, including viewpoint control, pointing, sketching, and digital human models. The goal was not to measure performance but to observe how these tools shape discussion, attention, and shared understanding.

The study (Paper V) was conducted using Gravity Sketch, a multi-user 3D environment that allows asymmetric participation across XR and desktop devices. Each session involved three participants: one in XR and two on laptops. Participants were asked to conduct a design discussion on a familiar product model. The sessions reproduced key aspects of a DR, including presenting, inspecting, and commenting on a design.

All sessions were captured using screen recordings of both XR and desktop views, along with audio of verbal communication, which was subsequently transcribed. Observations focused on how participants used the system to reference geometry, coordinate viewpoints, and express design ideas. The analysis followed qualitative content analysis (Mayring, 2014), where special attention was put on patterns of coordination, attention management, and spatial reasoning. Specific attention was given to how participants combined visual and verbal cues to maintain mutual understanding.

The main advantage of qualitative laboratory observation is the balance between control and realism. It allowed the study of natural collaboration behavior while ensuring that all participants had equal exposure to the same tools and tasks. The environment was controlled enough to permit detailed comparison between teams but flexible enough to accommodate authentic collaboration.

The main limitation is that the sessions, although realistic, were not tied to actual industrial decisions. Participants interacted with simplified product models and shorter sessions than those typical of real DRs. The data therefore, represent behavioral patterns rather than full organizational processes.

Nevertheless, this method provided a crucial bridge between controlled and field research. It revealed how participants adopt and combine collaborative capabilities in practice and how these features influence communication flow. The observations informed the refinement of the proposed framework, confirming that the value of advanced collaboration tools lies in their ability to enable continuous, embodied interaction around shared 3D representations, rather than in their immersive properties alone.

3.3.4 INDUSTRIAL WORKSHOPS

Workshops with industrial partners were a key method for connecting empirical findings to practical needs and for validating the framework and guidelines developed in this research. Workshops are widely used in design and human-computer interaction research as structured yet collaborative settings that support the co-creation of knowledge and the integration of diverse perspectives (Kensing & Blomberg, 1998; Sanders & Stappers, 2014). They provide a forum where practitioners can jointly reflect on current practices, evaluate emerging ideas, and refine tools or procedures based on shared experience (Brandt & Messeter, 2004).

The motivation for using workshops was to ensure that the research remained aligned with industrial priorities and constraints, particularly concerning workflow integration and tool feasibility, and to validate whether the framework and guidelines derived from the studies could be realistically applied in industrial DRs. Workshops provided a setting where abstract findings could be discussed, challenged, and translated into tangible procedures.

The workshops followed an iterative process designed to move from exploration to validation (Paper VI). Initially, moderated discussions were conducted where participants from two companies systematically explored the tool's capabilities to identify necessary coordination routines. By actively testing actions such as following, pointing, and anchoring comments within the shared 3D environment, participants identified specific problems and proposed "candidate practices" to address them. These sessions established the link between the friction situations observed in field studies and the technological capabilities required to mitigate them. Subsequently, validation workshops were conducted to refine these candidate practices into a stable framework. In these sessions, participants engaged with physical guideline cards representing the proposed procedures. Through group discussion and direct interaction with the tools, participants edited the cards to clarify terminology, assign roles, and define boundary conditions, such as when to enforce specific view modes or how to handle documentation. This process ensured that the final guidelines, covering pre-read, attention management, documentation, integration, and modality, were grounded in the consensus of the industrial practitioners. This method follows established participatory design practices in which artifacts act as boundary objects to facilitate discussion and shared understanding (Ehn, 2008).

Data collection during the workshops included audio recordings, written annotations on cards which I primarily recorded, and field notes of group discussions. I analyzed the data using an iterative thematic analysis to identify consistent feedback, points of disagreement, and company-specific adaptations. The analysis resulted in a refined set of guidelines based on the participants' feedback, organized around five categories: pre-read, attention management, documentation and traceability, integration and handoff, and modality.

The main advantage of workshops is the ability to integrate multiple perspectives in a short time. They allowed researchers and practitioners to converge on actionable solutions and ensured that the resulting framework and guidelines remained both relevant and usable. Workshops also provided insight into organizational and procedural barriers to implementation that could not be observed in laboratory or field settings.

The main limitation is that workshops rely on participant availability and engagement. Discussions can be influenced by the composition of the group or by the immediate concerns of the companies involved. To mitigate this, the workshops were structured, and results were verified across different sessions.

Overall, workshops provided the bridge between empirical findings and practical application. They connected theoretical insights about collaboration and capability use with the everyday routines of industrial DRs, ensuring that the final guidelines were not only conceptually sound but also operationally feasible.

3.4 DATA ANALYSIS

3.4.1 QUALITATIVE ANALYSIS

Qualitative analysis was central in this research because the goal was not only to identify whether collaboration in DRs worked but to understand how people made it work. The focus was on the reasoning and coordination processes that occur during DRs. Qualitative interpretation was therefore necessary to analyze the meaning of those interactions, how participants oriented to each other, and how they resolved ambiguities in the design representations.

The decision to use qualitative approaches followed the logic of the research questions. RQ1 and RQ2 required an examination of how participants negotiated meaning and interpreted the design. Observation, transcription, and iterative interpretation allowed the reconstruction of what participants were trying to do in specific moments of interaction.

The analytical process was guided by the tradition of interpretive design research (Dourish, 2001; Suchman, 1987) and by the practical structure proposed by Miles et al. (2013). Coding, defined here as the process of labelling data to find patterns, was used as a way to think with the material. This involved moving back and forth between field notes, transcripts, and analytic summaries to refine emerging ideas such as “viewpoint drift” or “anchored remarks” until they were consistent across cases. Matrices, such as the Practice × Problem Matrix or the Role × Move Matrix from Paper VI, were used to make the patterns in the sessions easier to see and compare.

The main trade-off of this approach was between interpretive depth and consistency. A more formalized analysis protocol, for example, could have increased replicability by imposing rigid coding units, but it would have constrained interpretation to pre-defined categories and missed the evolving, context-dependent nature of collaborative reasoning. Grounded Theory was also considered early in the project, but its requirement for theoretical saturation conflicted with the multi-study design and the industrial confidentiality conditions that limited participant access (Glaser & Strauss, 2017). Instead, a pragmatic, theory-informed coding strategy was adopted, which allowed insights from Activity Theory and affordance theory to inform category construction while remaining open to emergent themes (Clarke & Braun, 2017).

To account for the influence of the researcher on the data, I adopted a reflexive stance by explicitly defining and managing my operational role across the different studies. For the field observations (Paper III), I maintained a strict nonparticipating observer

role to minimize intrusion and capture naturalistic behavior. In contrast, during the co-creation workshops (Paper VI), I shifted to a facilitator role. In this capacity, I managed the structure of the sessions and ensured balanced participation across device types, while consciously refraining from imposing specific workflows to ensure that the resulting guidelines reflected the participants' procedural reasoning rather than my own. Working closely with industrial partners also required navigating the professional sensitivities inherent in DRs, where engineers expose their work to peer critique. Recognizing these dynamics was essential for distinguishing between technical coordination issues and social hesitation. Field notes included reflections on situational factors, such as team hierarchy or meeting purpose, which were later used to contextually interpret the patterns of speech and control found in the data. Additionally, triangulation across datasets and cases helped to ensure that the categories represented recurring collaboration mechanisms rather than single context artifacts.

Overall, qualitative analysis in this research was an interpretive bridge between observation and design. It generated the analytical foundation and theoretical insights used throughout the thesis, including friction situations and core technological capabilities that informed the proposed procedures.

3.4.2 QUANTITATIVE ANALYSIS

Quantitative analysis was used to examine how specific collaboration configurations influenced participants' observable behavior and task performance. It provided a structured way to describe what changed when certain technological features, such as viewpoint control or asymmetric participation, were introduced. The purpose was to identify measurable behavioral patterns that could later be interpreted qualitatively.

There were two main motivations to use quantitative analysis methods. First, they provided a degree of objectivity that complemented the interpretive nature of the qualitative work. Second, they helped to detect patterns that might otherwise remain unnoticed, such as variations in navigation activity or differences in speech dynamics between control conditions. These patterns served as anchors for subsequent qualitative interpretation, helping to connect changes in interaction flow to specific tool configurations.

The design of the quantitative analysis followed the logic of controlled experimentation in human-computer interaction (Lazar et al., 2017). Data was collected through system logs, audio recordings, and coded transcripts. Measures included completion time, number of navigation inputs, viewpoint angle dispersion, speaker changes, and talk time per participant. These variables were chosen because they reflect coordination effort and communication rhythm.

A key methodological consideration was balancing ecological validity and analytical precision. Controlled studies allow comparisons between well-defined conditions but risk simplifying the complexity of real design work. The tasks used in this research were therefore designed to reproduce essential coordination demands of DR, joint spatial reasoning, reference to shared objects, and distributed control with a defined central leader or presenter, while remaining abstract enough to enable replicability. The resulting measures captured generalizable aspects of collaboration, even though they did not replicate the full context of industrial DRs.

In this analysis, descriptive statistics, such as means and standard deviations, were used to summarize the data and show general behavioral tendencies. In Paper IV, independent samples t-tests were used to compare completion time, navigation inputs, speaker changes, and viewpoint dispersion between shared and individual control. These tests provided a simple way to examine whether the two configurations produced meaningfully distinct behavioral patterns.

Across studies, the aim was to use statistical methods that clarified behavioral tendencies without reducing the complexity of the task dynamics. Traditional usability measures, for example, task duration, detection counts, and performance ratings, were included to provide a broader picture of each session and to situate the more interaction-focused metrics. All quantitative results were interpreted alongside observations and transcripts, maintaining continuity with the qualitative analysis and ensuring that numerical patterns were explained through the same conceptual lens. This quantitative layer was essential for the mixed-method approach (Paper II, Paper IV), as it provided objective structural evidence that complemented and validated the behavioral insights derived from the qualitative analysis.

3.5 ETHICAL CONSIDERATIONS

The research presented in this thesis involves human participants across various settings, including industrial workplaces and controlled laboratory environments. To ensure the integrity of the research and the well-being of participants, the work was conducted in accordance with the ethical guidelines established by the Swedish Research Council (Vetenskapsrådet, 2024) and the General Data Protection Regulation (GDPR).

For all empirical studies, whether field observations, workshops, or experiments, informed consent was a prerequisite. Participants were provided with clear information regarding the purpose of the study, the nature of the data being collected (e.g., audio, video, screen recordings, or interaction logs), and how this data would be used. It was explicitly stated that participation was voluntary and that they reserved the right to withdraw from the study at any time without penalty or requirement to justify. In the industrial studies (Papers III and VI), it was emphasized that the participants' individual performance was not being evaluated.

Given the qualitative nature of much of the data (videos, transcripts, and field notes), strict measures were taken to protect participant anonymity. All personal identifiers were removed or pseudonymized during the transcription and analysis phases. Raw data files, such as video recordings and signed consent forms, were stored securely on password-protected computers accessible only to the research team and will be deleted upon completion of the research project. In publications and presentations, data is presented in an aggregated format or using anonymized quotes to prevent the identification of specific individuals.

A significant portion of this research was conducted in collaboration with automotive companies, involving proprietary product data and internal workflows. To respect industrial confidentiality, non-disclosure agreements (NDAs) were signed where necessary. In the dissemination of results, including this thesis and appended papers, sensitive details regarding specific product features or strategic company decisions have been omitted or generalized to protect trade secrets without compromising the

scientific value of the findings. Images of CAD models or prototypes used in the studies were either cleared for publication or replaced with generic representations.

For the studies involving XR (Papers II and V), specific attention was paid to the physical and physiological well-being of participants. The potential risk of motion sickness was mitigated by designing sessions with appropriate durations, providing clear instructions on adjusting headsets for visual comfort, and allowing participants to take breaks or stop the experiment immediately if they experienced discomfort. The physical laboratory space was organized to prevent collisions with real-world objects during the experiment.

RESULTS

4 RESULTS

This chapter presents and summarizes the six appended papers that form the thesis results. Table 5 in Chapter 3 shows how each paper connects to the three research questions and the corresponding objectives. The research progressed in a sequence. Papers I and II clarified the problem and the capability space, primarily contributing to RQ2 and providing early input to RQ1. Paper III focused on current DR practice and challenges, addressing RQ1 by identifying recurring collaboration problems. Papers IV and V examined how specific configurations and features influence coordination, attention, and shared understanding, supporting RQ2. Papers V and VI then defined the main ideas for supporting DRs by formulating core affordances and linking them to existing technology research, bridging RQ2 and RQ3. Finally, Paper VI evaluated and refined practical support by turning the framework into guidelines and testing their clarity and applicability in industry, addressing RQ3. This is summarized in Table 7.

Table 7. Summary of key findings and research questions addressed per appended paper.

Papers	RQs	Key findings
I	RQ2 (mainly) and RQ1	Mapped XR technologies to collaborative design tasks using dimensions of physical, social, and self-presence.
II	RQ2 (mainly) and RQ1	Demonstrated that asymmetric XR setups and digital human models improve collaborative ergonomics assessment.
III	RQ1	Identified four core friction situations in distributed DRs.
IV	RQ2	Revealed that independent viewpoint control breaks bottlenecks but increases the need for verbal coordination.
V	RQ2 and RQ3	Observed that a small set of features can help to overcome friction.
VI	RQ3	Developed and validated actionable industrial guidelines for integrating advanced tools into DR workflows.

4.1 PAPER I

Paper I was presented at the 10th Swedish Production Symposium in 2022 and is published in the conference proceedings.

The study aimed to organize and conceptualize the variety of XR technologies proposed for product design collaboration, focusing on how their capabilities relate to the specific demands of DR tasks. Extending established research on the dimensions of presence in virtual environments, the paper introduced a framework to classify XR systems by how they support different modes of collaboration.

Paper I argues that XR technologies should not be understood as replacements for conventional 2D screen setups but as complementary tools whose usefulness depends on the task and the type of presence required. Social presence is important for teamwork and idea generation, physical presence supports geometric or spatial evaluations, and self-presence is necessary when bodily awareness is required, such as in ergonomics assessments.

By mapping how different XR configurations provide these forms of presence, Paper I shows that effective use of XR in DRs depends on aligning the level of immersion and interactivity with the task's communicative and evaluative needs. The resulting framework provides a foundation for identifying which XR capabilities, such as shared spatial scenes, body-based interaction, and co-presence features, are most relevant for supporting collaboration in DRs.

4.2 PAPER II

Paper II was presented at the 7th International Digital Human Modeling Symposium in 2022 and is published in the conference proceedings.

The study examined how XR and digital human models can support ergonomics evaluations during collaborative DR-like activities. The focus was on understanding how different ways of sharing control and viewpoint affect teamwork and communication in these sessions.

A shared 3D environment was developed, allowing participants to join either through a VR headset or a standard computer screen. The environment included the assembly of a vehicle component and a digital human model for assessing ergonomics. Each participant was assigned a specific role: one acted as the builder, handling and positioning parts, another as the assessor, checking ergonomics, and the rest as observers, suggesting improvements.

The experiment compared two setups. In the first setup, participants, one in VR and the rest on desktop devices, could move freely and explore the model independently. In the second setup, the flat screen participants could only see what the person in VR was seeing, without the ability to move the camera or change viewpoints (simulating a videoconferencing screen-sharing setup). Fifteen participants tested both setups.

The results show that when all participants were able to move freely, collaboration improved. It was easier to discuss design and ergonomic issues, and participants

reported less effort in explaining the spatial aspects of the model. The setup, in which flat-screen participants were limited to the VR user's view, often caused more errors and a worse user experience.

The study demonstrates that granting participants autonomy in movement in collaborative virtual environments can improve coordination and understanding during a DR-like activity focusing on ergonomics assessments. It also illustrates how combining VR and traditional interfaces can create a more balanced and efficient collaborative environment.

4.3 PAPER III

Paper III is published in the journal Empathic Computing in 2025.

The study investigated how distributed DRs are conducted in real industrial settings when teams use a combination of CAD software and videoconferencing applications. The goal was to identify recurring moments of friction, meaning instances where communication or coordination falters and the progress of the review is temporarily hindered.

The research was conducted at a Swedish automotive company over a three-month period using the field observation method described in Section 3.3.1. Fifteen distributed DRs of a fixture component were observed. The meetings were conducted using CATIA for CAD visualization and Microsoft Teams for communication. The meetings followed a structured format with defined roles: a meeting manager led the session, a product expert presented the design, and a multidisciplinary audience of engineers provided comments. Only the product expert controlled the CAD model while others followed via screen sharing.

Data was collected through systematic observation and note-taking during the meetings, with particular attention to the specific interaction barriers defined in the observation protocol. The qualitative analysis of these observation notes revealed four main categories of friction situations. The first was requesting specific viewpoints, when participants needed to see a particular angle of the model but had to rely on the presenter to find it. The second was indicating specific elements, referring to difficulties in pointing out exact components or features on the shared screen. The third was expressing design change ideas, where participants struggled to communicate or visualize proposed modifications. The fourth was evaluating ergonomics, where the limitations of 2D screens made it difficult to assess human interaction with the product. These findings were later validated in an industrial workshop.

The study provides one of the first systematic descriptions of how friction emerges in real distributed DRs using common industrial tools. By identifying where and how these interruptions occur, it offers concrete evidence of the limitations of current CAD and videoconferencing setups and establishes a problem set that guided the design of later studies on advanced collaboration tools.

4.4 PAPER IV

Paper IV is under review in the journal Theoretical Issues in Ergonomics Science.

The study tested how viewpoint control shapes collaboration in a controlled 3D task. Sixty participants, organized in groups of three solved a color-coded cube puzzle in a collaborative virtual environment. One participant was the builder, and two were guiders (simulating a DR situation). Two conditions were compared (ten groups per condition). In Individual Viewpoint Control, all participants could navigate their own view. In Shared Viewpoint Control, only the builder controlled the view and guiders saw the builder's screen.

Measures included completion time, accuracy, clicks for navigation and manipulation, viewpoint angle logs, number of speaker changes, talk time per participant, and transcript analysis.

Results showed no reliable difference in completion time between conditions, but clear differences in collaboration patterns. When guiders could move their own camera, they needed to coordinate their views with the builder before giving placement instructions. This led to more navigation talk and higher turn-taking as participants aligned perspectives. With a shared view, everyone saw the same image, so alignment talk was almost absent, and guidance became short, direct, and limited to placement commands.

The data supported these observations across several measures. The analysis showed more speaker changes and longer guider talk time in the individual control condition, indicating greater interaction effort to maintain shared understanding. In contrast, with shared control, the guider spoke in condensed, directive utterances. Navigation logs recorded many more left-click rotations and higher dispersion of camera angles for guiders under individual control, compared to near-zero variation when the view was shared. Content analysis of speech also showed a shift: in the shared view condition, utterances focused on object placement ("put the blue one on top"), while in the individual control condition they mixed navigation and placement ("rotate left, now put the green one on the red face").

Completion times did not significantly differ between conditions, suggesting that both setups were equally efficient for this task. However, the communication styles were clearly shaped by viewpoint policy. Individual control encouraged autonomy but required more coordination to maintain common ground, while shared control limited autonomy but simplified communication. The results show that viewpoint configuration has a strong influence on how teams coordinate attention and share spatial understanding, even when the overall task time remains the same.

4.5 PAPER V

Paper V was presented at the 27th International Conference on Human-Computer Interaction in 2025 and is published in the conference proceedings.

The study examined how specific collaborative features influence the flow of distributed DRs in a shared 3D workspace. It focused on four core capabilities: independent viewpoint control, collaborative pointing, 3D sketching, and digital human models for ergonomics evaluation.

A distinctive strength of the study was the participant group. All participants were final year undergraduates studying Product Design Engineering who had used the multi-user 3D environment software Gravity Sketch extensively during their course. They had received structured training in both XR and the desktop version of the tool's collaborative workspace, which allowed them to join the same 3D scene also from a flat screen. Over the preceding two months, they had practiced DR-style collaboration in this environment and were working on their own product design projects (headphone design) when participating in the experiment. This combination of tool proficiency, shared project context, and established team dynamics enabled the observation of realistic collaborative behavior without the typical learning curve that affects many XR studies.

The sessions, involving nine participants divided into three groups, involved one participant joining in XR and two others participating from desktop devices. This setup enabled observing how asymmetric collaboration unfolded in a DR-like activity and how participants used the available features to discuss design aspects. The XR participant could move freely, point, and sketch in 3D, while desktop participants used a pointer and added sketches or comments.

The results showed that participants used the available features to manage attention, clarify design ideas, and ground their discussions in the shared 3D scene. Independent viewpoint control allowed participants to explore the model individually while maintaining group awareness through the follow feature. Pointing and sketching were often combined with verbal references to specify design changes, and digital human models were used mainly to discuss proportions and ergonomic aspects.

The study shows that collaborative features such as pointing, sketching, and viewpoint autonomy support rich and continuous communication during DR-like activities. Rather than demonstrating direct efficiency gains, the findings highlight how these capabilities shape the rhythm and structure of discussion, enabling participants to coordinate naturally within the shared 3D environment.

4.6 PAPER VI

Paper VI is submitted to the journal Research in Engineering Design.

This paper develops and validates a set of guidelines for conducting DRs using a set of features. The work was carried out with two automotive companies through an iterative process of familiarization, observation, and validation sessions. The goal was to turn previously identified coordination problems and tool capabilities into practical, verifiable procedures that fit real industrial contexts.

The study began by reconstructing each company's DR case in Campfire, a shared 3D environment that supports independent navigation, pointing, and anchored comments. After basic training sessions, preparatory runs linked each feature to specific DR tasks such as preparing agenda views, capturing remarks, and launching models from the product data system. These sessions produced initial candidate practices that were then discussed in group workshops with engineers and managers.

The group discussions were analyzed to identify which practices solved recurring issues such as viewpoint drift, unclear ownership, or weak traceability. The first workshop produced preliminary routines, while the second focused on wording,

timing, and ownership, using printed cards for participants to edit directly. Confirmation across both companies established the final guideline set. Stable practices included starting reviews in follow mode, using “bring everyone” controls to regain focus, limiting simultaneous pointers, keeping remarks and snapshots with the geometry, and launching scenes directly from the system of record. Conditional practices, such as meeting in space (meaning that participants join the DR session inside the shared 3D environment rather than relying on a parallel video call) or webcam use, were kept as optional or discouraged.

The resulting guidelines are organized around five connected areas: pre-read, attention management, documentation and traceability, integration and handoff, and modality. Together they describe a continuous process that begins with sharing the correct product model in advance, continues with coordinated viewing and discussion during the session, and ends with traceable documentation and smooth transfer of outcomes to the company’s product data system. The study provides a grounded, transferable procedure that aligns with how DRs are prepared, conducted, and documented when using advanced collaboration tools.

DISCUSSION

5 DISCUSSION

DRs serve as formal governance checkpoints in the product development process, where cross-functional teams evaluate design proposals against product requirements. When conducted in distributed settings using traditional screen-sharing tools, these sessions often suffer from a set of specific friction situations caused, at least partly, by the limitations of the tools used. Overcoming these collaboration challenges is necessary to increase the effectiveness of DRs. By identifying and characterizing the recurring friction situations in distributed DRs, this thesis demonstrates how specific technological capabilities (e.g., scene deictic indication) can enable effective shared attention and coordination. Furthermore, by translating these core affordances into a validated set of guidelines, this work provides industrial organizations with practical guidance to integrate advanced collaboration tools into their established workflows, ensuring that DRs remain effective, traceable, and grounded in actual operational constraints.

This chapter discusses the main findings of the thesis and reflects on how they contribute to answering the research questions. It first connects the results of Papers I to VI to RQ1 to RQ3 and their related objectives. This is followed by reflections on the methodological approach applied in this thesis, discusses limitations and generalization of the results, and summarizes the main contributions of the work.

5.1 ADDRESSING THE RESEARCH QUESTIONS

As outlined in Section 3.1, this research addresses the overarching question: *How can collaborative technologies support more effective and traceable design reviews?* The following sections synthesize the findings from the appended papers to answer the three focused research questions (RQ1–RQ3) and their corresponding objectives defined in the Research Method (Chapter 3).

5.1.1 RQ1: WHAT COLLABORATION CHALLENGES ARISE IN DESIGN REVIEWS?

Paper III addresses RQ1: *What collaboration challenges arise in design reviews?* through a field study involving the observation of distributed DRs at an automotive company. This study was informed by the literature mapping in the Background (Chapter 2).

The analysis of the observations in Paper III indicates that the main difficulties in distributed DRs are structural. The recurring collaboration problems arise from how

control, attention, and access to the shared product model are distributed. The sessions followed a presenter-centred pattern. One person controlled the CAD model and screen sharing, while others observed and issued verbal requests. This structure simplified meeting management but made other participants dependent on the driver whenever they needed a specific view or wanted to test an idea themselves.

From this field study, four friction situations were identified as recurring patterns that interrupt the flow of work:

- Requesting specific viewpoints
- Indicating specific elements
- Expressing design change ideas
- Evaluating ergonomics

These friction situations have the same underlying constraints: access to the shared representation is centralized in one operator, and spatial information must be communicated verbally. These constraints lead to delays, the need for repeated clarification, and misunderstandings, especially in distributed settings where gaze, gesture, and physical pointing are not available.

The analysis also shows that similar interaction structures appear in hybrid sessions, where co-located participants gather around a single screen, and online participants join remotely through videoconferencing applications. Even when the tools differ, the interaction logic remains presenter-centred and sequential, as the shared visual focus is still controlled by a single driver, limiting direct interaction for other participants. This supports the view that many collaboration challenges in DRs are rooted in how work is organized around shared representations rather than in distributed work alone.

Together, these results answer RQ1 by providing a grounded description of how collaboration challenges arise in DRs and how they are linked to the structure of interaction around the shared model.

5.1.2 RQ2: WHAT TECHNOLOGICAL CAPABILITIES HELP OVERCOME COLLABORATION CHALLENGES IN DESIGN REVIEWS?

RQ2: *What technological capabilities help overcome collaboration challenges in design reviews?* is addressed through a combination of conceptual work, controlled experiments, and qualitative studies in collaborative virtual environments. Chronologically, the research began by exploring how XR solutions could be leveraged to address collaborative challenges (Papers I and II), a direction initially influenced by the scope of the funding project (research project PLENUM, supported by Vinnova in Sweden within the Advanced Digitalization program). However, these early explorations highlighted a critical gap: collaboration challenges in DRs had rarely been investigated systematically without a preconceived technological solution in mind. Consequently, the research stepped back to identify and characterize these friction situations (as detailed under RQ1). Building on the results of that research, the later studies (Papers IV and V) returned to the technological dimension to analyze how

specific capabilities, such as viewpoint control and deictic referencing, address those identified problems.

Paper I classifies XR technologies according to how they support different forms of presence and collaboration. It outlined that these technologies should be matched to the demands of specific DR tasks, rather than treated as replacements for traditional screen-based setups. In particular, it distinguishes the value of social presence for communication, physical presence for spatial and geometric judgement, and self-presence for tasks with strong ergonomic or bodily components. This mapping helps explain why certain configurations of devices and tools are better suited to specific DR activities. Rather than assuming that higher levels of presence universally improve outcomes; applying this mapping in practice shows that each dimension introduces concrete gains as well as necessary trade-offs in how teams coordinate (Table 8).

Table 8. Concrete examples of gains and trade-offs across presence dimensions.

Presence Dimension	Possible Gain in Design Reviews	Possible Trade-off / Implication
Physical Presence (Spatial and geometric judgement)	Allows intuitive, 1:1 scale assessments of depth and volume, breaking the presenter-centred bottleneck by enabling independent exploration.	High physical immersion makes interacting with traditional 2D PLM interfaces or taking formal meeting minutes highly cumbersome, necessitating asymmetric (desktop) support.
Social Presence (Communication and awareness)	Deictic indication (pointing/sketching) allows reviewers to anchor their feedback directly to the 3D geometry, drastically reducing verbal clarification time.	Because users can navigate independently, high social presence does not guarantee shared focus; it often requires significantly more verbal effort ("navigation talk") to re-establish a shared view.
Self-Presence (Ergonomics and bodily components)	Embodying avatars or manipulating digital human models enables highly intuitive assessments of reach, posture, and sightlines using natural movements.	High sensory fidelity and bodily tracking demand greater technical setup and can introduce cognitive overload, occasionally distracting from purely analytical reasoning.

Paper II focuses on collaborative ergonomics assessment in virtual environments. It shows how digital human models and multi-user presence support communication about posture and reach, and how independent navigation and shared scenes allow ergonomics discussions to be carried out more flexibly than in conventional screen-sharing setups. Although the use case focuses on ergonomics rather than general DRs, the study shows that digital human models and shared virtual environments can address the friction in evaluating ergonomics by making human-product interaction visible and shareable as part of the review.

To understand the trade-offs of moving from a presenter-view-centered workflow to an independent view workflow, Paper IV isolates viewpoint control as a central

capability. In a controlled 3D puzzle task, it compares shared and individual viewpoint configurations. While task completion time did not differ significantly, the communication patterns were clearly shaped by the viewpoint configuration. Shared viewpoints led to concise placement instructions with fewer clarifications, while individual viewpoints led to more exploratory but also more demanding navigation talk. This shows that viewpoint control affects both how participants talk about space and how they balance efficiency with exploration. Furthermore, it also showed that both navigation modalities are necessary, since they provided different advantages.

Due to time constraints and the inherent difficulty of translating the effects of isolated capabilities into complex collaboration situations, Paper V extends this logic to a richer collaborative environment using Gravity Sketch. In these sessions, one participant per group used a head-mounted display and two participants used desktop devices. All participants had access to independent navigation, follow options, pointing tools, sketching, and digital human models. The study shows how participants combined these features to manage attention and clarify design ideas. Independent navigation allowed participants to inspect details on their own. Deictic tools such as pointers and sketches allowed them to anchor proposals directly in the scene. Digital human models were used to reason about ergonomic aspects in a shared frame of reference. Friction around viewpoint requests was almost absent because each participant could control their own view.

Across these studies and the industry workshops, a consistent set of capabilities emerges as central to addressing the problems identified under RQ1:

- Shared spatial presence in a common three-dimensional scene
- In scene deictic indication through pointing and sketching
- Independent navigation with explicit options to follow a presenter

In summary, RQ2 is answered by showing that a small set of technological capabilities strongly shapes collaboration in DRs. The research followed a logical progression: first mapping the capability space (Paper I), then isolating specific mechanisms in controlled settings (Paper IV), and finally observing their use in realistic tasks (Paper V). These capabilities allow participants to control their own view, to indicate geometry directly in the scene, and to share a sense of presence in the product space. Their value lies not only in improved immersion but in how they change the structure of coordination and attention.

5.1.3 RQ3: HOW CAN THESE INSIGHTS GUIDE BETTER USE AND DEVELOPMENT OF DIGITAL TOOLS FOR DESIGN REVIEWS?

RQ3: *How can these insights guide better use and development of digital tools for design reviews?* is addressed by combining the findings from RQ2 with real DRs and by testing the resulting support with industrial partners. This was realized in Paper VI, which defines the concept of advanced collaboration tools and translates the identified capabilities into practical guidelines evaluated with two automotive companies.

To provide a consistent basis for these guidelines, Paper VI first establishes a functional definition of advanced collaboration tools. Rather than categorizing tools by hardware (e.g., VR vs. desktop), advanced collaboration tools are defined by the

affordances they provide: shared spatial presence, in-scene deictic indication, and independent navigation. This definition shifts the focus from specific devices toward the collaborative functions that the system must support, framing the technology as a means to enable specific group behaviors.

However, defining these collaborative functions is only the starting point. While the feasibility of such features is often demonstrated in laboratory settings, there is a critical need to understand how they integrate with product data systems, how decisions are captured, and how they support accountability across product development stages. Consequently, Paper VI addresses the integration of advanced collaboration tools into established DR workflows by developing and validating a set of guidelines. Real DR cases from two automotive companies were reconstructed in a shared three-dimensional environment. Engineers and managers worked with guideline cards that linked specific features, such as follow mode, “bring everyone”, pointers, and anchored comments, to concrete actions and responsibilities before, during, and after the DRs. The workshops were used to refine the wording, timing, and ownership of these practices and to test whether they were clear and feasible in everyday work. The resulting guideline set is organized around five areas: pre-read, attention management, documentation and traceability, integration and handoff, and modality. Together, they describe a continuous process that starts with preparing the correct product model and views, continues with coordinated use of viewpoints and deictic tools during the session, and ends with traceable capture and transfer of outcomes into the product data system.

RQ3 is answered by systematically bridging the gap between technical potential and organizational procedures. The conceptual definition of advanced collaboration tools first established the necessary technological baseline, which served as the foundation for the final guidelines (Paper VI). These guidelines directly address the need for process integration by linking the identified affordances into a validated workflow. This progression ensures that the contribution is not merely a set of tools, but a cohesive framework that demonstrates how advanced collaboration tools can be successfully embedded into the process of industrial DRs.

5.2 METHODOLOGICAL REFLECTIONS

The combination of field, laboratory, and participatory methods was essential for capturing the complexity of collaboration in design reviews. This approach accounted not only for the social complexity of these sessions, but also for the challenges of distributed settings and the mediating role of technology. Field observations grounded the research in actual industrial practice, exposing that coordination issues may not appear in structured laboratory and development settings (Blomberg et al., 2017; Randall et al., 2007). Laboratory experiments isolated the specific elements of the DR process to examine how different technological configurations shape behavior (Lazar et al., 2017). Qualitative observations in laboratory settings provided a bridge between both, providing insight into how features interact in practice, while workshops enabled translation and validation (Sanders & Stappers, 2014).

This mixed-method structure allowed theory and practice to evolve together. However, it also involved trade-offs. Laboratory experiments produced data on how technological interventions affected specific aspects of collaboration, but necessarily

simplified real-world variability (Goodman et al., 2011). Field studies offered rich context but limited generalizability. The synthesis depended on careful iteration between abstraction and detail, theory and observation.

Analytically, the integration of qualitative and quantitative reasoning was not only a matter of triangulation but of conceptual alignment. Quantitative data indicated what changed, while interpretive analysis explained why and how. This interplay helped move from describing isolated effects to formulating a coherent model of mediated collaboration (Creswell & Clark, 2017).

5.3 LIMITATIONS AND GENERALIZATION

The research was bounded by context and scope. Most empirical studies involved small groups working on simplified design tasks or simulated reviews. The industrial validation did not resolve this limitation, but helped assess whether the proposed guidelines were understandable and agreed upon in real practice. Large-scale adoption studies would be needed to evaluate the organizational impact of the proposed guidelines. Furthermore, the industrial collaboration was specific to the automotive sector, and the participating teams consisted primarily of manufacturing and production engineers. As a result, the findings reflect the specific coordination needs, toolchains, and workflows of that domain and may not fully transfer to other contexts, such as early-stage conceptual design or service-oriented sectors. Closely related to this is the CAD-centric nature of the research, as the 3D model served as the primary artifact around which the DRs were studied. While this focus aligns with the engineering contexts investigated, it means that the interaction principles identified may not directly generalize to other collaborative environments, such as DRs where physical prototypes are the primary artifact, or where non-geometric or abstract data is being reviewed.

A further methodological limitation is the absence of formal quantitative measurements for individual differences, such as spatial cognitive ability, including the Mental Rotation Test (Peters et al., 1995), prior experience with XR tools, or perceived immersion and presence questionnaires (Witmer & Singer, 1998), which can act as confounding variables in 3D collaborative tasks. In the controlled experiments, the risk of varying baseline proficiencies was mitigated through standardized onboarding and training tasks designed to establish a common operational baseline before data collection. While these individual differences can influence aspects of user performance and interaction speed, the primary analytical focus of this research was directed toward qualitative communication patterns and structural coordination mechanisms. Consequently, while this unmeasured variance limits potential claims regarding absolute efficiency or task completion times, it does not diminish the validity of the overarching collaboration patterns observed. Another limitation lies in the technological maturity of available tools. Many XR and collaborative platforms evolved during the research period, influencing what could be tested. For instance, early studies (Papers II and IV) relied on custom-built prototypes because commercial platforms with integrated collaborative capabilities were not yet available. This required the manual implementation of features that are now standard, potentially limiting the user experience compared to the commercial-grade software used in the later stages of the research. Results should therefore be interpreted as principles of

interaction independent of the technical constraints or feature sets of the specific tools employed.

Furthermore, the decision to maintain a general approach by collaborating with multiple companies, rather than conducting deep action research with a single partner, involved specific trade-offs. While working with two primary automotive company partners (and several other companies in a lesser capacity) helped ensure broader applicability, it also introduced significant barriers regarding intellectual property and data security. These constraints limited the depth of direct intervention, as gaining full access to live, confidential product development cycles was more restricted than it might have been in a single-company embedded setup where trust and integration could be established more deeply.

CONCLUSION

6 CONCLUSION

This thesis contributes to design research in three main ways:

First, it provides a grounded understanding of collaboration challenges in product development processes, with particular focus on DRs. It demonstrates that the friction in DRs is not only technical but also structural. It shows how recurring issues are linked to the centralized organization of interaction around shared product model representations.

Second, it defines key technological capabilities. It identifies the specific affordances that enable teams to coordinate DRs effectively in both immersive and non-immersive environments. Specifically, it establishes that shared spatial presence, in-scene deictic indication, and independent navigation are critical for effective DRs.

Third, it provides a validated framework for process integration. It translates these insights into a concrete guideline system. This system connects interaction-level affordances to procedural and organizational structures, showing how advanced collaboration tools can be successfully embedded in industrial practice.

Together, these three outcomes represent the core academic and industrial contributions of this research. Academically, the thesis shifts the understanding of distributed DR friction from a technical issue to a structural one, while identifying the specific affordances required to resolve it. Industrially, it provides organizations with the actionable guidelines needed to successfully integrate these advanced tools into formal workflows. Furthermore, on a societal level, this research contributes to more inclusive and sustainable engineering practices. By demonstrating how specific affordances lower the barrier to interacting with complex 3D data, this work supports the democratization of the DR process, allowing non-engineering stakeholders to participate more fully. Additionally, by enabling highly effective spatial evaluations to occur remotely, it reduces the reliance on business travel and physical prototyping, aligning with broader sustainability goals.

6.1 FUTURE WORK

As an outlook for future work, first, the proposed guidelines need to be validated and refined across broader contexts. Since this research was grounded primarily in the automotive sector, future work should test the framework across other industries, such as consumer electronics or construction, to understand how product complexity varies across industries and how it affects collaboration needs. Additionally, the guidelines

should be evaluated in different phases of the product lifecycle, including early-stage DRs, where creative flexibility is likely to be more central.

Second, future research should deepen the investigation into how DRs are documented and how outcomes are operationalized. While this thesis addressed the interaction during DRs, more work is needed to explore the technical and procedural mechanisms for capturing decisions. Research should focus on how advanced collaboration tools can be integrated with PLM systems to automate the creation of records, facilitate the assignment of tasks, and ensure that decisions made are effectively acted upon in the subsequent engineering work.

Finally, future studies would benefit from adopting more immersive methodological approaches. While the current work relied on observations and workshops, expanding the research through longitudinal action research or in-depth ethnographic studies would provide deeper insights. Embedding researchers within a design team for an extended period would reveal how these collaboration practices evolve over time, how teams appropriate the tools in unexpected ways, and how the guidelines can be sustained as part of the organizational culture.

6.2 CLOSING REFLECTION

Reflecting on the journey of this research, a central lesson is the importance of looking critically at technological potential. It is not enough to introduce a new tool and assume improvement. One must rigorously question why it improves the process. Without deeply understanding the mechanisms of interaction and knowing exactly what the tool is doing to the team dynamics, there is a risk of misinterpreting the benefits or solving the wrong problems. Technology must be interrogated, not just implemented.

In my opinion, looking at the broader landscape, the tools used to create hardware are undeniably evolving. While the development of CAD, PLM, and related tools currently outpaces industrial adoption, the integration of collaborative features into standard CAD systems appears inevitable. Slowly but surely, the definition of engineering tools is shifting from solitary workstations to shared digital environments. Eventually, likely, these features will become the standard rather than the exception.

Realizing this shift, however, requires navigating the inertia of established processes. It is important to acknowledge that existing workflows, however imperfect, are resilient because they function. They get the job done. Resistance to change is often a rational defense of a proven system, and new approaches bring new risks. Therefore, we must not look down on the current way of working; rather, we need to identify and understand the parts that are currently effective, as this knowledge is essential for moving on to a new and better way of working. Yet, relying solely on incremental improvements to legacy methods has its limits. To borrow a metaphor, one cannot invent the lightbulb by simply improving the candle. To move forward, the industry must eventually find the courage to stop optimizing the friction of the past and embrace a fundamentally different way of working.

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INCLUDED PUBLICATIONS

PAPER I:
THE SCHEMATIZATION OF XR TECHNOLOGIES IN THE CONTEXT OF COLLABORATIVE DESIGN

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The Schematization of XR Technologies in the Context of Collaborative Design

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Abstract. Recently, the concept of Industry 5.0 has been introduced to complement, among other things, Industry 4.0's focus on efficiency and productivity with a focus on humans in digital design and production processes. The inclusion of human interaction with digital realities, extended reality (XR) technologies, such as augmented reality (AR) and virtual reality (VR), can play an essential role in Industry 5.0. While rapid advances in XR technologies are solidifying and finding their place in the product and production development process, terminology and classification scheme remain under-determined. As a result, there have been numerous classifications of XR technologies from different perspectives, but little widespread agreement. They have been classified by their level of immersion or how well they meet a specific purpose (such as training). In addition to that, the classifications are usually made for one particular field (e.g. marketing, healthcare, engineering, architecture, among others). Therefore, to set the basis for future research, it is essential to identify and outline the dimensions that intervene in product and production design in regards to XR facilitated collaboration. With the ideas proposed in this paper, we want to identify basic concepts that classify a collaborative XR system by analyzing how users interact with the environment and other users. Our motivation is that collaborative design involves not only the physical dimension but also a social dimension. Defining when an XR system contributes to increasing social and/or physical presence could clarify and simplify its categorization.

Keywords. Virtual Reality, Augmented Reality, Extended Reality, Collaborative design

1. Introduction

Recently, the concept of Industry 5.0 has been introduced to clarify Industry 4.0's focus on efficiency and productivity with a focus on humans in design and production processes. According to the EU, Industry 5.0 complements the concept of Industry 4.0 but putting research and innovation at the service of the transition to a sustainable, human-centric and resilient European industry [1]. While there are many technologies involved in the manifestation of Industry 4.0 and 5.0, one cluster of technologies that are often discussed is extended reality (XR) technologies. XR is an umbrella term that has become used to refer to virtual reality (VR), augmented reality (AR), and mixed reality (MR) technologies. The lines that separate the individual XR technologies can be blurry, and several research groups have suggested definitions and classifications of these terms [1-4]. Some definitions focus on specific hardware or software features of XR systems. In contrast, others focus on how interactions mediate between users and virtual

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environments, as in the Virtuality continuum proposed by Milgram and Kishino [4] (Figure 1).

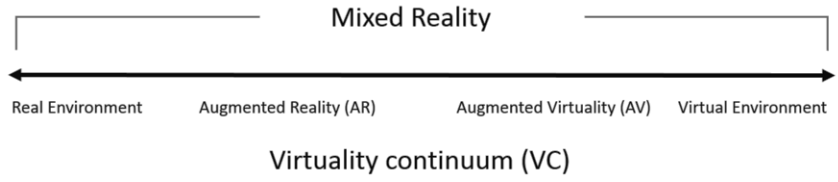


Figure 1. Virtuality continuum [2].

As industries adopt the goals of Industry 5.0, they might begin to introduce XR technologies to create a tighter interface between humans, digital tools and virtual environments. The application areas for XR are diverse, including training, meeting, collaboration, visualization, design, and testing. As XR technologies become increasingly accessible, it becomes essential to compare and decide which technologies fit a given application context and user group. Often recommendations regarding technologies are formulated in terms of expected use cases and general industry contexts. Focusing on specific applications can help avoid comparisons that are not valuable to stakeholders; as such, it is often preferable to explore categorizations and metrics of XR systems within candidate use contexts (e.g. healthcare or industry), although there are attempts to define more general ones

Categorizing hardware or software technologies into clusters can help to some extent but do not typically focus on the user's kind of interactive experience to accomplish a given task. Building a 3D digital prototype may differ between AR (where the real world is partially visible) and VR (where the real world is entirely obscured). However, this kind of distinction between systems only provides insight into what is visible to the user and the possible type of immersion. Immersion, in this case, is a measure of what percentage of a user's senses are filled with virtual information [3]. Immersion may indicate the number of digital interactions possible in a given technology but not the quality of those interactions. A metric that may provide insight into digital interaction experiences is *presence*. Presence is a concept that has been important in XR research since at least the 1980s [4]. With some oversimplification, presence is the experience of the virtual world *as if* it were the actual world. In the context of design, presence can assess how close interaction with a digital artifact in XR will align with similar interactions with the real artifact in the real world. Moreover, presence can provide insights into interactions and experiences of the design process. This latter point is crucial in collaborative design, where multiple stakeholders contribute to the design process.

Collaborative design can be understood as the decision process beginning with the conception of a product and moving through various design iterations where multiple designers work together to integrate the perspectives of multiple stakeholders. Collaborative design often requires regular meetings, user testing in various environments, and careful coordination of design phases and steps. XR can be a valuable tool for reducing the number of in-person meetings and allow for greater access to insights for diverse stakeholders in a wide range of contexts (simulated and real). In this paper, we expose the idea that presence is an essential dimension for classifying XR systems for deciding which XR tools and technologies are suitable and beneficial to specific collaborative design processes. We provide a preliminary overview of work defining presence and distinguishing it from immersion. We then discuss methods for

measuring different kinds of presence. We conclude with a discussion of how presence can be used to compare XR technologies and, importantly, how the concept of presence could also be applied to collaborative design processes to characterize the needs of designers and other stakeholders in the design process.

2. Presence and immersion

Many definitions for presence have been proposed, though there is no specific, widely accepted definition. The concept of presence was introduced in 1980 in the context of telepresence [4]. Presence in this context was related to the feeling of the operators when operating machinery remotely, often using cameras and screen-based presentations of the remote machinery's context. Witmer and Singer defined presence in 1998 as relating to this initial concept of presence and teleoperators: "the subjective experience of being in one place or environment, even when one is physically situated in another″ [5]. Steuer gave one early definition of presence in the context of VR: "the extent to which one feels present in the mediated environment, rather than in the immediate physical environment″ [6]. A more comprehensive definition of presence was developed by approximately 200 scholars in an online discussion about presence:

> Presence (a shortened version of the term "telepresence") is a psychological state or subjective perception in which even though part or all of an individual's current experience is generated by and/or filtered through human-made technology, part or all of the individual's perception fails to accurately acknowledge the role of the technology in the experience. Except in the most extreme cases, the individual can indicate correctly that s/he is using the technology, but at "some level" and to "some degree," her/his perceptions overlook that knowledge and objects, events, entities, and environments are perceived as if the technology was not involved in the experience. Experience is defined as a person's observation of and/or interaction with objects, entities, and/or events in her/his environment; perception, the result of perceiving, is defined as a meaningful interpretation of experience [7].

This more extended definition helps clarify what many researchers are reaching for when discussing presence and how difficult it can be to capture what that feeling is in words. While each of the definitions so far has its strengths and weaknesses, we will follow the definition of presence introduced by Kwan Min Lee in 2004: "psychological state in which virtual objects are experienced as actual objects in sensory or non-sensory ways″ [10]. This definition is helpful for XR collaborative design because it is extended to apply to three dimensions of presence which are:

- *Physical presence*: psychological state in which virtual physical objects are experienced as actual physical objects in sensory or non-sensory ways.
- *Social presence*: psychological state in which virtual social actors are experienced as actual social actors in sensory or non-sensory ways.
- *Self-presence*: psychological state in which virtual self are experienced as actual self in sensory or non-sensory ways.

Understanding a collaborative design process and what tools and technologies best support that process requires consideration of more than just the interaction between product and designer. The collaborative design also includes interactions between designers that can interact with (and between) stakeholders. Thus, both the physical and social presence dimensions should be considered when considering XR technologies for collaborative design. Moreover, some XR design activities, such as testing products in a virtual environment, would likely benefit from a higher feeling of self-presence.

Presence should not be confused with immersion. As mentioned above, immersion measures how much one or several senses are "taken up" by a digital experience. Immersion can be related to the technological features that cause a flow state between the virtual environment and the user. More sensation from the digital environment may result in a greater connection to the digital environment. Immersion is mainly related to senses, and it is an objective measure of sensory feedback provided by technology. Presence and immersion are often related and may even be proportional to one another, in that greater immersion often accompanies a greater feeling of presence and vice versa.

3. The measure of presence and factors that influence the presence

While presence has been discussed since the early 1980s and many definitions have been suggested, few measures of presence have been developed and validated. Further, a lot of the research into presence in XR was carried in the early 2000s, when even the best VR technology was significantly limited compared to many of today's consumer-grade systems (i.e., narrower field of view, lower frame rate, lower resolution, higher weight, less content, less comfortable, etc.). In the early 2010s, XR technologies underwent a paradigm shift that resulted in a significant technological advance and cost reduction. While XR research on presence prior to 2015 should not be ignored, it should also be carefully considered in its original context and thoughtfully applied and re-validated with modern XR systems. Thus, while presence is a valuable dimension of XR for collaborative design, measuring and validating presence is ongoing and rapidly developing.

Physical presence has been measured in various ways, including questionnaires, interviews, and physiological responses. Research regarding measures of physical presence date from at least the '90s. One of the first studies developed a 6-item questionnaire to measure the feeling of physical presence and a 7-item questionnaire to measure the fidelity of interactions in VR when the refresh rate was changed [9]. Witmer and Singer [6] also designed what they called a "Presence Questionnaire" to measure the physical presence in VR as well as an "Immersive Tendencies Questionnaire" to measure the differences in behavioural tendencies of the participants when they experience different levels of physical presence. Witmer and Singer's questionnaires seemed to indicate correlations between physical presence and immersion. They also observed a weak but consistent correlation between task performance and their measure of physical presence, indicating that task performance improved with a physical presence. Slater, McCarthy and Maringelli also developed questionnaires to explore physical presence [10], finding a positive correlation between physical presence and amount of body movement. They also identified a correlation between task complexity and gender, with women reporting a higher feeling of physical presence when performing more complex tasks in a virtual environment. There is criticism around using questionnaires to measure physical presence. Some researchers argue that measuring physical presence with

questionnaires post-experience can lead to a significant bias and inaccurate data [11]. One way to address this concern is to compare participant responses to the questionnaires during a VR experience and after a VR experience. This approach indicates that responses during and after are similar, although there is less variability when questionnaires are used during the experience [12]. There is also some research relating to physiological responses. This research aims to avoid subjectivism in measuring physical presence by tying physical presence to physiological measures such as heart rate, blood pressure and galvanic skin response [13]. Grassini and Laumann did a systematic literature review about the relation between questionnaires measures and physiological correlations of presence [14]. They concluded that most of the academics had chosen questionnaires, with Witmer and Singer's Presence Questionnaire being the most prevalent. Among the physiological measures, electroencephalography (EEG) was the most frequently used. Grassini and Laumann also mention that at the current state of research, no physiological measure has collected enough evidence to be considered "good enough" to be reliably used alone, without the user giving their subjective evaluation of the experience [14]. Generally, presence studies find a positive correlation between the realism, interactivity level of the virtual environment, technology level, etc., and the feeling of physical presence

There has been some research into social presence in XR, but this area is underexplored compared to physical presence in XR. Research has shown that a higher level of immersion or physical presence does not necessarily relate to a higher feeling of social presence. The main difference between physical and social presence is that it requires a co-present entity that one can readily take the intentional stance towards. When there is another intentional agent, a "person" of some kind, one can experience a social presence in the digital environment. The other agent may be a natural person, either in the same room or joining remotely, or an artificial agent of some kind. The research into social presence indicates that the experience of social presence may be correlated to both physical presence and psychological distance between the subjects. Regarding virtual environments, qualities that lead participants to perceive that they are together with another person seem essential to achieve a social presence [15]. There is less research into the experience of social presence across XR technologies and how the specific hardware affects or relates to social presence.

There is less research relating to self-presence and its measure, though questionnaires seem to be the primary method for measurement as with other types of presence. A related concept of embodiment is likely related and, in some cases, the same as self-presence, though the similarity between these concepts is not guaranteed. Rabindra Ayyan Ratan and Béatrice Hasler developed what they called the Standard Self-presence Questionnaire [16]. This same questionnaire was used to measure self-presence in collaborative environments [17]. They found that one's virtual body seems not to affect self-presence. In addition, they also concluded that social presence seems to be related to self-presence.

4. Presence in XR collaborative design

Due to the numerous collaborative situations involved, collaborative XR design systems can be very complicated to put together and develop. For instance, a meeting using XR to discuss comfort aspects of vehicle seats may involve different considerations than a

meeting using XR to discuss the impacts of changes to steering wheel configurations on driver visibility. The discussion of comfort may include looking at seat shape in detail, perhaps looking inside the seat with some form of virtual x-ray or cutaway visualization. A physical prototype, combined with AR or MR, may be helpful, but it should have features relevant to comfort. Driver visibility, on the other hand, may require a model of the entire vehicle, which the meeting participants can sit inside of and walk around. These two meetings might require different levels of presence to have a successful outcome. Moreover, not all the participants in each meeting might require the same level of either physical, social or self-presence to play their role in the meeting successfully.

Moreover, collaborative XR systems are usually asymmetric (i.e. different users across the team use different XR equipment [18]). In design teams, not everyone in the collaborative process has the same task or equipment. Moreover, different tasks usually involve different equipment (especially if we consider that XR technologies have a specific purpose). For example, an ergonomics team that assesses a product's physical and cognitive efficiency likely requires various types of body tracking systems and high-fidelity VR equipment. In contrast, the mechanical engineering team who design 3D models of mechanical parts might require versatile VR visualization equipment to seamlessly step in and out of virtual meetings to discuss design aspects. This asymmetry in the technology used also implies an asymmetry in the feeling of presence. At the same time, some users might have a high feeling of presence (both social and physical), others might have a lower feeling of presence. This is not undesirable; it can be desirable if each role participating in the design process has what is needed to perform that task most efficiently and effectively. Following the ergonomics team and the mechanical designers' example, the ergonomists can require a high level of physical and self-presence to experience the product as a user would do to run usability tests and ergonomics analyses. On the other hand, the mechanical designers' team does not require those physical or self-presence levels. However, they would need certain levels of social presence to have productive meetings in the virtual environment.

Therefore, different types of collaborations will have different kinds of necessities. Meetings require a higher feeling of social presence to be engaging [19], and design tasks require a higher sense of physical presence to be beneficial [20]. Therefore, if we identified what is needed to engage a higher feeling of social and physical presence and to which extent each type of presence is required in each situation, the development of XR collaborative systems could potentially be more efficient. They could be classified in a better way. To the best of our knowledge, there are very few XR collaborative systems developed or conceptualized with the concept of presence as a basis.

4.1. Example of tackling a situation with the proposed idea

In practice, identifying the right XR technologies for a collaborative design process should involve considering the type and amount of presence required for the design process. In order to further clarify how presence may relate to different XR design scenarios and how thinking in terms of presence leads to some important questions, we will briefly consider two common types of multi-user XR setup.

Figures 2 and 3 show two different collaborative situations that could occur during a collaborative process. In situation one, the users are together in the virtual environment. They can see a representation of each other, and they hear each other.



Figure 2. Collaborative situation 1.

The set-up in situation two is similar, except that the users share physical space and virtual space. They also can see a representation of each other in the virtual environment; however, they hear each other through the real world.

While in situation one, all interaction is mediated through a virtual environment, in situation two, the visual interaction occurs in the virtual environment, and the auditive interaction occurs in the physical environment.



Figure 3. Collaborative situation 2.

The feeling of presence will be different in each scenario, and most likely, each user's outcome and experience will be different. How does each type of situation affect the feeling of presence of each user? How does the sense of presence of each user affect their performance in the virtual environment? A collaborative design system should handle these two situations in a way that feels natural to the users to collaborate efficiently.

5. Conclusion

In this paper, we propose that in deciding on XR technologies to implement in collaborative design contexts, physical, social and self-presence should be considered. Translating a collaborative design process to an XR environment can be challenging, especially considering the number of interactions that can occur. Identifying what kind of presence is needed for a given collaborative design context to appropriate technologies according to the type of presence they create could be a valuable idea to approach this problem.

While there is still no widely agreed upon and validated measure of presence, we suggest that identifying and validating a measure of presence will be useful for collaborative XR design as presence might be a critical dimension for deciding which kind of design context or technology that might be more beneficial. In essence, we hypothesize that the requirements on a given collaborative design context, the factors of the virtual environment that need to be adapted, or the equipment that would be best may be well supported by the concept of presence.

Much research was carried in the field of presence in the 90s and early 00s. Current research incorporates many of the ideas and theories formulated back then. However, more research and unification around the concept of presence is needed [14]. Further conceptualization and measures on the feeling of physical, social and self-presence can be especially beneficial to apply the ideas proposed in this paper.

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PAPER II: IMPROVING THE EFFICIENCY OF VIRTUAL-REALITY-BASED ERGONOMICS ASSESSMENTS WITH DIGITAL HUMAN MODELS IN MULTI-AGENT COLLABORATIVE VIRTUAL ENVIRONMENTS

Improving the efficiency of virtual-reality-based ergonomics assessments with digital human models in multi-agent collaborative virtual environments

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Abstract

Often new digital tools are introduced alongside existing tools and workflows to augment and fill gaps in current processes. Virtual and augmented reality (XR) tools are currently being deployed in this way within design processes, allowing for interactive visualization in virtual environments including the use of DHM tools. Currently, the focus is on how to implement XR as a stand-alone tool for single user scenarios. However, in collaborative design contexts, screen-based and XR tools can be used together to leverage the benefits of each technology maximizing the potential of multi-user design processes. XR allows for an immersive exploration of designed objects in 3D space, while screen-based tools allow for easier notetaking and integration of additional non-3D software and meeting tools. Ensuring that these technologies are integrated in a mutually beneficial manner requires a framework for determining the best combination of technologies and interfaces for diverse design teams. This paper presents a framework for performing collaborative design reviews in a digital environment that can be accessed using both XR and 2D screen devices simultaneously. It enables asymmetric collaboration to provide each design team member with the technology that best fits their workflow and requirements.

Keywords: Collaborative design, Asymmetric collaboration, Extended Reality

Introduction

Developing a product is a complicated process that often requires collaboration from various stakeholders to succeed (Redante et al., 2019). Often, stakeholders in large organizations collaborate with globally dispersed teams in the product design process. Moreover, recent events, such as the Covid-19 pandemic and awareness of climate change, have further boosted interest in reducing the air travel for in-person meetings (Becken & Hughey, 2022). As a result, there is increasing demand for tools that can effectively mediate remote collaboration in all stages of the design process.

Design reviews are one stage of the design process that can often require collaboration among multiple stakeholders and have traditionally depended on in-person meetings and videoconferencing tools. In design reviews, several stakeholders come together to consider and critique a current design before production plans are finalized. The design review process is an essential control point for any design project to transit

from one stage to another and can include not only assessing the physical properties of designed artefact but also ergonomics and human factors through the use of DHM tools (Huang, 2002). Often, the designers and production managers performing these reviews are not geographically co-located, and remote collaboration tools can improve workflows when travel is not possible or not prefered (Tea et al., 2021). For many companies, videoconferencing apps and their corresponding screen-sharing capabilities constitute state-of-the-art for this kind of remote collaboration. Extended reality (XR) tools have recently been incorporated into collaboration workflow (Cooper et al., 2021). However, they are typically integrated alongside existing videoconferencing workflows and treated as another screen to be shared with online collaborators (Chen, 2021). While this approach is functional, it limits and misses many key benefits that XR can introduce into a design process. While a more integrated XR solution is needed to access these benefits, it is also important to consider that not everyone will benefit from using XR technologies in a collaborative design process (Burova, Mäkelä, et al., 2022). As a result, we anticipate that future collaborative design processes will involve a mixture of technologies resulting in various asymmetries between collaborators. Because this kind of intentional mixing of technical systems is new in the context of collaborative design, more frameworks are needed for organizing technologies, users, and user roles (Reski et al., 2022). In turn, such a framework can support determining user needs and assessing implementations for multiuser asymmetric (i.e., collaboration using different devices).

This paper presents an initial framework for asymmetric XR collaboration and applies it to a simple collaborative design review in a digital environment using both XR and 2d screen devices simultaneously. The central idea of this framework is to enable asymmetric collaboration to provide each design team member with the technology that best fits their workflow and requirements. We ran a small pilot study to validate the framework using an existing workflow at the Chinese-Swedish company CEVT.

Method

While some tools for multiuser XR and asymmetric design exist (Burova, Palma, et al., 2022), there is little guidance regarding how these tools might be best used for design teams. Discussions with CEVT gave us insights into their current workflow and needs, and we developed a prototype software accordingly. After that, we evaluated the software in a controlled environment to check the viability of the proposed framework. This process is illustrated in Figure 1.

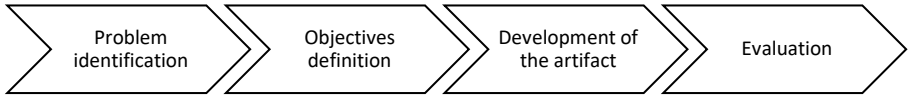


Figure 1. Development process

Problem specification and initial prototype

Some companies, including CEVT, have started including XR in their design review processes to support assessment of ergonomics and assembly feasibility within DHM tools. The expectation is that the use of digital tools and XR technology will allow for more efficient multi-user remote collaboration. However, current XR applications in DHM have limited multiuser interfaces, minimal screen sharing options, limited interaction options and underdeveloped workflows from CAD to XR tools.

CEVT currently implements XR-based design reviews by having one person in XR share their screen over a videoconferencing app with different stakeholders (see Figure 2). These stakeholders (the audience) watch as the person in XR (the VR navigator) assembles the product component by component. After each component is assembled, the VR navigator places a DHM manikin in an expected position and pose for that assembly step. Once the appropriate stakeholders have evaluated an assembly step and identified issues and risks for the current component, they move on to the new component.

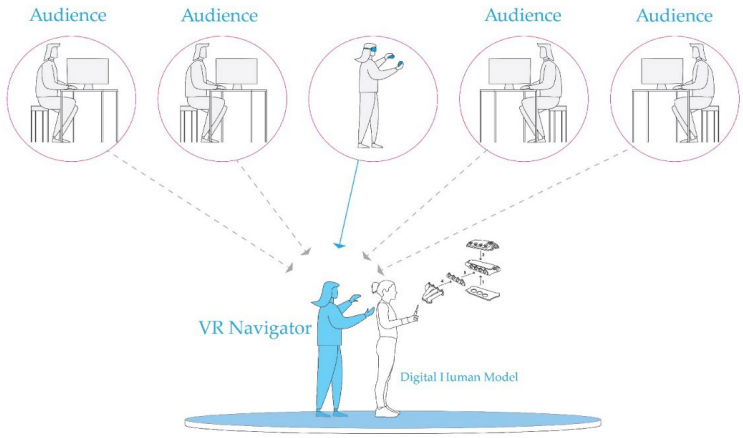


Figure 2. Illustration of current design review approach.

Having only one person able to navigate the environment (the VR navigator) and limiting the audience's view to the VR navigator's perspective are significant limitations in this workflow. The audience includes operators, manufacturing engineers, mechanical engineers, and ergonomists, among others. Not everyone needs the same kind of information or interaction capabilities, and their needs may not match the aims and

focus of the VR navigator. As an initial division of roles, we identified that some users needed more direct engagement with the 3D environment and that others needed to split their attention among other software tools that are not and may not be integrated into the 3D environment. As such, for some users XR tools are likely be beneficial but for others, the tools may be a hindrance, requiring frequent changes between the XR device and a 2D screen. In both cases, the ability to freely navigate the 3D scene appeared potentially valuable meaning that it may not be beneficial to limit free movement to only the XR users. Thus, it may be helpful if XR users can navigate the digital environment by using controllers or their bodies and screen-based users can use a keyboard and mouse to navigate the digital environment. Taking these insights, we developed a prototype multiuser asymmetric design review environment, as illustrated in Figure 3.

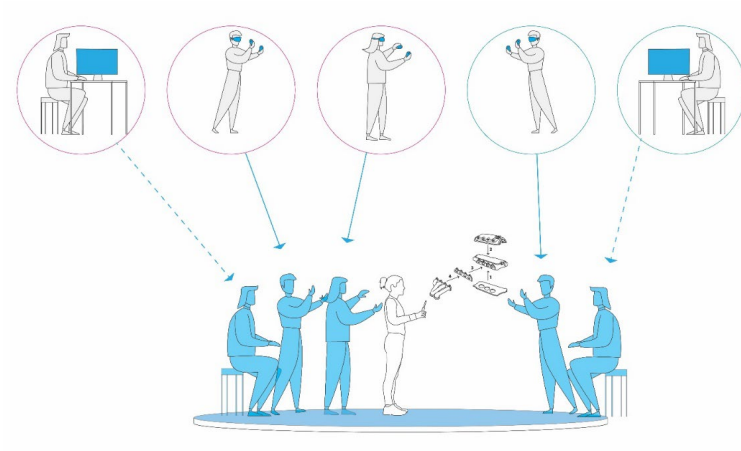


Figure 3. Illustration of software prototype roles and interfaces

As an initial solution, we developed a shared digital environment that the users can connect to either using a PC with a traditional monitor or an XR device (Figure 4). All users can navigate the environment, have avatar representations, and communicate using voice. The avatars allowed users to see one anothers location in the environment but were not intended to be a detailed representation of the users body state. Only users who access the digital environment using XR equipment can grab and move objects. Object models based on 3D CAD designs could be imported into the environment and simple interactions (grab and move) could be specified for the objects as needed. A digital human manikin is included in the environment which can be manipulated by moving the black spheres (visible in figure 4) attached to its hands and head allowing for visualization of static poses in an assembly process. In the current implementation, all the objects and

manikin movements are visible to all the users connected to the digital environment, though visibility could be limited based on user role or even clearence status within the company.

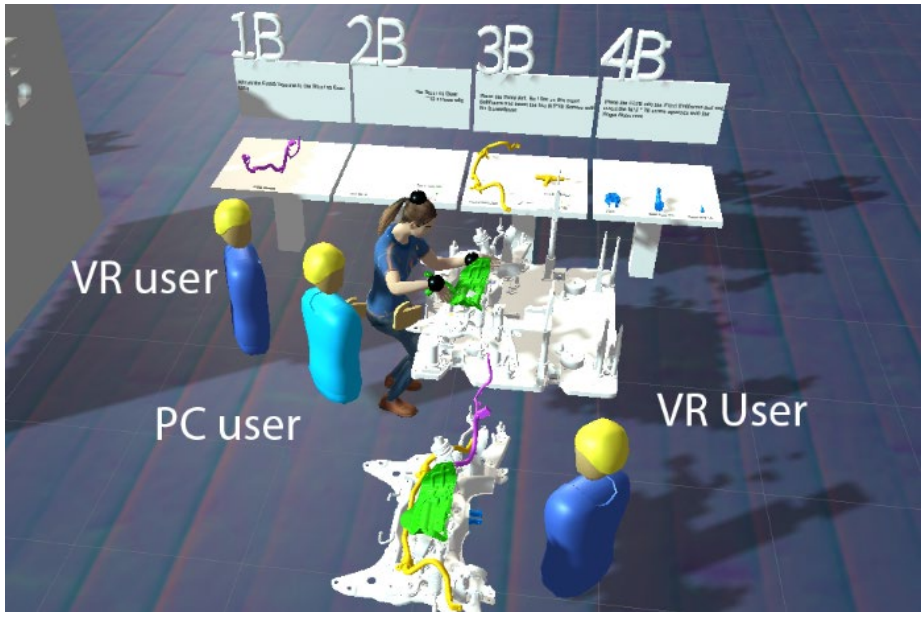


Figure 4. Digital environment with 2 XR users and one PC user

Material used in the development

We developed the prototype using Unity 2020.3 (Unity technologies, San Francisco, California). The software was built to run on either a Windows PC with a monitor or on the Oculus Quest 2, though the program could run on other XR platforms if needed. Users movements, object interactions and verbal conversations, were synchronized using Photon, a cloud service for multiplayer games (Exit Games, Hamburg, Germany). The manikin mesh was built in makehuman (*Make Human Community*, 2022), and its motions were driven by the FABRIK solver (Aristidou & Lasenby, 2011) in the Unity plugin FINALIK (*RootMotion*, 2022)

Evaluation

A pilot study was developed to test the design review tool and validate this framework implementation. The objective of this evaluation was to determine if collaboration in XR could be more effective when all

users can navigate the digital environment. In addition, we evaluated the user experience to identify the impact of free navigation and asymmetric tools.

Fifteen healthy subjects split into three groups were used as test subjects for the pilot study. All participants are current or recently graduated students at the University of Skövde. Ten participants are in the age range of 18-25 years, three participants are between 26-30 and one participant each is in the age range of 31-35 and 36-40. They were asked to perform five assembly tasks with the current approach used by CEVT (see Figure 4 condition A) and with the proposed for the prototype (see Figure 4 condition B). In condition A, there was an VR navigator, an observer and an assessor. The VR navigator assembled the components following instructions presented in the scene and posed the manikin. The observer and the assessor helped identify issues in the assembly process but were only able to see the scene from the VR navigators point of view. In condition B, the VR navigator's tasks remained the same. However, the assessor was in XR, and the observer used a PC with a 2D screen. The assessor and observer were allowed to navigate the digital environment and select viewing positions independently, but they could not interact with the assembly components or manikin.

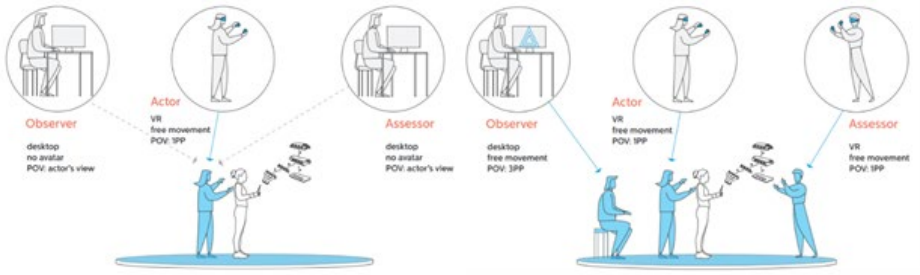


Figure 5. Illustration of condition A (right) and condition B (left)

Condition order was counterbalanced such that three groups started with condition A and two groups started with condition B. There were three issues planted in the sample assembly task: the posture of the manikin, an unfeasible sequence of assembly operations, and one required tool for the assembly not fit in the spot where it had to be used. Participants were instructed to note any issues they saw in the assembly process. We hypothesized that the participants would be more likely to find these issues when all of them could independently navigate the digital environment (Condition B). After the study, the participants completed a questionnaire (see Table 1) evaluating their experience comparing the two conditions (Schrepp et al.,

2017). The first four items evaluate pragmatic features of the system and the second four categories evaluate hedonic features.

Results

In condition A, the participants correctly identified issues 12 times and missed or mis-identified issues 11 times. In condition B, the participants correctly identified the issues 19 times and missed or misidentified issues 11 times. Figure 5 shows the data for the identified issues by the participants divided by correctly identified, missed, and misidentified. Correct identifications involve identifying a planted issue while misidentification involves indicating an issue exists that was not planted. In the case of misidentification, it is possible that an issue did exist, but later assessment suggests this was not the case. When an issues was planted but not identified it was counted as being missed. Note that in situations where one issue was planted and the participant reported a different issue the response was counted as both a miss and a misidentification. Results from the questionare are summarized in Figure 7.

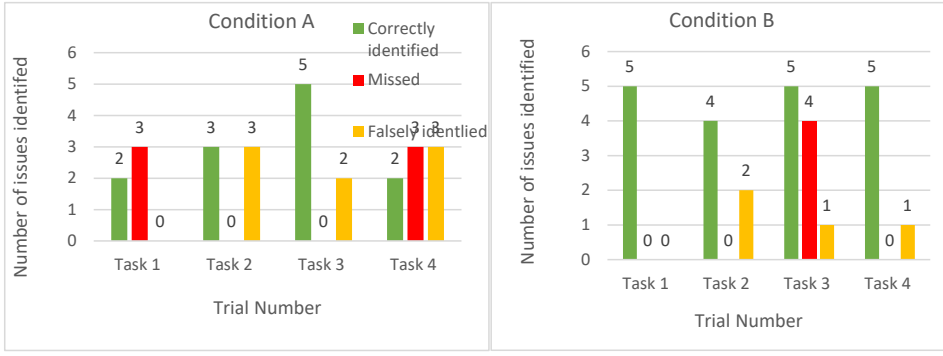


Figure 6. Identification of issues by the participants.

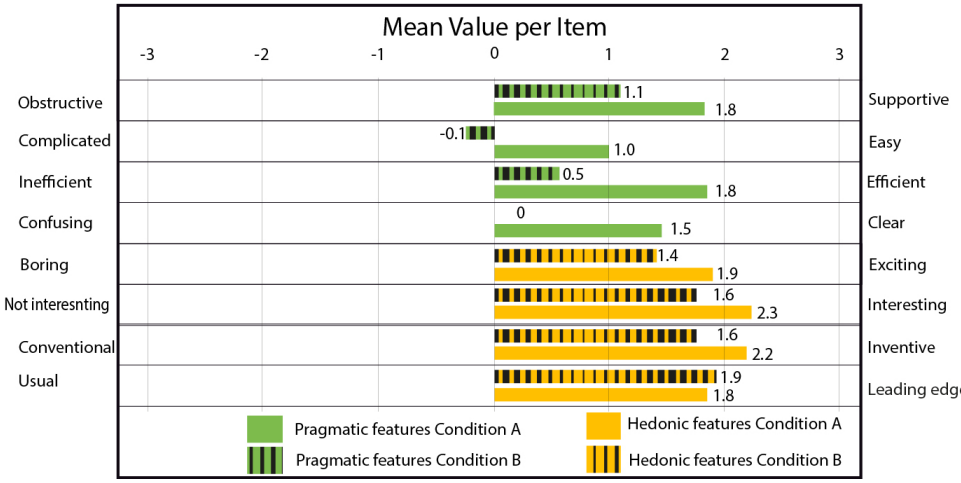


Figure 7. Mean values for the short user experience questionnaire conditions A and B

Discussion and Conclusions

In this pilot study, participants found the planted issues more easily when they could independently navigate the digital environment. In general the results show a consistent trend towards condition B in terms of correctly identifying planted issues. Note, in condition B task 3 the large number of missed issues is likely due to an error that resulted in there being 2 planted issues, so the spike may be due to the fact that users did not expect 2 errors in that case.

The user experience questionnaire also seems to indicate that the participants found condition B to be more practical and enjoyable. Given the size of the study and the limited scope of investigation, it is not clear if the enjoyment is related to the ability to navigate independently or the more streamlined inclusion of XR in condition B.

While the pilot and corresponding assessment is limited, the results are at least positive and indicative more work that needs to be done. Specifically, future work will need to investigate if there is any particular benefit due to XR integration or if the ability to navigate independently is sufficient for improving issue identification and creating an enjoyable process.

Collaboration and meetings using videoconferencing apps have become the norm. While design collaboration using screensharing and videoconferencing apps provides a functional solution that works

now, more work is needed to enable effective collaboration within DHM and CAD tools. Specifically, consideration of how different roles in a design process might benefit or be harmed from hardware and software constraints. Future XR design tools should consider not just multi-user XR collaboration but also how asymmetric collaboration can be incorporated. In this paper we have demonstrated an initial conceptualization of how asymetric tools can be enabled for remote collaboration.

Future work will focus on integrating more forms of XR while also making a more thorough and clear investigation of the benefits and potential challenges of using aysmetric XR systems in design processes.

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PAPER III: FRICTION SITUATIONS IN REAL-WORLD REMOTE DESIGN REVIEWS WHEN USING CAD AND VIDEOCONFERENCING TOOLS

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Open Access Research Article

Friction situations in real-world remote design reviews when using CAD and videoconferencing tools

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Abstract

Aims: Recent world events have resulted in companies using remote meeting tools more often in design processes. The shift to remote meeting tools has had a notable impact on collaborative design activities, such as design reviews (DRs). When DRs depend on computer-aided design (CAD) software, the lack of direct support for CAD functionalities in videoconferencing applications introduces novel communication challenges, i.e. friction. This study investigates friction encountered in real world remote DRs when using a combination of standard CAD and videoconferencing applications. The objective was to understand the main sources of friction when carrying out DRs using a combination of CAD and videoconferencing applications.

Methods: At a single Swedish automobile manufacturer, 15 DRs of a fixture component were passively observed. These observations were subjected to a qualitative thematic analysis to identify categories and sources of friction during these DRs. The DRs were carried out using a combination of CATIA CAD software and Microsoft Teams for videoconferencing.

Results: The analysis of the 15 remote DRs identified four recurring friction categories: requesting specific viewpoints, indicating specific elements, expressing design change ideas, and evaluating ergonomics. Each category highlights specific challenges that were observed during the DRs and emerged due to constraints imposed by existing methods and technologies for remote meetings.

Conclusion: This study provides a framework for understanding the current sources of friction in remote DRs using videoconferencing tools. These insights can support the future development of DR software tools, guiding the integration of features that address these friction points. Additionally, the results serve as a guideline for organizations to implement methods that reduce friction in remote DRs and improve DR quality and efficacy.

Keywords: Design review, product development, remote collaboration

1. Introduction

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Design reviews (DRs) are central to many organizations' product design processes (PDPs)[1]. DRs can be carried out at different stages of the PDP, and they serve as an evaluation activity where different aspects of design proposals are evaluated against defined requirements[2]. Hence, a core activity of DRs is to evaluate the design proposals to ensure they meet a pre-defined set of requirements. These requirements are typically formulated as specifications that must be fulfilled, i.e. constraints, and as criteria used to assess the superiority of design proposals. When this evaluation is successfully passed, a DR can act as a gate activity that allows product designs to advance to subsequent stages in the PDP[3]. DRs also serve as a platform for bringing together multiple stakeholders, who might have different roles, opinions, and interests, to make collective decisions on design proposals, thus facilitating early identification and resolution of design issues and saving time and costs in the long run[4].

Traditionally, DRs have been conducted in physical settings, where teams collectively gather to discuss and evaluate design proposals. Initially supported by physical drawings[1], the introduction of digital tools, particularly computer-aided design (CAD) software, expanded the scope of design factors considered during DRs, allowing more aspects of products to be considered and reviewed earlier in the PDP[5,6]. Recently, companies have begun transforming their work practices due to various factors, including economics, climate change, travel restrictions, and the growing tendency of remote work[7]. This shift has led to greater adoption of videoconferencing applications in the PDP, including Teams and Zoom[8-10]. These videoconferencing applications facilitate remote DRs by allowing designers to hold online DRs by screen-sharing 3D models of their design proposals, typically using their CAD software in combination with a videoconferencing app.

While a combination of videoconferencing applications with CAD applications enables remote collaboration in product design teams, there are also instances where technology gets in the way, hindering communication and causing interruptions[11]. Hindered communication and interruptions in DRs can be related to both the structure of remote DRs and software limitations introduced by the tools used to support remote DRs. While discussions of enhancing remote DRs often center around software and technical advancements, such as the creation of virtual reality (VR) tools for DRs[6,8], there is a lack of systematic studies exploring the challenges faced in remote DRs when utilizing current videoconferencing applications in conjunction with CAD software. While VR tools offer advantages, including increased immersion and even enhanced collaboration, they are not yet fully integrated into industry-standard software and most studies of VR tools in DRs focus on prototypes in experimental settings[12]. The lack of standard VR integrations with CAD common tools makes it challenging to investigate VR tools within established workflows at many companies, which heavily rely on traditional software solutions. Thus, we will identify challenges in existing workflows where videoconferencing tools are familiar to all involved. We expect that the results from this study will be useful in guiding future integrations of VR in real-world remote DRs to address existing challenges.

Along with proposals for technological solutions to solve challenges in remote collaborative design work, several frameworks have been proposed to describe and categorize what is needed for effective remote collaboration in design teams[13,14]. These frameworks are important tools that can guide the reorganization of existing remote design teams or the formation of new workflows. However, before identifying and developing a technological or organizational solution to real-world challenges in design workflows, it is necessary to understand the challenges as they emerge in typical workflows involving experienced design professionals. Thus, we engaged with the production design team of a Swedish automotive manufacturer to observe their remote DRs and evaluate where and how the flow of these DRs was hindered. These instances in which the flow of progress is interrupted may arise due to communication issues, difficulty conveying information, or misunderstandings and may depend on structural or technical issues in a DR. When these instances happen, there is "friction" in the communication between participants in the remote DR. Similar

instances have been referred to as workflow disruption, flow disruption, and communication friction[15,16]. In this paper, we introduce the term 'friction situation' to refer generally to situations that hinder remote DRs with a focus on moments where the flow of progress is impeded, resulting in delays or disruptions in the workflow. With the growing trend of remote work and the increasing number of product designers that use a combination of CAD with videoconferencing applications, there is a growing need to improve understanding of friction situations in remote DRs using these tools and provide guidance for structuring remote DRs and developing new tools to support remote DRs.

Friction situations are not exclusive to a specific combination of CAD and videoconferencing applications. They can transpire in any form of communication, including face-to-face interactions. In general, within design communication a common point of possible friction is in the presentation format of visual aids such as drawings, 3D models, or physical prototypes to communicate ideas[17,18]. The effectiveness of design communication is tied to both the type of visual aid and the technology used to present it[19]. It is essential to recognize that challenges may arise when the visual representation of a design fails to convey information comprehensively to all participants involved in the DR, as well as recognize when the technology used to mediate collaboration fails to do that efficiently[20,21].

Regardless of the communication medium and the specific visual aids employed, the primary purpose of DRs remains to identify potential issues in design proposals before proceeding to subsequent stages of the PDP[22]. To achieve this, there are common types of actions typically taken during DRs. These actions include observing the design proposals from different spatial perspectives, discussing different functionalities, taking notes, and engaging in discussions with others. In face-to-face interactions, these actions can be easily executed with tangible representations of the product[23]. However, when DRs are conducted online, all interactions and visualizations are mediated through software tools and not all users may be using the same set of software tools, even when viewing the same information. Moreover, many of these tools are not specifically developed with remote DRs in mind, increasing the risk of friction situations during DRs. In this paper we identify the recurring friction situations that arise when conducting remote DRs using a combination of CAD and videoconferencing applications.

2. Methods

Fifteen remote DRs within production design at a Swedish automotive company were attended by the first author. The first author, a researcher with a background in mechanical engineering, production, and human centred PDP, was invited to sit in and observe the DRs as a part of an ongoing research project aimed at developing new multi-user design tools for collaborative design processes. A qualitative research approach was applied, in which 15 DRs were studied by collecting data through observations, where findings were concluded in a validating workshop.

2.1 The design review case

The product that was the subject of the DRs was a fixture component that would be part of the assembly workstations in one of the company's future factories. The engineering objective of the DRs was to assess and refine a design proposal for the design of this fixture component. It is important to note that in this case there were not several design proposals being compared with each other, but rather one design proposal that was iteratively refined in detail over the course of the 15 DRs. The DRs were highly structured with three clearly defined roles: the meeting manager (1 person), the product expert (1 person), and an audience of 13 to 16 participants. The meeting manager, the product expert remained consistent throughout the whole duration of study. The meeting manager was responsible for initiating and leading the meeting, serving as a mediator between individuals as needed, and ensuring that the meeting's activities progressed smoothly. The product expert oversaw the design proposal and presented relevant functional components, referred to as

units, to the audience. The audience involved 13 to 16 participants, and it was made up of experts from diverse areas, such as maintenance and factory layout, who assessed the design proposal based on their competence and responsibility. The audience remained consistent throughout the study, with minimal variations due to occasional absences. The participants were predominantly experienced engineers responsible for various aspects of the fixture, including maintenance and ergonomics.

Invitations to participate in a DR were sent out by the meeting manager to relevant participants a few days before the DR was to take place. The meeting invitation included the relevant documentation and a link to the virtual meeting. Microsoft Teams (Microsoft Corporation, Redmond, United States) was used for videoconferencing and screen sharing. The CAD tool used was CATIA (Dassault Systèmes SE, Vélizy-Villacoublay, France), which was displayed to meeting participants via screensharing functionalities in Microsoft Teams. Notes were taken in Microsoft Excel (Microsoft Corporation, Redmond, United States). Occasionally, two to three participants gathered physically at the company's offices, although most of them always attended online. The meeting started with an introduction from the meeting manager outlining relevant details about the fixture component, objectives, and remarks specific to each DR. The product expert then proceeded to present the design proposal unit by unit. The product expert was the only person controlling the CAD view. No edits in the CAD model were made during the DR. Throughout each unit's presentation, the audience had the opportunity to provide comments or raise concerns. Both comments and concerns were typically addressed toward the presentation's conclusion. In some instances, urgent or specific issues were addressed during the presentation.

Every comment was evaluated for its impact on the fixture component and, if necessary, formulated as an actionable item. Specifying these actionable items involved identifying required changes, assigning responsibility for each task related to the change, and documenting the details in notes taken by the meeting manager. **Figure 1** provides a basic representation of the flow within the DR process.

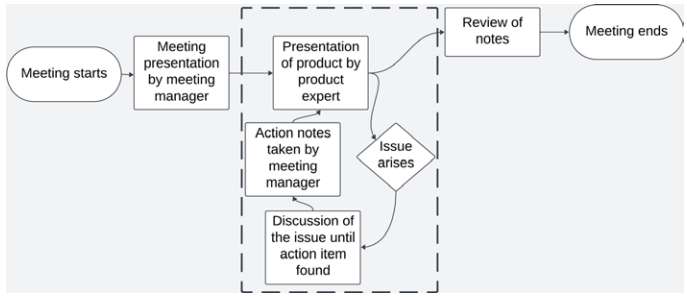


Figure 1. Observed DR flow. DR: design reviews.

2.2 Data collection

The first author passively attended the 15 DRs over a period of three months. The researcher did not engage in any conversation or discussion during the DRs. The participants were informed of the researcher's presence in the meetings through their manager. Moreover, a prior agreement with the automotive company specifying data confidentiality and privacy obligations for all parties was followed throughout the project. The data collection process primarily focused on identifying friction situations during the remote DR process. All the communication with the participant groups was mediated through a contact person at the company. Data was collected in the form of the researcher's notes taken during the DR using the note-taking template illustrated in **Table 1**. During the DR, any friction situations were recorded as separate items in the template.

Table 1. Template used for notetaking.

Date	Time	Number of Attendees			
Friction situation description			Impact of design review	Cause of friction situation	Attendees' roles involved in the friction

All recorded data was securely stored on a designated and protected computer system, with access restricted to authorized personnel. Confidential data pertaining specifically to the company was not disclosed to external parties. No audio or video data was collected in the meetings and meeting participant identities were not recorded. The decision not to capture audio or video data during the DR sessions was primarily influenced by confidentiality policies set by the research project that this study is part of, as well as compliance with the company's policies. Without video or audio recordings, data collection relied on notetaking. To increase consistency and decrease any potential loss of data, a systematic note-taking approach was implemented using the predefined template shown in **Table 1**. A validating workshop was also conducted with participants, providing an opportunity to corroborate the findings and ensure that the observed friction situations accurately reflected the challenges experienced during the DRs.

2.3 Final workshop

After observing 15 DRs, a summary of initial results was produced and presented by the first author at a physical three-hour workshop held with the participants involved in the DRs, at the company's building. All participants in the workshop were located physically in the same room. The workshop aimed to provide an opportunity for knowledge sharing, discussion, and corroboration of the findings. The workshop's main aim was to collect feedback on the observed friction situations from the perspective of the DR participants. The preliminary findings were presented to the participants with the aim of finding out if the findings were aligned with their experiences. After the presentation of the different friction situation categories, the participants were asked to express their ideas, whether they agreed with the findings, and if according to their judgement something was missing. This conversation was carried out in an unstructured setting to allow for more freedom to the participants to express their ideas. Notes were taken about possible areas of improvement and current challenges expressed by the participants. The notes were taken by the contact person at the company. The participants engaged actively in the workshop, offering their perspectives, experiences, and insights related to the identified friction situations presented to them. Furthermore, participants expressed their agreement with the identified friction situations after the presentation. Participants shared their experiences with their current setting on the remote DRs, and a long discussion on how the software used failed to deliver some key functionality was held.

2.4 Data analysis

The notes from the DRs were analysed using qualitative thematic analysis[24] to identify and derive recurring instances of friction from the collected data. After each DR, the notes were evaluated by the first author, and recurring friction situations were identified and grouped by comparing noted friction situations with the aim of identifying distinct categories of friction situations. After each meeting, the categorized notes were assimilated with the previous notes. An example of this method is illustrated in **Table 2** and **Table 3**. **Table 2** illustrates a friction situation as annotated during the DR meeting, and **Table 3** illustrates the progressive refinement of notes to convey to categories.

Table 2. Friction situation note taken during the DR meeting.

Friction situation description	Impact of design review	Cause of friction situation	Attendees' roles involved in the friction
Two people from the audience start discussing the design change idea. This discussion does not involve the product expert, and it requires continuous instructions to the product expert to move the camera around so they point at the desired units.	It is slowing down their discussion.	Not have the possibility to control the camera perspective independently.	Two members from audience and product expert.

DR: design reviews.

Table 3. Refinement of the note after the DR meeting.

Friction situation description	Impact of design review	Cause of friction situation	Attendees' roles involved in the friction
Two people from the audience start discussing about design change idea.	It is slowing down their discussion.	The people discussing the design change idea do not have the possibility to independently control the camera perspective.	Two members from audience and product expert.
Friction situation description (Refined)		Cause of friction situation (Refined)	
Discussions between members of the audience not involving the product expert which require constantly moving the camera view.		Inability for members of the audience to control the perspective that everybody sees.	

DR: design reviews.

After 10 DRs, no novel categories emerged from the data, as all friction situations could be appropriately classified within the pre-established categories presented in **Table 4**. To validate the suitability of these categories in classifying recurrent friction scenarios, the researcher participated in five additional DRs. After these DRs, no new categories emerged.

Table 4. Four categories of recurring friction situations.

Friction situation	Cause
Requesting specific viewpoints	Inability to freely navigate the 3D space, leading to delays in fulfilling requests. Lack of clear naming and visibility of components. Limited visibility and accessibility of component names, causing confusion.
Indicating specific elements	Verbal communication required to indicate the desired component, instead of enabling direct remote interaction.
Expressing design change ideas	Lack of a more efficient means to express design ideas graphically.
Evaluating ergonomics	Difficulty in understanding how humans will interact with the product.

3. Results

After observing the DRs and processing the notes taken during these sessions, four categories of friction situations were identified (**Table 4**). For each of these categories, a definition and a detailed description were established. Additionally, within a category, features of specific friction situations may moderate the impact of the friction situation on the meeting. These moderating factors are identified for each category and may be continuous or discrete, but typically account for variation in the impact of friction situations within the same category. Finally, for each category an example annotation of a specific friction situation is presented.

3.1 Requesting specific viewpoints

3.1.1 Definition

This category includes friction situations in which participants need to request specific viewpoints in the 3D model. In these situations, the product expert needs to interpret the requests and change the viewpoint accordingly, which may be relatively easy or may require several complicated procedures.

3.1.2 Detailed description

Participants in remote DRs often want to examine and discuss specific elements of the 3D models. A shared perspective on the relevant element(s) and an understanding of the spatial layout of the model can facilitate clear communication. As a result, participants frequently request specific viewpoints within the 3D virtual space that they believe would best facilitate communication of their ideas and insights. This process is often time-consuming and can lead to minor frustrations as participants have to guide the expert to the desired viewpoint. The necessity for specific viewpoints often arises from the need to visualize and understand complex design concepts. In the observed DRs, viewpoint was controlled at any given moment by a single product expert, thus requiring participants to communicate their preferred viewpoint to that individual and wait for the adjustment to be made. A raw data example of this friction situation is shown in **Table 5**.

Table 5. Examples of Raw Data.

2023-04-24	8AM	11			
Friction situation description			Impact of design review	Cause of friction situation	Attendees' roles involved in the friction
Someone from the audience asks the product expert to move to specific perspectives of the 3D model because the person doesn't see what they want.			Unnecessary delays 30 s per request plus the loss of focus by the audience.	Lack of possibility to freely navigate the 3D space.	Product expert and member of the audience.

In some cases, two or more participants discussing an element of the 3D model would desire different viewpoints. In these situations, each participant may redirect the product expert to their desired viewpoint, increasing the amount of time required to discuss the element and potentially introducing additional areas of miscommunication.

3.1.3 Moderating factors

This friction situation could be affected by three main factors. The complexity of the requested viewpoint, the frequency of the requests, and the number of participants requesting viewpoints. Some viewpoints were not easy to get right since they could require close-ups in sections where there were many components, making the instruction, normally from a member to the audience to the product expert, quite difficult. The extent to which participants request specific viewpoints may vary, with some DRs experiencing frequent

requests while others have fewer. Lastly the number of participants simultaneously requesting viewpoints greatly affected the level of this friction situation. When there were two or more participants requesting viewpoints simultaneously, the discussions could get quite inefficient due to the necessity of constantly directing the product expert to a specific viewpoint.

3.2 Indicating specific elements

3.2.1 Definition

This category includes situations in which a participant attempts to indicate or reference a specific element of a 3D model. With complicated 3D models, when the indicated element is obscured in the current view, and/or when multiple elements are indicated, it can be difficult to direct the attention of all participants to the element(s), slowing or halting the flow of the DR.

3.2.2 Detailed description

Within the context of remote DRs, participants often need to discuss and potentially redesign specific components within complex 3D models. When design changes involve multiple components, participants often suggest new ideas to improve compatibility between them. However, this can lead to friction situations as one participant's proposed change may require alterations in a different component, causing confusion. In complex products or areas with numerous components, participants may struggle to indicate and discuss specific elements. Additionally, there were instances where participants thought they were talking about the same component, only to realize later that they were referring to different ones. Indicating an element may be done to seek more information or clarification, adding to the complexity of discussions. The complexity and spatial arrangement of components within the 3D model and their visual similarities often contribute to this friction situation. Participants may struggle to distinguish between components with similar appearances or designations in a confined space, leading to difficulty indicating the intended objects. This friction situation is often related to the friction situation "Requesting specific viewpoints", since there were instances where participants requested specific viewpoints in order to indicate specific elements. A raw data example of this friction situation is shown in **Table 6**.

Table 6. Examples of Raw Data.

2023-03-03	8AM	14			
Friction situation description			Impact of design review	Cause of friction situation	Attendees' roles involved in the friction
2 members want to ask about a specific component. There is a difficulty on pointing that specific object.Issue of knowing what object audience wants the product expert to point.			Unnecessarily long conversation to just refer to a specific component.	Having to verbally communicate what component they want to highlight.	Product expert and 2 members of the audience.

3.2.3 Moderating factors

Indicating specific elements was highly affected by the number of additional components surrounding the indicated component. Moreover, if the nearby components were similar to the indicated component, indicating the right component could be difficult and impact the meeting flow. Familiarity with the product also contributed to the impact of this friction situation. When some participants are very familiar with the product, they refer to the desired component by its designation. However, participants who are less familiar with the component might struggle to know what that designation refers to in the product, making following discussions difficult for some people.

3.3 Expressing design change ideas

3.3.1 Definition

This category includes situations where a design change is proposed by a participant, but where at least one other participant does not understand the proposal or require additional clarifications. In these friction situations, the flow of the DR is disrupted as the participants attempt to clarify and understand the proposed design change in relation to the current design.

3.3.2 Detailed description

Participants in DRs may, at times, disagree with the current design choices and propose design changes. These changes may be motivated by a variety of factors, including the need for higher structural integrity, compatibility with other equipment in the factory, ergonomics considerations, or improved ease of manufacturing. Proposed design changes could range from a simple change in measurements to an entire re-design or complicated reconfiguration. Specifically for complicated changes, verbally communicating the proposed changes could be difficult or even impossible for some participants. Moreover, differences in the domain knowledge of participants, including how they evaluate requirements, can further complicate communication because other participants may not immediately see the relevance or importance of the proposal. This category of friction situation is closely related to the "Indicating specific elements" category, as participants often find it necessary to indicate which component they wish to redesign before articulating their design change idea.

During this type of friction situation, participants may introduce additional tools, such as drawing tools in an attempt to communicate their proposed design change. A raw data example of this friction situation is shown in **Table 7**. For example, a participant may take screenshots of specific design views and draw on them to visually represent their design change ideas. These supplementary tools are employed to bridge the gap between verbal communication and the conveyance of spatial design concepts. However, this often introduces other friction in the DR process. For example, the screenshot may require them to request that the CAD view be modified by the product expert or that they open the 3D model on their own computer to achieve a specific perspective, thus taking their attention off of ongoing discussions in the meeting.

Table 7. Examples of Raw Data.

2023-04-17	8AM	14			
Friction situation description			Impact of design review	Cause of friction situation	Attendees' roles involved in the friction
Someone has a disagreement with the structural integrity of one of the pillars. He expresses a design idea but the product expert is not understanding him.			Delayed the design review with an unnecessarily long conversation. Screen shot and drawing over it. Even the drawing has not made it clear. The note that was taken might not reflect what the participant was trying to express.	Lack of possibility to express design ideas more easily.	Product expert and member of the audience.

3.3.3 Moderating factors

The impact of these friction situations may be moderated depending on the complexity of the design change

ideas being conveyed. Complicated or intricate design changes typically had a greater impact on the meeting flow. When design modifications were difficult to communicate verbally, visual aids may need to be introduced, introducing additional communication and technology challenges. An additional factor that moderated the impact of expressing design changes on the meeting flow was the individual participant's familiarity with the problem, the component, or the particular unit under discussion. Familiarity often included knowing the names of surrounding components to denote positions, making it easier to specify where the change would happen and what it might affect. When a knowledgeable participant communicated a design change idea to another knowledgeable participant, the communication tended to be smoother compared to when one of the relevant participants was not as familiar with the unit or problem under discussion.

3.4 Evaluating ergonomics

3.4.1 Definition

This category includes situations related to evaluating physical ergonomics with the 3D model. These situations happened when there was a difficulty understanding how end users, e.g. workers at the factory, would interact with the product. As a result, participants could not accurately assess ergonomic factors including reachability, visibility, and work posture.

3.4.2 Detailed description

Ergonomics evaluations play a crucial role in product design, ensuring that workstations are conducive to user comfort, well-being, safety, and productivity. However, in the context of remote DRs, participants encounter friction situations when attempting to assess aspects related to physical ergonomics, particularly when constrained to 2D screens. This problem partly arises from the limitations in assessing the spatial information needed for ergonomics evaluations. Participants find it challenging to determine if certain spaces are too narrow, require awkward work postures, or accommodate various body types. While there was a possibility to represent a digital human model within the CAD software being used, the functionality of the digital human model was quite basic, and therefore it did not convey all the nuances of ergonomics evaluations. A raw data example of this friction situation is shown in **Table 8**.

Table 8. Examples of Raw Data.

2023-04-17	8AM	14			
Friction situation description			Impact of design review	Cause of friction situation	Attendees' roles involved in the friction
Someone is asking about the usability of the product but it is difficult to get a grasp of how people would use it.			Time is spent on subjective assessments when there could be objective data that could allow for a quicker and more accurate assessment.	Not being able to experience the workstation or at least have the possibility to make an ergonomics study.	Member of the audience and product expert.

Furthermore, variations in workstation usage among individuals with different anthropometric dimensions add to the complexity of ergonomic evaluations. Participants often struggle to grasp how considerations related to ergonomics may differ for different users and find it challenging to express these variations effectively.

3.4.3 Moderating factors

This category was related to the complexity of the questions being considered. In some instances of these friction situation, the participants expressed difficulty in assessing whether the space allocated for users to move comfortably is sufficient. Other questions included usability, sequences of manual tasks, and how the design proposal could work for people with different anthropometries. These later questions were more complex, often making the impact of the friction situation more significant.

4. Discussion

In this study, we observed, identified, and classified friction situations as they occurred in a series of DRs within a Swedish automotive company. These friction situations emphasize the complexity of remote collaborative DRs. While several friction situations have been identified, the remote DRs fulfilled their objective of finding issues with the current state of the design proposal. However, as expressed by DR participants and observed in this study, friction situations could still be addressed to improve the remote DR process. One notable observation is that many instances of friction involve more than one source of friction. As a result, reducing one source of friction may reduce friction situations due to other sources of friction. For instance, reducing friction situations related "Requesting specific viewpoints" could also reduce the friction situations related to "Expressing design change ideas." The relations between friction situations should be considered to address them efficiently, and therefore it is important to understand the root problems that are causing the friction situations. For instance, verbal communication is required to indicate a desired component, which can also affect "Indicating specific elements" and may also influence "Expressing design change ideas." Recognizing these interdependencies in the causes of friction situations is important for understanding the root issues that impact the primary friction situations. Additionally, friction situations were nested in some instances, meaning that a friction situation could happen within another friction situation. For instance, "Requesting specific viewpoints" was a recurring friction situation that happened as a part of other friction situations. In some cases, participants wanted to "indicate a specific element," and for that, they needed to request a specific viewpoint, meaning that the fact that a participant wanted to highlight a specific component created the friction situation of "Requesting specific viewpoints." Similarly, there were instances where participants wanted to request specific viewpoints that would help them express their design change idea. This suggests that addressing the problem of having a fixed viewpoint that is controlled by one person in the meeting could have a positive impact in many of the friction situations, and not only the ones under the category of "Requesting specific viewpoints". Similarly, as mentioned in the results, there were a few instances where, as a part of a friction situation of "Expressing design change ideas", there was a friction situation of "Indicating specific components" since participants needed to indicate specific components to express their design change idea. This connection suggests that having an additional way to refer to components other than verbally, could alleviate not only the friction situation of "Indicating specific components" but also the friction situation of "Expressing design change ideas" or even "Requesting specific viewpoints". Furthermore, addressing these friction situations could also mitigate the effects of videoconferencing fatigue by making the DRs more streamlined, which has been shown to cause cognitive and physical exhaustion in remote settings[25].

5. Limitations and Future work

This study focuses narrowly on a set of DRs at a single Swedish automotive manufacturer focused on the design of one component. Data collection in a real-world DR provided a close look at real challenges that emerge in ordinary workflows with professional design teams. However, we acknowledge that the focus on a single company and DR team means that there may be additional friction situations or causes of friction situations in other contexts. Given the challenges of collecting real-world data in an active company, e.g. data collection agreements and time constraints, we decided it was best to focus our resources on a single company in this initial study to ensure timely communication of results and insights in a fast-moving

technological landscape. We believe that while these limitations are unfortunate, the insights into challenges in a real-world workflow are incredibly valuable for current and future researchers and technology developers. Future work will aim to collect data over longer periods of time, from more teams/companies, and with more designed products to provide validation and refine understanding of friction situations in remote DRs.

The current study took a user-focused approach to understanding challenges in remote DRs, focusing on challenges that emerged with technology users in existing workflows[26]. As a result, the friction situations are not defined with the assumption of a specific technology solution. Future work should focus on identifying technologies, hardware, and software that can best address and minimize friction situations in DRs. These solutions may include more advanced solutions, e.g., VR/AR. However, they may also involve simpler software solutions, such as allowing participants to control their own viewpoints or annotate specific components during remote DRs. This could alleviate several interconnected friction situations, such as "Requesting specific viewpoints," "Indicating specific components," and "Expressing design change ideas" in a way that is easy for companies to integrate into existing workflows. Evaluating the effectiveness of these features in reducing friction and enhancing collaboration would provide valuable information for designers and technology developers.

Exploring organizational strategies to mitigate friction situations is also important. This could involve establishing best practices for facilitating remote DRs, training programs to improve communication skills in virtual environments, or guidelines for effectively utilizing available technologies. The extent to which existing organizational frameworks for remote collaborative design shape current real-world design practice should be further investigated, and the effect of implementing these frameworks studied[13,14]. These frameworks emphasize key factors such as shared mental models and workspace awareness, which are critical for effective collaboration. However, recent research suggests that additional factors, such as activity awareness and social presence, are equally important in remote collaborative environments[27]. Understanding how organizational factors that can alleviate friction may lead to more efficient and productive remote DRs.

Finally, further research will examine the impact of reducing friction situations on overall efficiency, participant satisfaction, and the quality of DRs and design outcomes. By quantifying these effects, organizations can better assess the value of investing in solutions to address friction situations. Moreover, future work should investigate how friction situations are related to cognitive and physical fatigue during remote DRs and how possible connections can be leveraged to improve work environments. This could include exploring alternative communication modalities, optimizing meeting structures, or incorporating breaks to enhance participant well-being and engagement.

6. Conclusion

This study provides insight into friction situations as they occurred in real-world remote DRs using a combination of CAD and videoconferencing applications. In this study, the main friction points in DRs meetings have been identified and analysed. Additionally, friction situations were matched to software functionalities, providing useful insights for developing and integrating critical software features into the remote collaborative design process. Improving remote DRs requires technological innovation, careful controlled research, and regular observation and evaluation of real-world practices in workplace settings. The current study provides critical insights grounded in the latter, and contributes a framework for understanding communication challenges in remote DRs as they emerge in design practice.

Declarations

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Authors contribution

Rivera FG: Data collection, data analysis, writing, reviews.

Lamb M, Alenljung B, Högberg D: Supervision and guidance on study framing, manuscript revision

All authors reviewed the final manuscript.

Conflicts of interest

The authors declare no conflicts of interest.

Ethical approval

Not applicable.

Consent to participate

Consent to participate was obtained from all participants. They were fully informed about the study's purpose, procedures, risks, and benefits before consenting to participate.

Consent for publication

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PAPER IV: SHARED VERSUS INDIVIDUAL VIEWPOINT CONTROL IN REMOTE DESIGN REVIEW MEETINGS

Shared versus Individual Viewpoint Control in Remote Design Reviews Meetings: Effects on Communication and Collaboration

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Abstract

Remote Design Reviews (DRs) are often conducted through videoconferencing with screen-sharing, enforcing an asymmetric Driver-Navigator interaction in which a single presenter controls the view while other stakeholders verbally instruct navigation. This Driver-Navigator arrangement introduces an interactional bottleneck: stakeholders must spend effort describing how to orient the shared viewpoint of a 3D model before they can discuss or raise issues with the model. This study investigates how different viewpoint control configurations affect communication and collaboration. Sixty participants completed a collaborative 3D puzzle assembly task in a collaborative virtual environment, in which one builder (Driver) manipulated the 3D model and two guiders (Navigators) provided verbal instructions. In a between-subjects design, groups used either Individual Viewpoint Control (all participants navigated independently) or Shared Viewpoint Control (only the builder controlled the view). Task completion times did not differ significantly between conditions. However, differences in collaboration dynamics were identified, with guiders speaking more and increased speaker changes in the Individual Viewpoint Control condition and fewer navigation-related clarifications in the Shared Viewpoint Control condition. Behavioural and conversational data show a trade-off between the cost of grounding and the cost of instruction. The findings suggest that viewpoint autonomy should be matched to the phase and purpose of the DR to improve cognitive ergonomics of collaborative DR tools: shared control may be better suited to presentation and consensus-building, whereas individual control supports exploratory, discipline-specific inspection.

1. Introduction

Design reviews (DRs) are an essential component of the product design and development process, serving as critical checkpoints where engineers and stakeholders evaluate suggested product designs' performance and functionality at various stages of the process (Adams et al., 2016; Ostergaard et al., 2005; Patil et al., 2017). The primary objective of DRs is to identify potential design flaws or issues that need to be addressed before advancing to subsequent phases, thereby reducing the risk of future problems and enhancing product quality (Yabuki, 2011). DRs often involve stakeholders with diverse opinions, roles, and interests, collaborating to identify and develop successful product designs that meet agreed-upon requirements (Gräßler et al., 2022). Such collaborative efforts facilitate early detection and resolution of design issues, ultimately improving the final product design, saving time and reducing costs (Adams et al., 2016).

While the cognitive objective of a DR is evaluation, the collaborative mechanism relies on a centralised interaction model. This dynamic aligns with the structural constraints of the 'Driver-Navigator' paradigm described in Pair Programming research (Bryant et al., 2008; Williams & Kessler, 2000). Although originally defined for collaborative coding, where participants frequently switch roles to construct software, this framework effectively captures the asymmetric interaction inherent in remote DRs. In this context, the single presenter acts as a permanent Driver, maintaining exclusive control of the CAD interface and handling tactical execution. Simultaneously, the stakeholders act as Navigators, who must verbally articulate their intent to explore, critique, or propose new solutions for specific aspects of the product model. Despite the difference in task domain, spatial manipulation versus code generation, the potential for cognitive ergonomic friction remains the same: the group's efficiency depends on negotiating an interactional bottleneck where the strategic perception of the product model is separated from the manual action required to manipulate it (Rivera et al., 2025).

In co-located teams, participants can rely on social cues to navigate Driver-Navigator bottlenecks. Social cues such as physical pointing, body language, gaze direction, and group behaviour can facilitate communication and establish a shared context in person (Bly et al., 1993). However, driven by the necessity of collaboration among physically distributed teams, whether due to international supply chains, increased work-from-home arrangements, or the need to reduce the environmental impact and costs of travel, the adoption of videoconferencing tools (e.g., Microsoft Teams and Zoom) has surged. Consequently, DRs are increasingly conducted in remote interaction contexts. In these remote meetings, computer-aided design (CAD) tool-based visualizations are shared via screen-sharing technologies and interactions are mediated through standard video and audio systems, e.g., computer screens, webcams, and headsets (J. Chen et al., 2019). According to Clark & Brennan's (1991) theory of grounding, the loss of social cues increases the "cost of grounding". While videoconferencing preserves facial expressions and some body language, the lack of a shared physical environment increases the effort required to establish that all participants are looking at and addressing the same thing. Thus, the shift to remote DRs and the corresponding loss of shared social cues increases the cost of grounding. Moreover, the common Driver-Navigator scenario (Williams & Kessler, 2000), creates a

bottleneck through which all grounding efforts must pass. Meeting stakeholders cannot simply point on the product model to address a design flaw, they must verbally describe the location, orientation, and operation required to take the desired viewpoint of the product model. The Driver must then accurately interpret this description and translate it into a shared viewpoint on the CAD product model for the entire group. Consequently, the cognitive focus of the meeting frequently shifts from the primary objective of evaluating the design proposal to the secondary task of instructing the Driver on how to manipulate the view of the product model.

The constraint of a single shared viewpoint raises critical questions regarding remote collaboration: Does enabling Individual Viewpoint Control, allowing stakeholders to break the Driver-Navigator dependency, enhance the collaborative process of establishing common ground and identifying issues? Or does it fracture the shared focus required for structured discussion? While Individual Viewpoint Control is technically possible in modern web-based CAD tools, there is a limited understanding of how this shift affects the fundamental communication loops that constitute the collaborative design tasks that make up DRs. Therefore, while DRs involve a wide range of tasks, including design critique, problem solving, and model construction, understanding how viewpoint control impacts communication in general, and the Driver-Navigator bottleneck specifically, is essential. This study investigates these effects by investigating a collaborative puzzle-solving 3D task under both shared viewpoint (Driver-Navigator) and individual viewpoint (everyone drives) control paradigms. The aim is to develop an initial understanding of the effect of different viewpoint control configurations in communication dynamics during collaborative tasks where there is a central driver and multiple stakeholders. We expect insights from this research to support improved cognitive ergonomics in common collaborative design processes, such as in remote DRs, providing a basis for interaction paradigm selection for collaborative tasks.

1.1. Previous research

Effective collaboration depends heavily on the group's ability to maintain common ground, a shared understanding of what is being referred to, and what actions are being suggested or performed (Clark & Brennan, 1991). In design-oriented tasks, grounding relies on both verbal references and what groupware (collaborative software) research describes as 'consequential' communication within workspace awareness frameworks. Consequential communication is the collaborators' ability to deduce others' intentions by seeing their cursors, controllers, or avatars act on shared artefacts in real time (Assaf et al., 2024; Gutwin & Greenberg, 2002). When participants share the same view, verbal references to commonly visible elements can reduce the cognitive effort required to coordinate attention and interpret contributions.

In collaborative work involving digital spatial representations, such as 3D models or virtual environments, maintaining this shared understanding becomes significantly more challenging (Liu & Kaplan, 2018). Navigation and object manipulation introduce challenges related to depth, occlusion, and orientation. Research on collaborative virtual environments indicates that shared viewpoints help users stay aligned and avoid spatial confusion (Kim et al., 2017). In contrast, independent viewpoints

increase personal flexibility but often require more explicit verbal communication to re-establish common ground (Chen et al., 2021; Lee et al., 2020; Valin et al., 2001). Recent studies in Extended Reality (XR) highlight that independent navigation can break the shared mental model unless the system provides strong "mutual awareness cues", such as visual tethers or view frustums (Assaf et al., 2024; Piumsomboon et al., 2019; Teo et al., 2019). Without these cues, independent users struggle to deictically reference objects, leading to breakdowns in spatial understanding (Piumsomboon et al., 2019). At the same time, equipping every participant with individual navigation capabilities can introduce coordination overhead and the risk of “losing each other”, as well as extra attentional effort devoted to navigation by all group members (Weissker & Froehlich, 2021).

Currently, most research in this area evaluates individual reviewers or small teams in conditions where all participants have direct control over navigation and inspection tools, without explicit role differentiation (Lukačević et al., 2021). In contrast, real-world engineering practices (specifically DRs) often operate under an asymmetric structure where a single "Driver" controls the view for passive "Navigators”. In this dynamic, the Navigators must dedicate cognitive resources to 'navigational instruction', verbally guiding the Driver to the area of interest, rather than focusing purely on central design tasks. Hove & Watson (2022) link the lack of shared spatial agency and nonverbal cues to increased cognitive load, noting that the lack of nonverbal cues in remote meeting contexts forces users to compensate with additional explicit verbal cues. This transforms the collaboration from a joint interaction activity into an interface management task where the processes of interface management induce specific friction situations where disconnects between stakeholder intent and driver execution disrupt the workflow (Rivera et al., 2025). Consequently, despite the potential benefits of independent exploration, the technical limitations of standard screen-sharing tools restrict the group to a single, driver-led perspective (Jordan & and Adams, 2016).

Although modern web-based tools now make it technically possible for each participant to navigate independently, there is limited research on how changing viewpoint autonomy affects collaboration. Existing work has not yet examined how independent viewpoint control influences grounding behaviour, shared focus, or communication patterns during collaborative interaction tasks in the context of design. This study addresses this gap by isolating the effect of different viewpoint control configurations within a controlled collaborative 3D task, explicitly designed to reproduce the interactional constraints of the Driver-Navigator dynamic.

2. Method

The experiment was designed to isolate viewpoint control's effect on a simple collaborative task, focusing on common base elements of collaboration in DRs. The task involved assembling a 3D puzzle in a collaborative virtual environment (CVE) using color-coded cubes, also referred to as *pieces* (Figure 1, and Figure 2). The CVE is presented on 2D computer monitors for participants. This task was chosen because it replicates a key aspect of collaborative DRs, i.e., viewpoint control, while remaining controlled and repeatable. While DRs include other more complex communicative and cognitive processes, the experimental task allows for isolation of common basic interaction activities that occur

in a collaborative DR. The specific task requires multiple participants to coordinate, communicate spatial information, and interpret partial representations of a shared design objective.

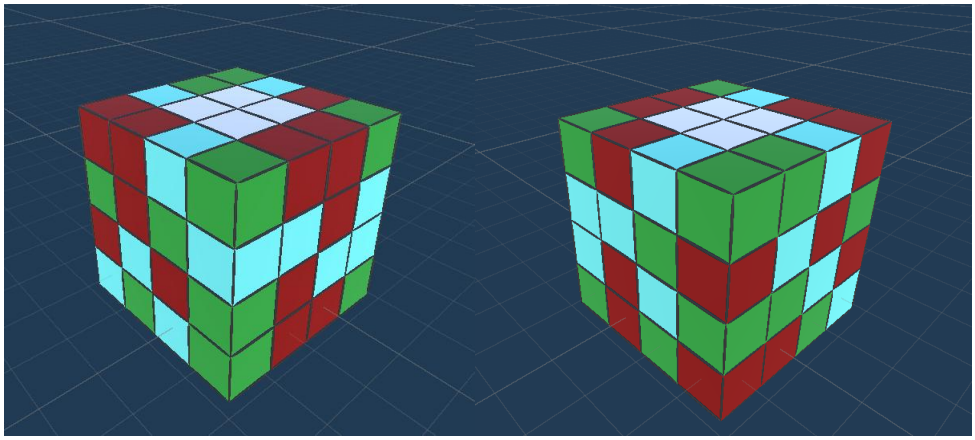


Figure 1. One of the cubes assembled during the experiment, seen from two viewpoints.

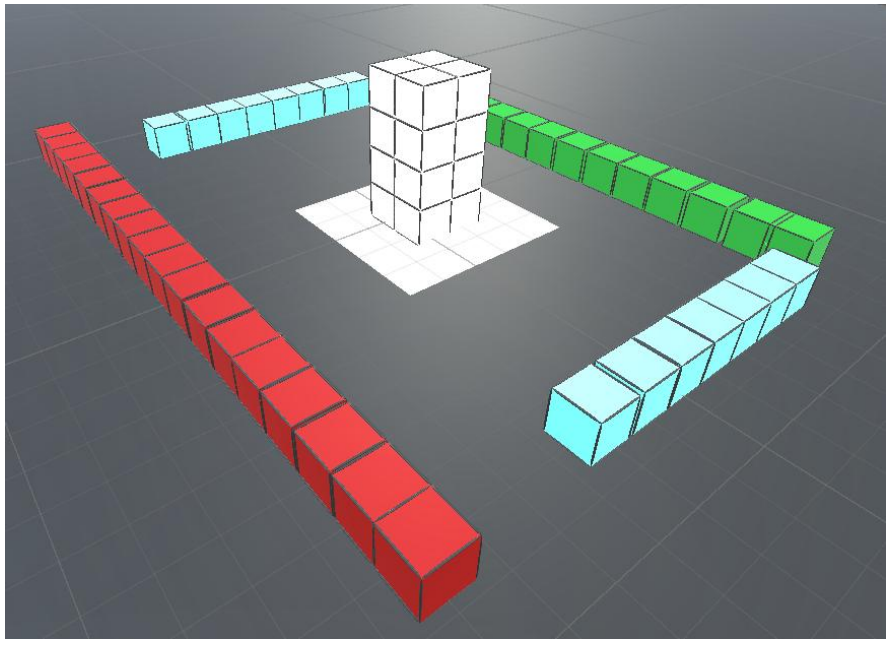


Figure 2. Initial setup to start assembling cubes. Coloured pieces to be correctly assembled on the white column to complete the 3D puzzle.

Participants collaborated in groups of three, each group consisting of one *builder* and two *guiders*. The builder was responsible for manipulating coloured cubes (puzzle pieces), not knowing the solution of the 3D puzzle, while each guider only had partial information about the solution of the 3D puzzle. This role distribution intentionally reproduces a Driver-Navigator interaction structure: the builder (Driver)

manipulates the shared model, while the guiders (Navigators) must verbally direct the required operations. The pieces were assembled on a central white column as a reference to facilitate the operation (Figure 2). Figure 2 shows what the disassembled cube looked like before the experiment started. The study employed a between-subjects design with two different viewpoint control conditions:

Condition A (Individual Viewpoint Control): All participants could control their own viewpoints, allowing the guiders to navigate and view the puzzle from different angles and aiding in guidance. A basic avatar showed each participant's position within the environment for improved spatial awareness (Figure 3).

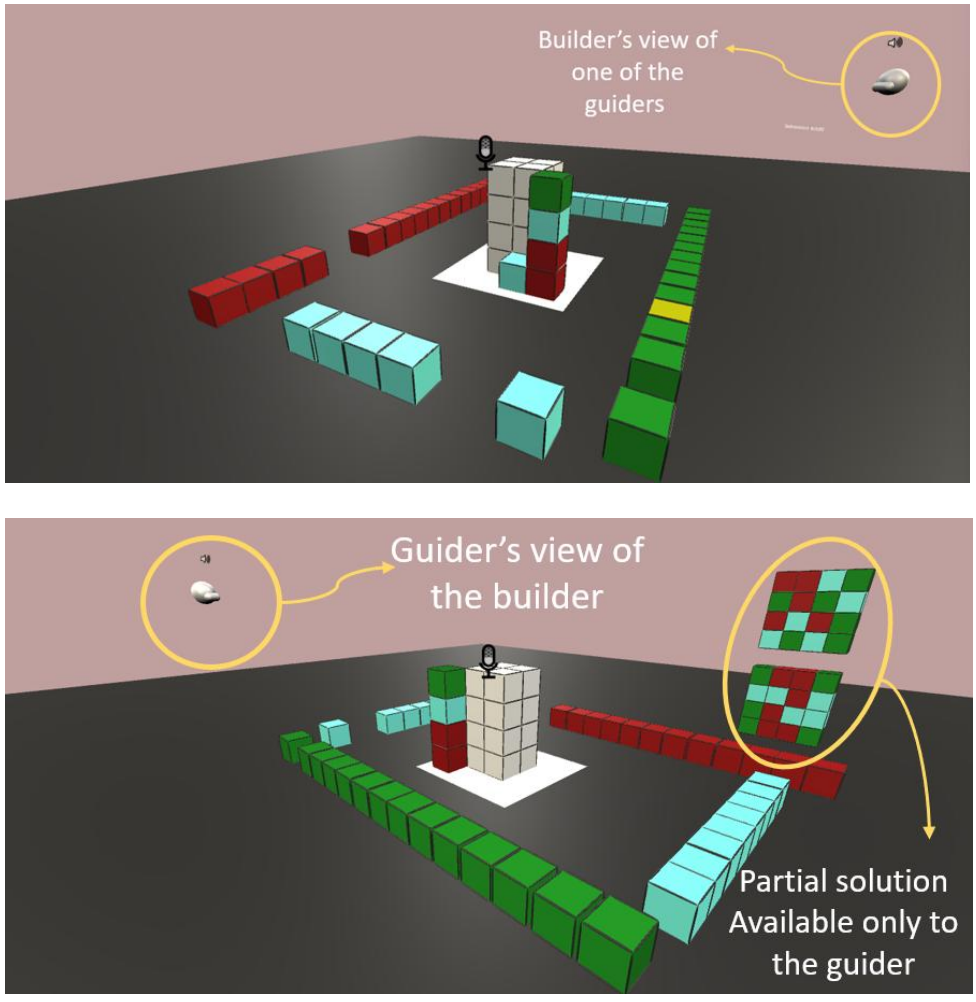


Figure 3. Builder's view (top) and guider's view (bottom) of the Individual Viewpoint Control condition. In the bottom image, each guider is provided partial solution information, which is available at all times in the upper right corner.

Condition B (Shared Viewpoint Control): Only the builder controlled the viewpoint, and guiders observed the puzzle solely from the builder's viewpoint, mimicking a screen-sharing scenario common in remote collaboration through videoconferencing tools (Figure 4). This condition most closely reflects the traditional Driver-Navigator workflow observed in remote DRs, where stakeholders depend on a single presenter's view.

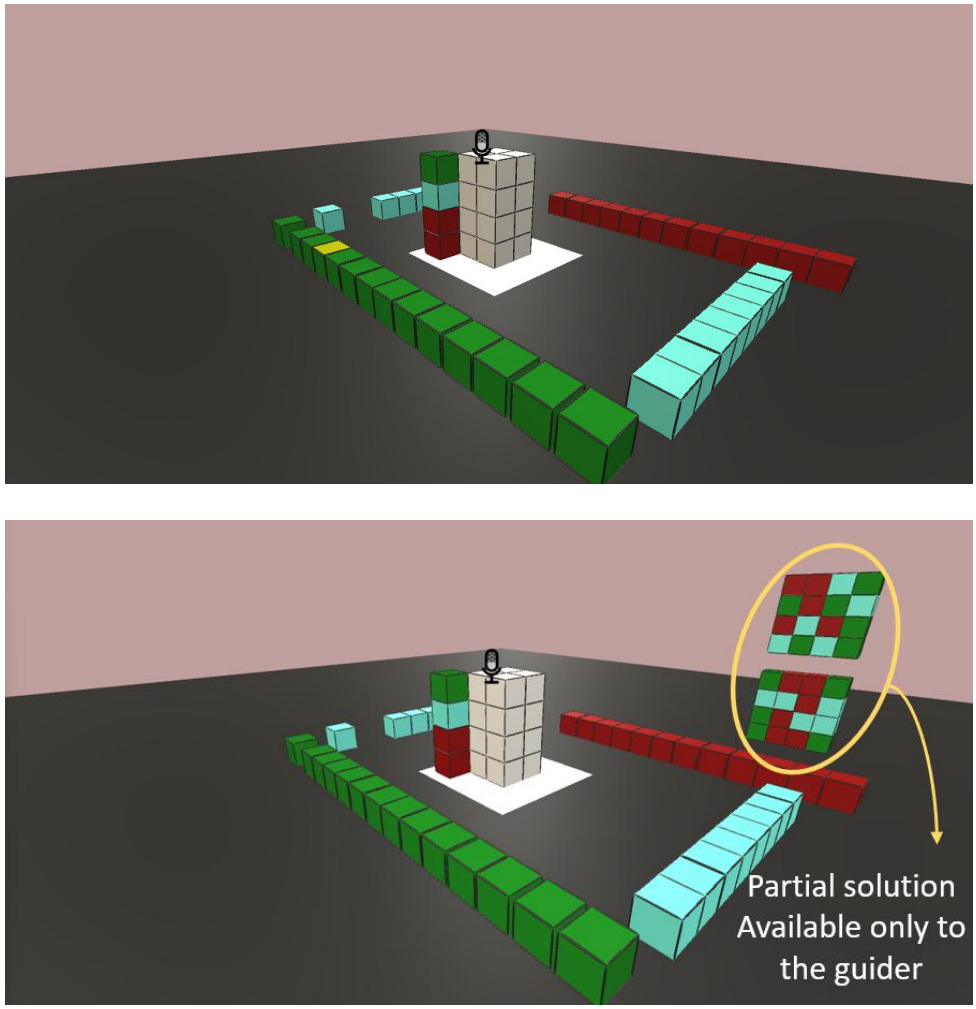


Figure 4. Builder's view (top) and guider's view (bottom) of the shared viewpoint condition. In the bottom image, each guider is provided partial solution information, which is available at all times in the upper right corner.

The study involved 60 participants, aged 18 to 35 (mean = 22.5, SD = 3.5), with 36 males and 24 females with basic computer proficiency. They were organised into twenty groups of three, each group was assigned randomly to one of two experimental conditions to have an even distribution (ten groups per condition). Participants were recruited from a university engineering student population. While they were not professional engineers, they all had familiarity with standard desktop navigation interfaces.

To accommodate participants' linguistic backgrounds, groups conversed in either Italian (for groups where all participants had Italian as native language) or English (for groups with mixed native language). Language use was consistent within each group to ensure no disparities in communication latency. Two different puzzles were made to reduce the effect the puzzle could have had on the task. For half of the groups in each condition, one puzzle was used as training, and the other one was used for the experiment.

2.1. Experimental Procedure

Upon arrival at the laboratory, participants were briefed on the experiment's objective and randomly assigned to their respective roles as guiders or builders. They were seated at individual workstations in the same room, but the configuration prevented them from seeing each other's screens or interacting visually (Figure 5). Participants could only view their own screen while hearing each other's voices and communicating verbally.

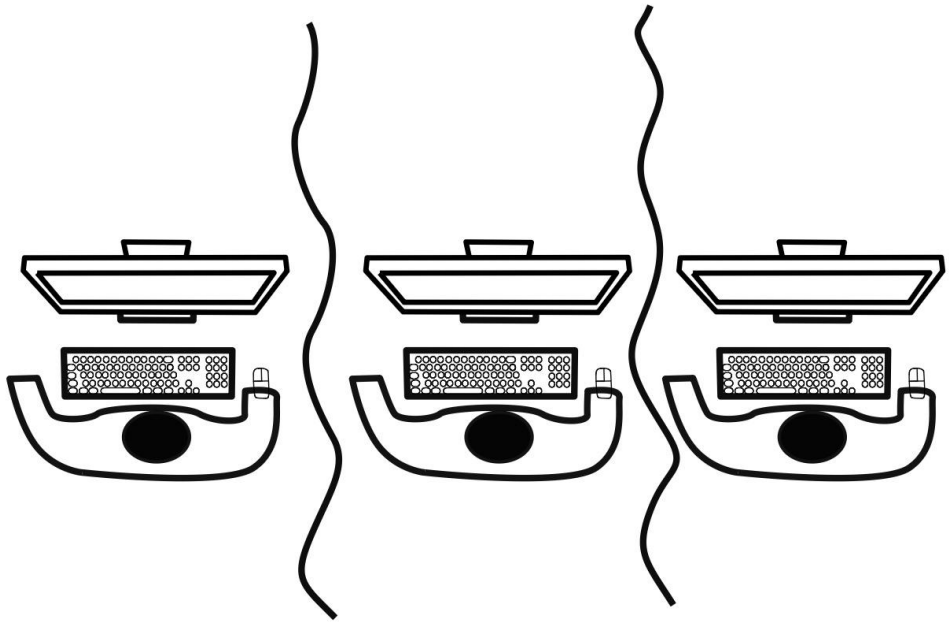


Figure 5. Layout in the experiment.

Before starting the main task, participants completed a brief training session to help them understand the mouse controls and their specific roles. The controls in this experiment were similar to those commonly used in CAD software. The viewpoint control was simplified by limiting navigation to rotating around the puzzle's vertical axis at a fixed distance, height, and camera orientation. Participants could rotate their viewpoint by pressing the left mouse button while moving the mouse left or right. This caused their viewpoint to rotate horizontally around the puzzle at a fixed distance.

Thus, there was no possibility of panning or zooming. To place a puzzle piece (only the builder could place pieces), the builder would first select the piece they wanted to assemble by right-clicking on one of its faces, then right-clicking on an adjacent face of an existing piece to position it accordingly. This process is visually illustrated in Figure 6. The pieces could be rearranged after they were positioned. The white pieces, collectively representing the column in the centre (Figure 2), could be used as a reference to place other pieces but could not be moved. While builders and guiders shared the same navigation controls, only the builder could interact with the cubes. Note that in the Individual Viewpoint Control condition, both the builder and the guiders had control over their own view, whereas in the Shared Viewpoint Control condition, only the builder controlled the view. Hence, in the Shared Viewpoint Control condition, the guiders' views were synchronised with the builder's view, so they could only observe the puzzle from the builder's viewpoint and did not navigate independently. Therefore, in the Shared Viewpoint Control condition, the guiders did not have to interact with the computer manually, but rather only gave instructions verbally.

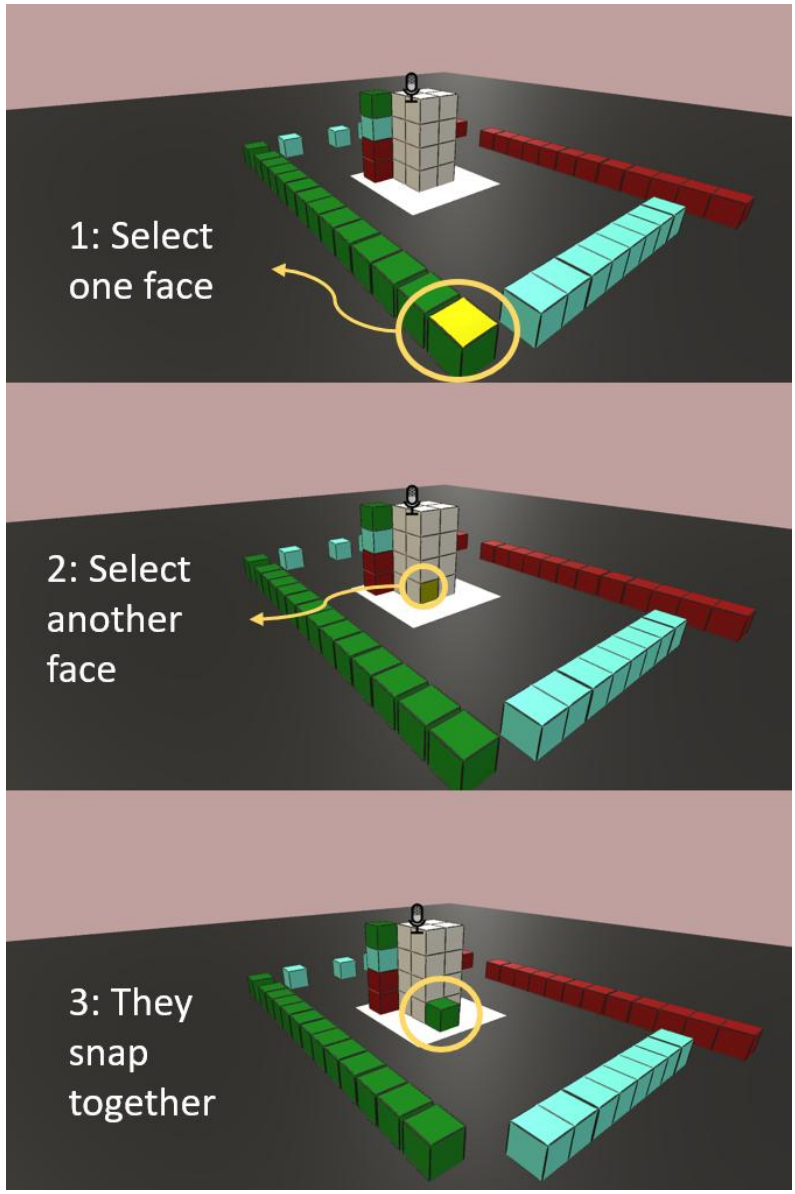


Figure 6. Process of snapping two pieces together.

Throughout the experiment, each guider had continuous but only partial access to the solution of the 3D puzzle. Each guider had half of the solution always visible on each their respective screens, displayed in the upper right corner, as shown in the bottom half of Figure 4. On the other hand, the builder did not have access to the solution and depended solely on the guiders' verbal directions to complete the task.

During the experiment, participants communicated verbally to collaborate on solving the puzzle. The builder manipulated the pieces based on the guiders' instructions. The group dynamics were generally similar across all teams. Initially, the two guiders discussed and agreed on how to direct the builder, and once they reached a consensus, they proceeded to complete the puzzle. Typically, one guider would lead the builder through assembling a full face of the cube before the group moved on to the next. In some teams, one guider directed the builder through both of their assigned faces and then handed over the task to the other guider for the remaining faces. These approaches largely depended on they agreed upon at the beginning of the task.

2.2. Data collection

Data was collected to capture differences in task performance, behavioural patterns, and self-assessed user experiences across different conditions. Task performance data was measured by cube completion time. Accuracy was not analysed as a dependent variable because the experimental protocol required all groups to assemble the puzzle with 100% accuracy, matching the solution to complete the trial successfully. Therefore, accuracy was a constant constraint, and time served as the only performance metric. To examine behavioural patterns, the number of right-clicks the builder performed, indicating the movement of pieces, was recorded. Left clicks from both the builder and guiders were also recorded to capture the number of times they changed the viewpoint. Although individual clicking habits may vary, the between-subjects design and the number of groups per condition mitigate such variability, making these measures suitable for comparing interaction patterns across conditions. Moreover, while some accidental or non-functional clicks may have occurred, they would most likely be few and evenly distributed across groups. Excessive accidental clicks would indicate failure to understand the task/technology and serve as grounds for exclusion. This was not observed in the data. Viewpoint position data were collected by tracking participants' rotational angles around the cube. Communication behaviour patterns were analysed by recording the number of speaker changes and the total speaking time for each participant. Furthermore, the conversational data were recorded as audio files and transcribed. Since the experiment was conducted in Italy, some groups conversed in Italian (when all participants were Italian), while others used English (when one or more participants were not Italian). The conversational data were transcribed in the language of the conversation. Participants also completed a self-assessed user experience questionnaire (Laugwitz et al., 2008) at the end of the experiment to reflect on their satisfaction and perceived user experience.

Except for the questionnaire and the conversational data, all data were collected using the software employed for the experiment, with information from all participants recorded on the computer used by the builder. The data structure included timestamps, click actions, speaking status, and rotation angles for each participant. As mentioned, the builder's data included indicators for right-clicks (rightClickBuilder), left-clicks (leftClickBuilder), speaking status (talkingBuilder), rotational position (RotationBuilder), and whether a puzzle piece was moved (SnapBuilder). Similar data were recorded for each guider. Table 1 shows a summary on the data structure for each group.

Table 1. Recorded data.

Data Tag	Time Stamp	Right Click Builder	Left Click Builder	Talking Builder	Rotation Builder	Snap Builder	Left Click Guider1	Talking Guider1	Rotation Guider1	Left Click Guider2	Talking Guider2	Rotation Guider2
Data Type	Integer	Boolean	Boolean	Boolean	Float	Boolean	Boolean	Boolean	Float	Boolean	Boolean	Float
Cond-ition	Angle around the cube	True if clicking	True if clicking	True if talking	Angle around the cube	True if piece was moved	True if clicking	True if talking	Angle around the cube	True if clicking	True if talking	Angle around the cube

Finally, participants completed the self-assessment User Experience Questionnaire (UEQ), evaluating the system's pragmatic and hedonic qualities (Laugwitz et al., 2008), where each item could be answered from 1 to 7 (Table 2. User Experience Questionnaire (Laugwitz et al., 2008).

Table 2. User Experience Questionnaire (Laugwitz et al., 2008).

Negative	-3	-2	-1	0	1	2	3	**Positive**
Obstructive								Supportive
Complicated								Easy
Inefficient								Efficient
Confusing								Clear
Boring								Exciting
Not interesting								Interesting
Conventional								Inventive
Usual								Leading edge

2.3. Data analysis

The analysis centred on task performance metrics and behavioural metrics to explore how viewpoint control impacts task completion efficiency and the dynamics of team interactions within the CVE.

For task performance, we measured the puzzle completion time. We compared the Individual Viewpoint Control condition and the Shared Viewpoint Control condition using independent samples t-tests to determine whether Individual Viewpoint Control impacts teams' efficiency in the task.

Speaker changes and speaking time provide insights into the communication patterns among participants, indicating whether Shared Viewpoint Control induces more balanced and dynamic interactions. A higher number of speaker changes and more evenly distributed speaking times could suggest that all participants were more engaged and the resolution of the task is less centred around a central presenter (in this case, the builder). The number of clicks, especially the builder's clicks, provides useful insights into how the builder engages with the environment. The left click count, from now on referred as *navigation clicks*, indicates the number of viewpoint changes, and the right click counts, from now on referred as *placement clicks*, indicate the number of piece placements/removals. Participants' movements within the CVE were analysed by tracking their rotational angles (from 0 to 359°) around the puzzle. This positional data provided insights into how participants navigated and oriented themselves in the virtual space, both on their own and relative to one another. Viewing

duration and viewpoint changes could provide insights into the differences across conditions. These behavioural metrics help understand the collaboration dynamics within the CVE. Independent-samples t-tests were used to compare these metrics between the two conditions, with $p < 0.05$ set as the criterion for statistical significance. The selection of these behavioural metrics aimed to investigate the collaboration dynamics aspect when performing 3D tasks in CVE.

Additionally, the conversation data were transcribed and categorised into three criteria: instruction to change the viewpoint, instruction to place, and clarification. Clarifications were defined as instances where a participant asked for or provided additional information to confirm or refine an instruction. For example, if a guider said, “Go to the other corner,” the builder could go and respond, “This one?” to confirm, followed by the guider replying, “Yeah.” This exchange would be coded as an instruction to change viewpoint followed by one clarification. Similarly, a guider might instruct, “Put the green on top,” and after the builder proceeds to place it, the guider clarifies, “No, not that one—on top of the green one,” prompting the builder to ask, “This one?” and receiving a “Yeah” in response. These exchanges distinguish clarifications from direct instructions, as they involve resolving misunderstandings or ensuring accuracy. By examining the frequency and context of these types of interactions, we could assess how viewpoint control influences the need for certain communications. For example, more clarifications might indicate that participants experience more confusion or misunderstandings, affecting collaboration efficiency.

The study adhered to ethical guidelines for human subject research throughout the process. All participant data were anonymised to protect confidentiality, individual responses could not be traced back to any participant. Participation was entirely voluntary, and participants were informed of their right to withdraw from the study at any time without penalty and signed a participation consent agreement. After the experiment concluded, a debriefing session was held to explain the study’s aims in greater detail and to address any questions or concerns the participants might have had.

3. Results

3.1. Task performance and behavioural data

Table 3 presents a comprehensive summary of the statistical comparisons between the two experimental conditions. It details the mean values and standard deviations (SD) for task completion time, verbal interaction metrics, and input frequency (clicks). The results of the independent samples t-tests are presented to indicate statistical significance between the Individual Viewpoint Control condition and the Shared Viewpoint Control condition groups.

Table 3.Summary results from task performance and behavioural data analysis.

Metric	Individual Viewpoint/ Condition A		Shared Viewpoint/ Condition B		p value
	Mean	SD	Mean	SD	
Task completion time (s)	612	90	582	102	0.34
Speaking time builders (s)	210	42	312	54	0.0002
Speaking time guiders (s)	390	48	258	36	0.0001
Number of speaker changes	122	15	86	12	0.001
Number of navigation clicks	92	10	105	12	0.003
Number of placement clicks	124	15	127	18	0.48

The average task completion time across all groups was approximately 600 seconds (10 minutes). An independent samples t-test was conducted to compare the mean completion times between the two conditions. As shown in Table 3, the mean completion time in the Individual Viewpoint Control condition was 612 seconds (SD = 90 seconds), while in the Shared Viewpoint Control condition, it was 582 seconds (SD = 102 seconds). This difference was not statistically significant ($p = 0.34$).

The number of speaker changes within each group was recorded to assess the dynamics of verbal interaction. As shown in Table 3, groups in the Individual Viewpoint Control condition had a higher number of speaker changes than those in the Shared Viewpoint Control condition. Specifically, the mean number of speaker changes per group was 122 (SD = 15) in the Individual Viewpoint Control condition and 86 (SD = 12) in the Shared Viewpoint Control condition. This difference was statistically significant ($t(38) = 7.88$, $p = 0.001$).

The total speaking time for builders and guiders was measured to assess participation levels. Builders in the Individual Viewpoint Control condition talked significantly less (Mean = 210 seconds, SD = 42 seconds) than those in the Shared Viewpoint Control condition (Mean = 312 seconds, SD = 54 seconds), as shown in Table 3. This difference was statistically significant, $t(38) = -7.14$, $p = 0.0002$.

For guiders, in the Individual Viewpoint Control condition, the mean speaking time was 122 seconds (SD = 15 seconds), compared to 86 seconds (SD = 12 seconds) in the Shared Viewpoint Control condition. This difference was also statistically significant, $t(38) = 9.19$, $p<0.0001$, as shown in Table 3.

The number of navigation clicks performed by the builders relates to the number of times the builders took the active decision to change viewpoint in the CVE. As shown in Table 3**Error! Reference source not found.**, builders in the Shared Viewpoint Control condition clicked significantly ($t(38) = -3.16$, $p = 0.003$) more (Mean = 105, SD = 12) than those in the Individual Viewpoint Control condition (Mean = 92, SD = 10).

The number of placement clicks relates to the number of times that the builders clicked on a piece to change its position. There was no significant difference in the number of placement clicks (Individual Viewpoint Control condition, Mean = 124, SD = 15; Shared Viewpoint Control condition, Mean = 127, SD = 18, $t(38) = 0.71$, $p = 0.48$), as shown in Table 3.

The extent of rotational movement around the puzzle by the builders was analysed to assess navigation behaviour. Plotting the distribution of angular positions (Figure 7) indicates that in the Individual Viewpoint Control condition, the builders tended to observe the puzzle from positions near two of the corners. Meanwhile, in the Shared Viewpoint Control condition, the builders tended to move around the puzzle uniformly. Participants started the task at 0 degrees and shortly after they tended to move clockwise to a position of around 15 to 30 degrees. There was no restriction in the movement when rotating around the puzzle.

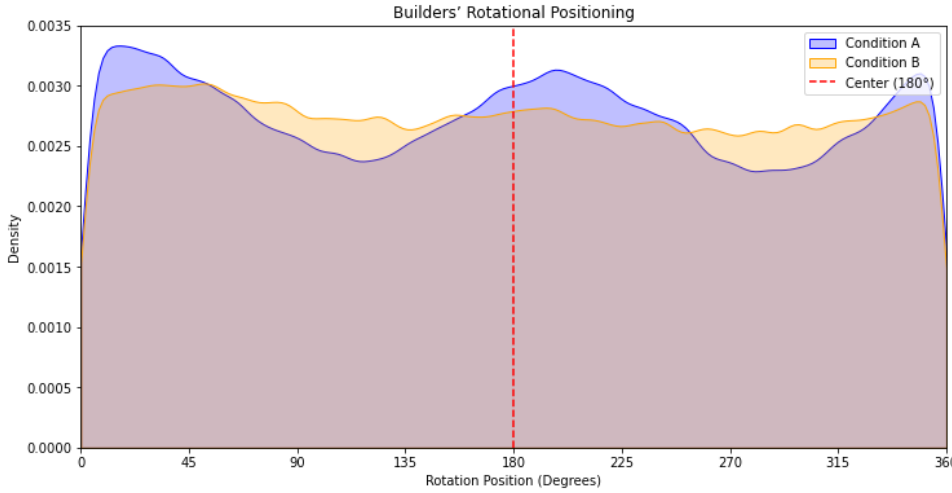


Figure 7. Distribution of the builders' rotational positioning around the puzzle in the individual viewpoint condition (Condition A) and the shared viewpoint condition (Condition B).

3.2. Conversational Data

The transcribed conversational data was categorised into three distinct types: *Instructions to change the viewpoint*, *Instructions to place*, and *Clarifications*. The analysis showed key differences between the two conditions in how participants communicated, especially regarding the nature and frequency of these conversational categories.

Instructions to change the viewpoint

The number of instructions to change the viewpoint from the guiders to the builder ranged between 6 and 8 per group across both conditions (Individual viewpoint control condition: Mean = 6.8, SD = 0.5. Shared viewpoint control condition: Mean = 7.3, SD = 0.6). These instructions involved guiders directing the builder to shift their focus to a different puzzle face or viewpoint position.

An important difference between the conditions was the nature of these instructions. In the Individual Viewpoint Control condition, guiders often relied on visual cues to direct movement. The verbal commands to direct the builders to a specific place often had the form of: "Come here" or "Come

where I am," as builders could see the guider's position as avatars on the screen. In contrast, in the Shared Viewpoint Control condition, guiders had to be more precise, frequently issuing directional instructions like "Go left" or "Go right" since their view was tied to the builder's navigation.

Instructions to place

The total number of instructions to place puzzle pieces had a mean of 67 instructions per group with no significant difference across both conditions. The Individual Viewpoint Control condition had a mean of 66.5 instructions (SD = 5.8), while the Shared Viewpoint Control condition had a mean of 67.5 instructions (SD = 6.2). For context, to perfectly complete the puzzle with as few instructions as possible, participants needed to issue 56 instructions to place, corresponding to the 56 pieces that had to be correctly positioned.

Clarifications

The number and nature of clarification requests varied between the two conditions. In the Individual Viewpoint Control condition, clarifications were more likely to follow instructions to place puzzle pieces, with a mean of 30 clarifications per group. In contrast, in the Shared Viewpoint Control condition, clarifications were more often associated with instructions to change the viewpoint, with a mean of 25 clarifications per group.

3.3. Self-assessed Questionnaire

There were no significant differences in ratings of pragmatic quality as measured by the first four items on the UEQ (Table 2). Pragmatic quality relates to how the participants perceived the system's efficiency for completing the task. While there were no significant differences between conditions, the Shared Viewpoint Control condition showed a trend towards higher ratings in items related to efficiency and clarity. Specifically, the Inefficient/Efficient and Confusing/Clear items in the questionnaire approached statistical significance towards efficient and clear ($p = 0.071$, $p = 0.083$, respectively).

Similarly, no significant differences were found in ratings of hedonic quality as measured by the final four items on the UEQ (Table 2). Hedonic qualities reflect the enjoyment of the participants relative to the collaborative tool. Although no significant differences were found, the Individual Viewpoint Control condition showed slightly higher ratings for excitement and inventiveness, suggesting that participants may have experienced more engagement in the Individual Viewpoint Control condition.

Overall, as shown in Figure 8, the results show that pragmatic quality tended to be perceived as slightly higher in the Shared Viewpoint Control condition, while hedonic quality showed minor differences favouring the Individual Viewpoint Control condition. However, these trends were not statistically significant.

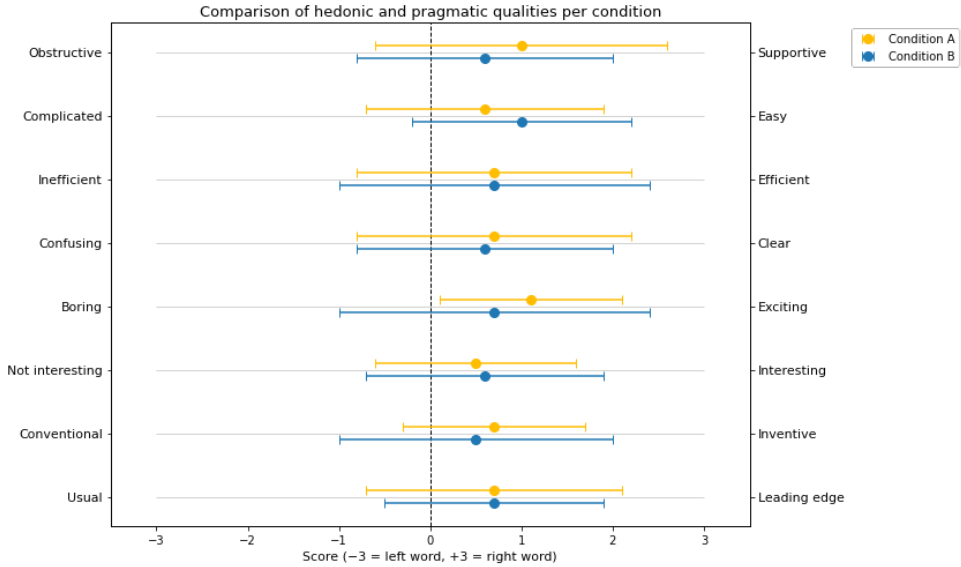


Figure 8. Results from the User Experience Questionnaire.

4. Discussion

The primary aim of this study was to determine how viewpoint control impacts the interactional bottleneck inherent to the Driver-Navigator interaction paradigm commonly present in DRs, especially in remote DRs. While task completion times were similar across conditions, the collaboration mechanics differed significantly. These results suggest a theoretical trade-off regarding the Driver-Navigator dynamic: Shared Viewpoint control minimizes the "cost of grounding" but not the "cost of instruction," whereas Individual Viewpoint reverses this dynamic.

In the shared viewpoint condition, the Builder spent significantly more time speaking than in the individual viewpoint condition. This aligns with the Driver-Navigator dynamic described in the literature (Bryant et al., 2008; Williams & Kessler, 2000), in which the driver must constantly verbally confirm navigational instructions. The shared viewpoint condition forces the Driver to act as the central processing unit for the group's spatial reasoning, increasing their cognitive and verbal load. Conversely, the individual viewpoint condition resulted in significantly more speaker changes, indicating a more distributed conversational structure. By decoupling the viewpoint, the "Driver" load was reduced, but the group was forced to engage in more frequent verbal exchanges to maintain mutual awareness of who was looking where. Furthermore, in the individual viewpoint condition, Builders tended to cluster their rotational positioning around specific "anchor points" (15-30°). Rather than exploring freely, this suggests that Builders prioritised stability to make it easier for Guiders (represented by avatars) to locate the Builders' perspective in the space. In contrast, Builders in the shared viewpoint condition performed significantly more navigation clicks (viewpoint changes) and explored the puzzle more

uniformly. Knowing that the Guiders were aligned with their view, Builders in the shared condition appeared to feel greater freedom to dynamically manipulate the perspective without fear of breaking the group's shared context.

This divergence in grounding strategies also shaped the nature of verbal clarifications. In the shared viewpoint condition, "Instructions to place" were followed by fewer clarifications because all participants shared a common perspective and a stable frame of reference. However, this came at the cost of increased navigation-related clarifications, where Guiders had to verbally instruct the Builder to rotate without being able to inspect the model independently. In the individual viewpoint condition, this pattern was inverted: independent viewpoints allowed Guiders to direct Builders using deictic references grounded in their own viewpoint (e.g., “Come here”), which reduced navigation clarifications but introduced discrepancies in reference frames that required additional clarifications for object placement. Although self-assessed user experience scores did not differ significantly, the observed pattern, where the individual viewpoint condition tended to score higher in hedonic qualities and the shared viewpoint condition in pragmatic qualities, suggests a balance between the engagement afforded by autonomy and the ease of coordination.

These findings suggest that greater individual freedom in viewpoint control is not uniformly beneficial in collaborative tasks and should be matched to the purpose of the DR. Based on the current findings, for tasks focused on presentation and consensus, shared control may be preferred because it reduces the need for navigational clarifications and ensures the group stays aligned, though it places a higher verbal burden on the presenter. For tasks focused on exploratory, discipline-specific inspection, individual control significantly increases stakeholders' verbal engagement, transforming them from passive observers into active participants. This suggests that enabling independent viewpoint control in remote DRs is best suited to tasks that require active examination and evaluation of the design of the product model, such as error detection, identifying opportunities for improvements, or detailed confirmation of compliance, rather than passive approval.

Furthermore, individual and Shared Viewpoint Control does not need to be treated as mutually exclusive configurations. These findings can be applied in hybrid systems that allow for adaptive viewpoint autonomy based on task demands and stakeholder roles. Many modern CVEs already include a follow mode, where participants can voluntarily (or automatically) adopt the presenter’s view and then decouple when wanted or needed. Our results suggest that DRs could benefit from a more structured use of such features and clearer guidelines for use. For example, participants responsible for documentation, requirements verification, or decision logging may be best kept in a persistent follow state (i.e., corresponding to the shared view) to maintain a stable shared context. In contrast, stakeholders performing discipline-specific inspections, e.g., concerning ergonomics, packaging constraints, clash searches, manufacturability, and serviceability, may be allowed or even encouraged to temporarily decouple, conduct independent exploration, and then rejoin the shared view. Similarly, teams may “force-follow” during presentation phases to maintain common alignment and focus. The

goal of a selected interaction paradigm should be to reduce individual and group cognitive effort by considering task, role, and impact of the selected paradigm on collaborators.

This perspective reframes viewpoint control not as a binary choice but as a configurable interaction resource that can be dynamically assigned based on the DR's structure, participants' expertise, and the discussion's phase. Effective DR systems might therefore incorporate role-based permissions, between shared and independent modes, and smooth transitions between following and free exploration.

5. Limitations and Future Research

This study focused on a controlled puzzle-solving task to isolate the mechanics of viewpoint control. While effective for measuring interaction dynamics, this task lacks the semantic complexity of a real engineering assembly or of a real DR. In a professional DR, the primary cognitive challenge often lies in understanding complex engineering constraints rather than in the mechanics of navigation. Consequently, the findings regarding communication overhead may differ in scenarios where the "what" (the design issue) is more complex than the "where" (the location). Future research should explore the impact of viewpoint control in more realistic, domain-specific contexts to validate the observed trade-off between the grounding efficiency of shared control and the increased active engagement facilitated by individual control.

Another limitation concerns the participant pool. The study utilised university students with basic computer proficiency. Professional engineers with years of experience in the Driver-Navigator interaction for DRs may have developed specialised verbal shorthands (e.g., "Section view, Y-axis") that reduce the friction observed in the Shared Viewpoint condition. Future research should examine whether expert teams can overcome the Driver-Navigator bottleneck through standardised communication protocols.

It is also important to note that it is unclear how these findings relate to longer-duration tasks. The relatively short time required to complete the puzzle in this study (~10 minutes) may not have been sufficient to cause fatigue or reveal the long-term effects of constantly re-orienting oneself in an independent viewpoint setting. In tasks that require prolonged attention (for example, extensive DRs that can last for hours), differences in viewpoint control might have a more pronounced impact on user fatigue, cognitive load, and attention. Future research could explore these dynamics by extending the duration of tasks to better understand behaviour and user experience over time.

Regarding the conversational analysis, this study prioritised a high-level assessment of communication friction by aggregating all 'clarifications' into a single metric. Consequently, the quantitative scope of this study is limited to the magnitude of communication breakdown rather than the specific semantic category of the error (e.g., navigation vs. placement). Future research could extend this work by employing a more granular coding scheme to determine if the nature of the misunderstanding shifts alongside the frequency.

Future research will also explore how features such as pointing annotations, advanced viewpoint management systems, and note-taking affect collaboration dynamics and outcomes in engineering DR efforts in CVE.

6. Conclusion

This study found that task completion time did not significantly differ between conditions, and collaboration dynamics varied depending on viewpoint control. In the individual viewpoint condition, participants had more distributed verbal exchanges, as shown by the higher number of speaker changes. In the shared viewpoint condition, collaboration was more centralised, requiring fewer clarifications after placement instructions but more explicit navigation instructions from guiders. Additionally, builders in the shared viewpoint condition changed viewpoint more frequently, as shown by their significantly higher number of navigation clicks and their more distributed positioning around the puzzle. In the individual viewpoint condition, participants relied more on visual information, as indicated by the conversational analysis, than in the shared viewpoint condition. Additionally, the self-reported user experience assessment did not show a significant difference between conditions, with the individual viewpoint condition rated slightly higher for hedonic features and the shared viewpoint condition rated slightly higher for pragmatic features.

These findings have practical implications for designing and using collaborative tools for remote DRs in collaborative virtual environments. While task performance metrics did not differ significantly, the collaboration dynamics differed across different conditions. This suggests that the specific context and objectives of the collaborative task should guide the choice between individual and shared viewpoint control.

Furthermore, these findings support the implementation of hybrid systems that allow for adaptive viewpoint autonomy based on task demands and stakeholder roles. For example, participants responsible for documentation may benefit from a persistent "follow" mode to maintain a stable context, while those performing specific inspections should be encouraged to temporarily decouple for independent exploration. Ultimately, viewpoint control should be treated as a configurable interaction resource, dynamically assigned to reduce cognitive effort.

Declarations

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Competing and financial interests

The authors declare no competing or financial interests.

Ethics approval and consent to participate

The study was conducted under institutional and national research ethics standards for research involving human participants. Participation in the study was entirely voluntary, and all participants provided written informed consent prior to participation. Participants were informed of their right to withdraw from the study at any time without consequence. No sensitive personal data were collected, and all data were anonymised to ensure confidentiality.

Data Availability Statement

The data supporting this study's findings are available from the corresponding author upon reasonable request. However, personal data will not be shared to protect participants' privacy.

Consent for Publication

All authors have read and approved the final manuscript for publication.

Authors contributions

F.G.R. led the study, conducted the research, and was the main author of the manuscript. M.L. and D.H. contributed to the experimental design; M.L. also assisted with the software used in the study. D.H. supported the theoretical positioning of the research within the design field. N.D. and F.F. assisted with the experimental setup. D.H. and M.L. reviewed and edited the manuscript for clarity and rigor. N.D. and F.F. also contributed to the revision process. All authors reviewed and approved the final manuscript.

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PAPER V:
BEYOND VIDEOCONFERENCING: HOW COLLABORATIVE TOOLS MAKE VIRTUAL DESIGN REVIEWS WORK

Beyond Videoconferencing: How Collaborative Tools Make Virtual Design Reviews Work

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Abstract. This study examines how specific collaborative features for independent viewpoint control, collaborative pointing, sketching, and manikin representations support design reviews in a virtual environment. Participants conducted design reviews using the collaborative design tool Gravity Sketch while engaging with these features. Results show that friction situations were minimal, with viewpoint control eliminating the need for perspective adjustments. Collaborative pointing and sketching effectively clarified design modifications, while manikin representations aided discussions on ergonomics. The findings highlight how the evaluated features facilitate communication in remote design reviews.

Keywords: Design Reviews · Remote Collaboration · Product Development

1 Introduction

Design reviews (DRs) are important checkpoints in the product design and development process. During a DR, engineers, designers, and other stakeholders evaluate various aspects of product design proposals, including performance, functionality, and manufacturability at various stages [1, 2]. The primary goal of DRs is to identify potential design flaws or issues before advancing to subsequent phases [3], thereby enhancing overall product quality and reducing the risk of downstream problems. Additionally, DRs serve as collaborative platforms where stakeholders with diverse roles and interests work together with engineers and designers to refine design approaches that meet agreed-upon requirements [4]. Although DRs have traditionally been conducted in physical settings, they are increasingly transitioning to remote and hybrid formats due to the rise of distributed work environments [5]. While these new formats enhance accessibility and scheduling flexibility, they also introduce challenges, particularly in sustaining the continuity of discussions. Factors such as technical difficulties reduced non-verbal cues, and the potential for disengagement and delays in remote distributed settings can hinder the natural flow of conversation, making it more difficult to maintain effective collaboration.

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A key aspect of DRs is visual support, which plays a crucial role in driving discussions and decision-making [6]. Visual elements, including drawings, 3D models, and physical prototypes, provide participants with essential reference points and something to "*look at*" during the DR. However, in remote DRs and hybrid settings, effectively integrating and interacting with these visual supports requires new approaches to ensure discussions remain productive and well-structured. When participants are remotely located, the most common method for providing visual support involves using computer-aided design (CAD) software in combination with screen-sharing in videoconferencing tools [7]. However, this approach often disrupts the flow of communication [8]. Usually, a single user (the presenter) controls the shared viewpoint, while other participants provide verbal instructions to engage with the content. This constraint can hinder collaboration because the presenter must interpret instructions and manually adjust the shared view, leading to potential delays and misunderstandings, especially when requested views are complex or need attention to complicated details. In this context, misunderstandings and delays create friction situations, disrupting the continuity of interaction and impeding smooth collaboration.

Friction situations are interruptions in workflow that arise from technical or procedural barriers during remote DRs. Through a previous study, we identified four types of recurring friction situations that disrupt collaboration and decision-making in distributed DRS [8]:

- Requesting specific viewpoints: The need for specific perspectives on the design, requiring the presenter to adjust the view manually.
- Indicating specific elements: The challenge of pointing to or highlighting specific design features without direct interaction.
- Expressing changing design ideas: Difficulty in conveying modifications and adjustments efficiently.
- Evaluating physical ergonomics: The inability to assess human interaction with the proposed design due to limited visualization tools.

Because friction situations disrupt remote DRs and because remote DRs have become common, there is a need to identify methods for reducing friction situations in remote DRs. A commonly proposed solution is implementing extended reality (XR) technologies, which can improve certain aspects of collaboration in DRs, such as real-time spatial awareness, intuitive interaction with 3D models, a better understanding of design change ideas, and enhanced remote collaboration [7, 9–11]. When conducting DRs with XR technologies, DRs occur in collaborative virtual environments (CVEs), digital spaces where users can gather and interact. CVEs can be accessed through a range of devices, from standard laptops to immersive augmented reality (AR) and virtual reality (VR) systems [12, 13]. As such, unlike traditional screen-sharing solutions, which assume 2D screen-based interactions adapted to 3D CAD tools, CVEs are designed assuming 3D navigation and interaction by default. Thus, while XR devices offer advantages by facilitating natural interactions with 3D content, much of the improvement in collaboration in DRs may come from the CVE itself and not the specific technology used to access the CVE. Unlike traditional screen-sharing in video conferencing applications, a CVE provides participants with a greater sense of agency and autonomy, allowing

them to explore and manipulate visual content independently rather than relying on the presenter's-controlled viewpoint.

Building upon our previous findings [8], this study investigates the dynamics of friction situations within a CVE during DRs. Specifically, we examine a commercially available CVE design tool not explicitly made for DRs, Gravity Sketch, to investigate how users leverage various collaborative features that are potentially important to mitigate or eliminate friction situations in DRs, including:

- Advanced Viewpoint Control: This feature allows users to navigate the 3D model independently or to follow another participant's perspective. Additionally, users can request that others follow their views. In the Gravity Sketch implementation, the decision to control one's viewpoint or to follow another's is left to the user rather than being centrally managed, and users can only have one viewpoint at a time. Furthermore, while VR users cannot continuously follow others, they can move to a specific location in space to align with another participant's view. Users can also request others to follow their viewpoint, and this can be accepted or rejected by others in the CVE.
- Collaborative Pointing Tools: This feature enables users to point out elements or specific locations within the CVE so everyone's attention can be drawn to the same location or elements. In Gravity Sketch, a pointer indicates a specific location in space, though it does not highlight individual elements. The pointer is available to users on 2D devices, while VR users can either use a laser pointer or extend their hand or controller towards an object, others in the virtual space can follow their gesture.
- Manikin Representations: Manikins facilitate ergonomics evaluations of product design proposals by integrating digital human figures. Gravity Sketch offers male and female manikins whose postures can be adjusted (in VR only). Although the models do not reflect anthropometric diversity, they can be uniformly rescaled, meaning that all body measurements adjust proportionally. There are also 3D assets of specific body parts, including the torso and head.
- Sketching Tools: These tools support real-time drawing directly within the CVE. VR users can sketch in 3D and have various options for quick sketching, while 2D users can only do basic sketches in a 2D place.

While remote DRs offer increased accessibility and flexibility, they also introduce friction situations that can disrupt collaboration and impede decision-making. Shifting to a CVE can provide a viable alternative to conducting DRs by facilitating more natural, interactive engagement with 3D content. Our previous research identified the aforementioned friction situations and these key features that have the potential to reduce them or eliminate them. In this study, we investigate how the implementation of these features affects the occurrence and resolution of friction during DRs.

2 Method

This study utilized Gravity Sketch[1], an XR-native design tool, as the platform for conducting DRs. Gravity Sketch integrates key collaborative features and supports simultaneous use of XR devices (e.g., Meta Quest 3) and 2D screens, making it a suitable choice

[1] Gravity Sketch Limited, London, UK: https://gravitysketch.com.

for this experiment. Its capabilities allowed us to observe how participants exploited its features to overcome friction situations [8].

In Gravity Sketch, the CVE (referred to as "CollabRooms" by Gravity Sketch) is accessible via invitation on both XR devices and 2D devices. Users on 2D devices must install the *Gravity Sketch ScreenCollab* software to join. Within the CVE, XR users are represented as headsets with two controllers, as shown in Fig. 1, while 2D users are represented as small squares with a line underneath, as shown by the red squares in Fig. 2. For this experiment, one participant used a Meta Quest 3, and two participants joined using laptops connected to an external display.

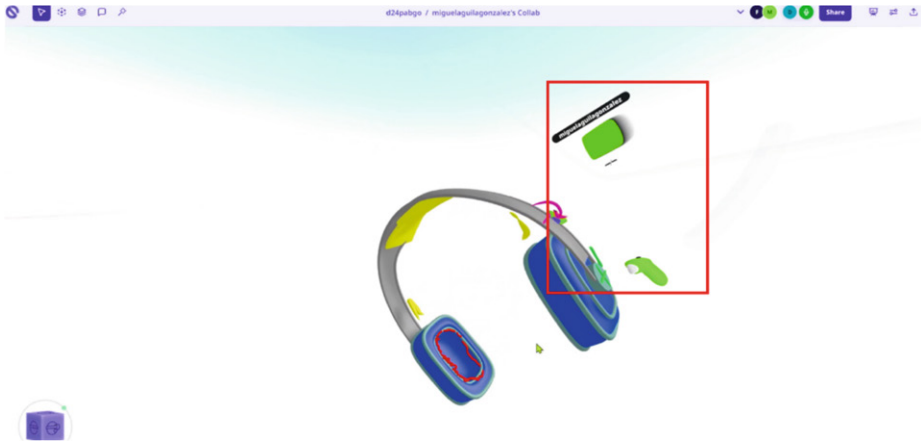


Fig. 1. 2D participant view of the CVE. The XR participant is represented inside of the red square. (Color figure online)

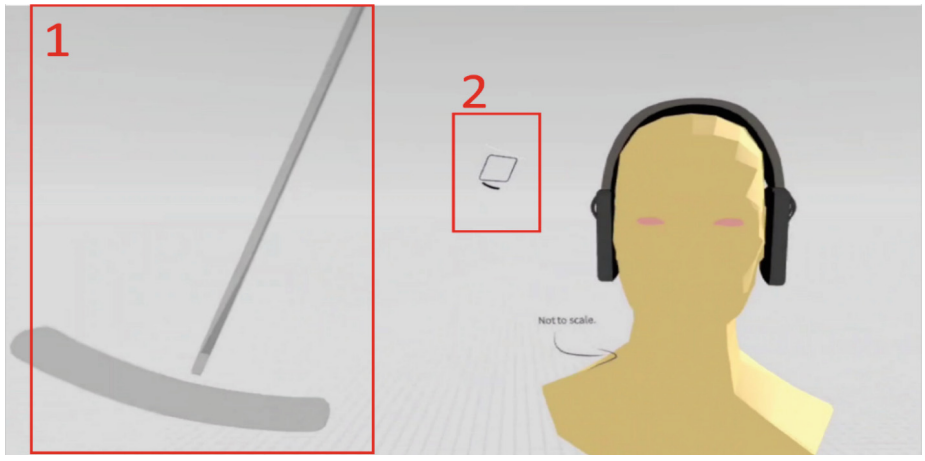


Fig. 2. XR participant view of the CVE. The 2D participants are represented inside of the red squares. (Color figure online)

2.1 Participants

A common challenge in XR experiments is recruiting participants with both hardware knowledge and software proficiency. In this study, participants were final-year BSc Product Design Engineering students enrolled in the *Digital Tools in Product Development* course at University of Skövde. As part of their coursework, the authors provided Gravity Sketch training, which began two months before the experiment. Each group of four students was assigned a Meta Quest 3 headset, requiring them to share and take turns using it. Students were trained in Meta Quest 3 and ScreenCollab for Gravity Sketch to ensure active participation. Students were encouraged to use *ScreenCollab* throughout the course for collaboration while alternating headset use. Additionally, students practiced DR-style collaborative tasks in the CVE, gaining hands-on experience in using the software for teamwork.

By the time of the experiment, all participants were proficient in Gravity Sketch, enabling them to fully utilize its features during the DRs. As part of their coursework, they were tasked with designing over-ear headphones, providing a relevant context for the DRs. Out of 18 students in the course, nine students (five males and four females, aged 20–45) volunteered for the experiment, forming three groups of three. Participating students were informed that participation in the study would not affect their course grade and written consent was obtained. Students were informed that they could end the study task at any time, though all groups completed the task. To maintain familiarity with the design and ensure established team dynamics, groups remained the same as in coursework, ensuring prior experience in CVE collaboration.

Each DR focused on reviewing the group's headphone design, aiming to collaboratively identify around five key action items for improvement. The role of the XR participant was assigned through voluntary selection within each group.

2.2 Procedure

Each session began with an introduction, during which the three participants were briefed by the first author on the purpose of the study and the task they were about to undertake. The experiment took place with everyone in the same physical room, seated, as shown in Fig. 3 One participant in XR drove the conversation, and two others sat each with laptops connected to an external display. The 2D participants were facing each other while the XR participant stood next to them. Participants could not see each other screens.

Following the briefing, participants were introduced to their task: conducting a DR of their headphone designs realized as 3D models within their course. They were instructed to evaluate the design from various perspectives, focusing on key elements such as the comfort of the headband, the positioning of the ear cups, the adjustability of components, and the distribution of pressure across the head and ears. They were also reminded of the collaborative features available in Gravity Sketch, both in XR and in the ScreenCollab version, especially the possibility of following someone's view, the collaborative pointing tool, the manikin representation, and the sketching tool. They were not specifically told which features they should use but were reminded that the features were available.

A researcher was present in the physical room (sitting between the XR participant and the 2D participants in one of the corners to not enter the XR safe area), observing

the discussions but not directly intervening in the selection of action items. While the researcher did not intervene in the discussions or influence the specific action items, they guided the conversation when suggestions were too general. This was necessary because overly broad action items often lacked references to specific shapes, locations, components, or perspectives—factors that typically contribute to friction situations. For instance, if a participant suggested, "*make the headphones more comfortable"* or "*make the headphones foldable"*. The action items had to have a narrow focus that could be actionable (i.e., clear instructions on what to do). One accepted action item could be, "*Implement an axis of rotation at the end of the headband so that the headphones can be folded around the longitudinal axis when wearing them around the neck"*, or "*Add foam material to the locations marked in red in the headband to distribute weight across the skull better*". It was also necessary that participants understood how the action item had to be implemented.

After the session, participants were given additional information about the study objectives, and there was a debriefing for questions and concerns.

2.3 Data Collection

The data collected during the DR sessions comprised multi-source recordings to allow for a comprehensive analysis of participant interactions and software feature usage after the sessions. Data was collected through screen recordings which included audio. OBS Studio[2] was used for screen recording on the laptops. The Meta Quest 3 viewpoint was recorded using the Meta Horizon companion app for smartphones.

Once all recordings were collected, they were imported into DaVinci Resolve software for synchronization. The three video streams were aligned (using the common audio data since all participants were in the same room) to visualize the session from all perspectives, as shown in Fig. 3. Furthermore, the audio recordings were transcribed using a locally hosted instance of OpenAI's Whisper model for the analysis. The combined video and audio datasets were the primary data used to analyze the DR session and code the friction situations.

2.4 Data Analysis Method

The data analysis aimed to determine whether the friction situations identified in previous research [8] emerged during these DRs and, if so, to understand how participants used specific collaborative features to overcome the friction situations to ensure an efficient design review process. In this context, friction situations are defined as disruptions or barriers to smooth communication and workflow that arise from technological or procedural constraints during a DR. To provide a structured analysis, we focused on four main categories of friction challenges:

- Requesting Specific Viewpoints: This friction occurs when participants need to view particular perspectives of the 3D model to understand better or discuss design elements. In traditional remote DRs, a central presenter typically controls the viewpoint, meaning that participants must request adjustments to see the 3D model from

[2] OBS Studio, https://obsproject.com/.

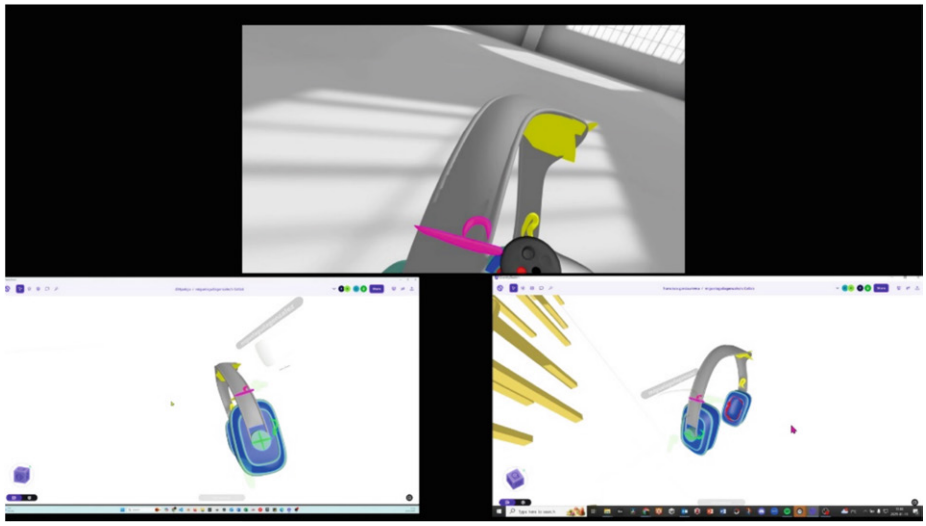


Fig. 3. Synchronized view for analysis. The top panel is the view from the XR participant. The bottom panels are the views from the 2D participants.

the desired location in the CVE. Such requests can lead to *delays* and *interruptions*, especially when the desired viewpoints are complex, multiple participants request different perspectives simultaneously, or the shared control mechanism limits immediate access.

- Indicating Specific Elements: This category refers to situations where participants encounter difficulties pointing out or referring to particular components within the 3D model. In complex designs, elements can be obscured or appear similar to surrounding parts, leading to *misunderstandings* about which component is under discussion. The friction situation can be worsened when the indication is done verbally or without adequate visual support, thus hindering the natural flow of the discussion.
- Expressing Design Change Ideas**:** Friction in this category arises when a participant proposes modifications to the design, but the idea is not immediately clear to the rest of the team. The challenges here come from the inherent difficulty of conveying spatial and design nuances through verbal descriptions alone. When participants *struggle to communicate* their design change ideas effectively, additional tools, including real-time sketching or screenshots, may be needed to bridge the gap, although their use can sometimes introduce further complexity or delay.
- Evaluating Ergonomics: This friction situation involves difficulties in assessing how well a design accommodates human use. Ergonomics evaluations require understanding spatial relationships and human interaction with the design. These assessments can be challenging on 2D screens or when using simplified digital manikins, leading to *prolonged discussions* as participants attempt to reach a consensus on issues such as comfort, reachability, and usability. Limited level of accuracy provided by the manikin representations, e.g. in terms of representing anthropometric details or diversity within the targeted user population, can worsen these challenges.

To understand how these friction situations arose or were avoided, we examined the use of four collaborative features within Gravity Sketch described earlier: advanced viewpoint control, collaborative pointing tools, manikin representations, and sketching tools. A coding protocol was developed to track friction situations and use of the target collaborative features in Gravity sketch. This included recording the following information for identified friction situations:

- Time Stamp: The onset and offset times (using the [mm:ss] format).
- Friction Category: One of the four types: Requesting Specific Viewpoints, Indicating Specific Elements, Expressing Design Change Ideas, or Evaluating Ergonomics.
- Brief Description: A summary of the situation to provide context, including what the user was trying to communicate and how the other participants understood (or misunderstood).
- Collaborative Feature Utilized: Documentation of any tool use (e.g., advanced viewpoint control, collaborative pointing, manikin representations, or sketching tools), employed to address or mitigate the friction.

The data analysis process started by visualizing the videos several times to get familiar with the data. After that, the videos were watched in more detail to transcribe the friction situations according to the template. Every time a friction situation was identified, the video was paused, and the specific timestamps limiting the friction event were documented. All screen recordings were visible simultaneously, allowing for an analysis of what each participant was doing when the friction occurred and why it happened. If there was a misunderstanding, the recorded footage provided insights into what one participant intended to convey versus how others interpreted it. These friction instances were reviewed multiple times. Each section was replayed to confirm its category, provide a detailed description, and identify features used during the friction situations. The first author conducted all the coding. Additionally, the footage was reviewed to understand tool usage during smooth interactions, whether the features were part of the workflow or used only in cases of misunderstanding, and to clarify the purposes for which the different features were employed. Although these instances were not documented in the template and were gathered in the form of unstructured notes, this review provided insight into how features were used proactively, beyond simply resolving friction situations.

3 Results

Each DR session lasted approximately 15 min. A total of 10 friction situations were observed, primarily related to expressing design change ideas, indicating specific design elements, and evaluating ergonomics. Two sessions recorded three friction situations each, while the third session had four. The recorded friction situations are presented in Table 1. These friction situations were relatively rare and when they did occur, they were quickly resolved using the available collaborative features.

The collaborative pointing tool was the most frequently used feature for resolving friction situations (see Table 2). It was particularly effective when participants struggled to indicate specific elements or express design changes. Additionally, it played a crucial role in confirming proposed modifications, enabling users to quickly validate

Table 1. Recorded friction situations.

Friction Situation	Total Instances	Effectiveness of Collaborative Features
Expressing design changes	5	Quickly resolved in most cases using collaborative pointing and VR sketching
Indicating elements	3	Occurred mainly when participants forgot to use the pointing tool; quickly corrected when used
Evaluating ergonomics	2	Manikin representations helped clarify usability concerns immediately
Requesting viewpoints	0	No issues reported, indicating viewpoint control tools worked as intended

whether they were referring to the same component or spatial information. In five separate instances, collaborative pointing effectively clarified intended changes, swiftly resolving misunderstandings. The sketching tools were used three times to address friction situations, primarily when participants needed to illustrate modifications that were difficult to describe verbally. Manikin representations were employed twice, exclusively in ergonomics evaluations. While independent viewpoint control was frequently utilized throughout the sessions, especially when identifying new action items, it was not directly tied to resolving friction situations. Table 2 provides a breakdown of how these features contributed to overcoming friction situations.

Table 2. Feature used to overcome friction situations. Note that the total usage per feature only refers to total usage to solve friction situations, the features (especially independent viewpoint) were constantly used during the DRs.

Collaborative Feature	Total Usage (to resolve friction)	Primary Role
Collaborative pointing	5	Helped clarify intended changes and indicate specific design elements
Sketching tools	3	Used for visually explaining complex design changes, preventing miscommunication
Manikin representations	2	Addressed ergonomics concerns, reducing ambiguity in usability discussions
Independent viewpoint	0	Not tied to friction resolution but enabled participants to explore models autonomously

The following section provided further detail on how each collaborative feature was used.

3.1 Collaborative Pointing

Proactive vs Reactive Use. This tool was used both proactively and reactively. When used proactively, participants directed attention to specific elements or directions, preventing friction situations from arising in the first place. In these cases, the tool was an intuitive means of communication, seamlessly integrating into the workflow without causing disruptions. In contrast, when used reactively, collaborative pointing solved misunderstandings. Participants employed the pointing tool to visually clarify their statements if verbal explanations did not convey an idea effectively. This helped resolve confusion and allowed for smoother communication.

Friction Situation Addressed. The friction situations addressed are explained in Table 3.

Table 3. Friction situations addressed by collaborative pointing.

Friction Situation	Example
Indicating Specific Elements: Collaborative pointing was mostly used to refer to the specific element being discussed, especially in cluttered zones with multiple components	During one session, a participant stated, "*We should make this button smaller…"* while discussing a control panel with four buttons. Initially, they hovered over the button with their mouse without using the pointing tool, which led to the other two participants responding, "*I don't know what button you are talking about."* Once the pointing tool was employed, the intended button was immediately identified, and the discussion moved forward without further confusion
Expressing Design Changes: Collaborative pointing was also used to illustrate design changes, particularly those involving movement or rotation	In one instance, a participant said, "*We should make the headphones foldable like this…"* while pointing in a specific direction. This allowed the group to visualize the intended adjustment and confirm agreement before proceeding with further refinements. In most cases, this usage did not introduce friction. However, if the design idea was too complex, other features (mostly sketching) were used in combination to clarify the concept

3.2 Sketching Tools

Proactive vs Reactive Use. XR participants initiated multiple sketches throughout the sessions, often drawing freehand annotations, highlights, or rough modifications to support their verbal descriptions. These sketches were mostly created spontaneously, and

in many cases, XR participants would make a quick sketch and then erase it once the point was communicated.

2D participants used sketching reactively, mostly in response to confusion arising during discussions. When a design change idea was unclear, 2D participants would attempt to use the sketching tool to clarify their thoughts. However, this usage was far less frequent than among XR participants, as 2D participants typically relied more on verbal explanations and pointing. Moreover, there were instances where the 2D participants began to sketch something, but the XR participant took over sketching based on the vague explanations that the 2D participants provided. Sometimes, rough feedback was needed to refine the sketch, but the XR participants mostly did the sketch.

Friction Situations Addressed. The friction situations addressed are explained in Table 4.

Table 4. Friction situations addressed by sketching tools.

Friction Situation	Example
Expressing Design Changes: Sketching tools were primarily used to illustrate modifications that were difficult to describe verbally	A 2D participant was discussing the placement of padding in the frame of the headphones. However, the other participants did not properly understand the proposed change This led to a prolonged exchange where the other participants misunderstood how the padding would be distributed. The 2D participant began using the sketching tool to clarify their idea and the XR participant also began to draw a rough outline of the intended placement and shape. The 2D participant deferred to the XR participant. Although some feedback was needed to capture the 2D participant's idea, the sketch was fully done by the XR participant
Indicating Specific Elements (XR Participants Only): While the sketching tool was used several times as a pointing tool, one friction situation was recorded when the feature was used to indicate specific elements	During a discussion concerning the structural integrity of a hinge for folding the headphones, an XR participant attempted to verbally indicate a specific weak point but was met with uncertainty from the team. To clarify, they circled the joint using the sketching tool

3.3 Manikin Representation

Proactive vs Reactive Use. Manikins were not proactively used; their use was exclusively reactive when ergonomics concerns could not be solved with verbal information alone. Because of the nature of the product, participants only used partial human representations (in this case the head) as opposed to the full manikin.

Friction Situations Addressed. The friction situations addressed by manikin representations are addressed in Table 5.

Table 5. Friction situations addressed by manikin representation.

Friction Situation	Example
Evaluating Ergonomics: Manikin representations were effective in resolving usability, mostly related to the fit on the ears or the curvature of the frame around the head	A 2D participant raised concerns about whether the headband of the headphones would distribute pressure evenly. The other participants did not share this concern, leading to uncertainty about the design. Moreover, the others were unsure if a proposal of additional padding material on the sides would impact overall fit. A manikin representation was introduced to simulate a human head, allowing the group to observe contact points and confirm that adjustments were necessary

3.4 Viewpoint Control

Proactive vs Reactive Use. All of the viewpoint control usage was proactive, as participants freely adjusted their perspectives without requiring intervention. Some 2D participants relied heavily on the follow view feature, almost always following someone else's viewpoint. In contrast, others rarely, if ever, followed anyone, preferring to navigate independently. There were instances where a conversation was ongoing, but a (participant who was not involved in the discussion) might have been focusing elsewhere, possibly not paying attention. These moments did not cause misunderstandings but indicated variations in engagement strategies.

4 Discussion

The study's findings demonstrate that CVE can significantly enhance remote DRs for both XR and 2D screen-based users by reducing communication breakdowns and workflow disruptions. However, beyond their immediate functionality, a broader examination of their implications on collaborative design, participant behavior, and system design principles offers valuable insights for future development.

4.1 Rethinking Remote Collaboration: Beyond Passive Viewing to Active Participation

Traditional remote DRs often limit participants to passive viewing of shared screens, requiring verbal exchanges to navigate and discuss designs. In contrast, CVEs redefine

remote participation by enabling direct engagement with 3D content, enabling active participation rather than passive observation. The ability to freely navigate, point, sketch, and manipulate objects shifts the role of participants from spectators to co-creators. Active engagement in CVEs is not limited to XR participants with many features of CVEs benefiting active engagement for 2D screen-based users as well. Future collaborative system designs, such as the one explored here, can focus on facilitating active, multi-user engagement, ensuring that all participants can contribute equitably rather than relying on a single presenter to dictate the discussion.

4.2 Balancing Autonomy and Synchronization in Collaborative Workflows

Another key takeaway is the importance of flexibility in viewpoint control, allowing participants to choose between independent exploration and synchronized viewing. However, the occasional disengagement of participants when navigating independently suggests that unstructured autonomy can sometimes lead to fragmented discussions. One way to address this would be the design of CVEs that integrate adaptive synchronization features, where viewpoints can be temporarily anchored or guided during critical discussions without restricting participants' ability to navigate freely when needed. For instance, drawing inspiration from the creative practices of artists who guide the attention of XR users toward specific tasks or virtual objects [14], a soft-guidance system could be developed to provide non-intrusive cues. This system would subtly direct users' focus toward relevant discussion points while still allowing for autonomy when appropriate. Such an approach could enhance collaboration by ensuring that critical aspects of the design review receive collective attention without imposing rigid viewpoint restrictions.

4.3 Intuitive Tool Discoverability and Proactive Assistance

While the study found that collaborative tools like pointing and sketching effectively resolved friction situations, their inconsistent usage suggests that participants may overlook available tools in dynamic discussions. As the modeling of objects in CVEs and their digital representation becomes more complex—such as mechanical assemblies with interdependent moving parts—the technical challenge of rendering and delivering these environments to XR devices in real time increases. Ensuring that participants can manipulate objects seamlessly while maintaining a coherent sense of movement and causality becomes particularly critical in XR, where the temporal alignment of actions differs from traditional 2D interfaces. Unlike users interacting with virtual objects on a 2D screen, XR participants may experience delays or preemptions when interacting with the same object. These discrepancies arise due to the need for real-time spatial rendering, affecting both responsiveness and the perceived continuity of actions.

However, while latency is a challenge in both 2D and XR in CVE, XR introduces additional complexities due to its spatial and interactive nature. In XR, slight delays in rendering or synchronization can create a disconnect between user input and system response, leading to interaction breakdowns. These timing discrepancies are particularly relevant when multiple users interact with the same digital object in a CVE, as maintaining a shared, temporally consistent experience requires precise real-time updates [15]. Thus, usability challenges arise not just from ensuring tool availability but also from

making tools discoverable and seamlessly integrated into workflows. Addressing these challenges requires strategies such as context-aware tool suggestions to prompt users when verbal descriptions cause ambiguity, shortcut-based activations like gesture-based or multimodal interactions to reduce cognitive effort, and subtle system nudges that highlight underutilized tools without disrupting the workflow.

4.4 Equitable Collaboration Between XR Users and 2D Users

The observation that XR participants predominantly performed sketching tasks suggests that CVEs need to address, or at least make explicit, role imbalances between device types. While XR provides a more immersive interaction mode, non-XR users must be equipped with equally intuitive input methods to contribute effectively. For example, Gugenheimer et al. [16] have previously highlighted both the opportunities and challenges of designing a shared, collaborative mixed-reality game for XR users and 2D users. Potential solutions for DR could include enhanced 2D sketching interfaces, such as stylus or touchscreen support, to enable 2D users to contribute more fluidly; cross-platform consistency to ensure that tool accessibility and interaction remain device-agnostic rather than XR-centric; and shared control dynamics, allowing 2D and XR users to co-edit sketches simultaneously, fostering truly collaborative workflows instead of hierarchical interaction patterns.

On the other hand, the reactive use of manikin representations suggests that ergonomics considerations are often overlooked until explicitly needed, highlighting a broader issue in design workflows. Usability and human-centered factors tend to emerge late in the design process rather than being proactively integrated from the outset.

One way to address this is by embedding ergonomics evaluation prompts at relevant points in the discussion, encouraging teams to consider human factors earlier. Additionally, real-time ergonomics simulation overlays can allow designers to visualize ergonomics constraints dynamically rather than treating them as a separate step. Another approach involves scenario-based analysis tools, where usability tests within the CVE simulate real-world interactions based on predefined ergonomics benchmarks, ensuring that ergonomics considerations become an integral part of the design review process rather than an afterthought.

4.5 Limitations and Future Work

While this study provides valuable insights into the effectiveness of CVEs in DRs, some limitations should be acknowledged. One consideration is the relatively small number of participants. While more participants would be valuable, the use of recently trained final-year product design students with substantial experience using Gravity Sketch, as well as direct familiarity with the product design they were reviewing, allows for observation of realistic behaviors of trained professionals.

Another limitation is that the study focused on a single phase of the design process, specifically an early-stage design review of over-ear headphones. This phase involved identifying key improvements and potential modifications rather than refining final design details. While this is a relevant and necessary stage of product development, later phases (where designs are more constrained by manufacturing and engineering

considerations) may present different friction situations and collaboration needs. Future research could explore how CVEs support more advanced design stages, where precision and technical feasibility play a larger role in decision-making.

Additionally, the study was conducted exclusively within Gravity Sketch. While this platform is one of the most advanced available for collaborative design in XR, it represents only one possible implementation of a CVE. At present, few alternatives offer comparable feature sets, particularly in terms of supporting both VR and 2D collaboration. However, as XR collaboration tools continue to evolve, future studies could examine whether different CVE implementations introduce unique interaction patterns or if the results observed here are consistent across multiple platforms. Finally, the study lacks a comparison group to represent the traditional DRs without Gravity Sketch CVEs features. This could lead to less qualitative/quantitative comparison to specify which friction situations are eliminated the most. But with our previous study and references for traditional DRs, [8] we could still have a preliminary assessment of the CVEs with practical study and observations.

Despite these limitations, this study provides strong evidence that CVEs, when equipped with well-integrated collaborative tools, and utilized by trained users can be a viable alternative to carry out DRs.

5 Conclusion

This study highlights the effectiveness of CVEs in supporting remote DRs involving both XR and 2D screen-based users, demonstrating that interactive tools such as collaborative pointing and sketching facilitate communication and decision-making in DRs. One of the most notable findings was the absence of friction related to viewpoint control. Participants could navigate the 3D model independently or follow another's perspective when necessary, ensuring that discussions remained focused on the design rather than on the logistical challenges of adjusting viewpoints. This suggests that giving participants autonomy on how to manage their viewpoint, makes smooth collaboration in DRs regardless of their use of XR or 2D screen-based technologies.

Friction situations were rare. When they did occur, participants resolved them efficiently using the available collaborative features. Occasional lapses in tool usage suggest that user interface guidance and training could further support users in fully using the available features. The findings reinforce that digital collaboration tools can play an important role in DRs, providing an interactive workspace where design discussions can take place.

Additionally, while CVEs provide tools that help structure discussions, they do not eliminate the need for organizational measures to ensure that DRs run smoothly. Clear facilitation, well-defined objectives, and structured participation remain essential to conducting effective sessions. The availability of collaborative features can support communication, but their impact is ultimately shaped by how they are integrated into the DRs.

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PAPER VI: GUIDELINES DEVELOPMENT FOR DESIGN REVIEWS WITH ADVANCED COLLABORATION TOOLS

Guidelines Development for Design Reviews with Advanced Collaboration Tools

Abstract

Design Reviews (DRs) function as formal governance checkpoints within the product development process where cross-functional stakeholders evaluate design proposals against product requirements to determine their readiness for progression. As design teams become increasingly global, they rely heavily on remote and digital workflows to conduct DRs. While research explores the implementation of new tools to enhance collaboration in DRs, studies often focus on technical feasibility or interaction effects in isolated settings rather than on how these technologies can be embedded in actual industrial processes. In this study, we examine the use of advanced collaboration tools, defined here as digital systems that provide shared spatial environments, in-scene deictic referencing, and independent navigation. Currently, organizations lack actionable guidance on integrating these tools into their DR processes. Thus, this paper develops an evidence-informed set of procedural guidelines for planning, conducting, and documenting DRs using advanced collaboration tools. Working with two automotive companies, we conducted a multi-phase study where two real industrial cases (one per company) were reconstructed to facilitate collaborative workshops aimed at mapping technological capabilities to procedural rules. By analyzing verbal data and interaction logs, we developed a set of guidelines grounded in the specific coordination problems identified by practitioners and the technological capabilities required to resolve them. The resulting guidelines identify five domains, including pre-read and preparation, attention management, documentation and traceability, integration and handoff, and modality and role distribution. The guidelines provide a transferable structure for organizations to transform advanced collaboration tools from isolated visualizers into reliable, managed elements of the DRs workflow.

1. Introduction

As design teams become increasingly global and remote work opportunities expand, Design Reviews (DRs) are undergoing rapid transformation, with teams relying more heavily on distributed, digital, and immersive collaboration tools. While existing literature extensively describes how collaborative technologies can benefit DRs (Bellalouna & Puljiz, 2023; Gebert et al., 2020; Horvat et al., 2022; Paes et al., 2021), there is limited guidance on how to incorporate collaborative technologies into established DR practices. This paper introduces a set of evidence-informed guidelines that link interaction capabilities to procedural expectations in DR planning, execution, and documentation.

1.1. Background and Motivation

DRs are formal checkpoints (often in the form of meetings) within the Product Development Process (PDP) where cross-functional stakeholders examine a design proposal against requirements, risks, and project objectives to determine its readiness for progression to subsequent stages of the PDP (Pahl et al., 2007; Ulrich et al., 2019). DRs are also structured evaluation assessments that integrates perspectives from different stakeholders, including engineering, manufacturing, ergonomics, quality, service, and management, to align understanding, raise issues early, and reduce downstream rework (Adams et al., 2016; Gräßler et al., 2022; Horvat et al., 2022). Within the PDP, DRs serve as governance points that link technical evidence to decision-making and record the reasoning behind progressing, redirecting, or postponing work (Chen et al., 2019; Huet et al., 2007).

A typical DR is often planned around a defined scope and a set of artifacts, for example, drawings, sketches, computer-aided design (CAD) models, simulation results, mock-ups, or physical prototypes, prepared to support the targeted evaluation of fit-to-requirements and overall design quality (Adams et al., 2016). DRs are typically led by a central presenter who guides the walkthrough. At the same time, domain specialists provide focused assessments, a decision authority confirms outcomes against product criteria, and the outcomes are then collected and documented (Adams et al., 2016). DRs may occur in person around shared visualizations or remotely via digital collaboration environments, but the procedural pattern includes guided joint inspection, expert discussions, and decisions on concrete actions for the next stage of the design proposal (Chen, 2021; Gebert et al., 2020; Yabuki, 2011).

1.2. Process Management Problems in Design Reviews

The expected outputs of a DR should be explicit and actionable. The decisions made, the reasoning supporting them, and any follow-up actions, including their respective owners and timing are typically documented and associated with a product data system, often a Product Lifecycle Management (PLM) system, to ensure traceability over time and across iterations (Bellalouna & Puljiz, 2023; Chen et al., 2019; Gräßler et al., 2022; Liu, 2020). When embedded in the organization's PDP and linked to configuration and PLM, DRs are essential to communicate technical and managerial judgment into documented, auditable progress on the product (Eckert et al., 2017; Puzović et al., 2025; Valeri et al., 2003). In practice, DRs are organized as scheduled meetings aligned with the progression of the PDP (Bedingfield & Clarkson, 2020; Chen et al., 2019).

Across co-located, hybrid, and fully remote meetings, DRs generally follow a similar procedural pattern. A designated presenter leads participants through a predefined agenda using selected design representations, most commonly CAD models, while domain specialists provide targeted assessments (Chen et al., 2019; Détienne et al., 2005). The artifacts used during the session, including drawings, bills of materials, simulations, and prototypes, are chosen to support the specific evaluation task and enable the group to jointly inspect the proposal and agree on required actions.

Despite their formal structure, studies indicate that DRs suffer from recurring process management problems, specifically regarding how outcomes are captured and acted upon. Frequently, meeting minutes fail to record key rationale and decisions (Elgh et al., 2018; Huet et al., 2007). Additionally, studies report that even when decisions are made explicitly during a DR, the resulting actions and design changes are often not implemented effectively because they are not captured or transferred into the organization's product data system (Kinneging et al., 2020). To address these disconnects, literature

proposes three main solution streams: (1) standardization of objectives and roles (Huet et al., 2007; *ISO 9001*, 2021); (2) higher digital integration with product data systems (Hall et al., 2018); and (3) the use of tools that consolidate meeting minutes, versioned artifacts, action logs, and decision rationales in shared environments (Liu et al., 2018). However, implementation lags behind theory. In practice, many teams struggle with inconsistent objectives (Conway et al., 2007), and tool fragmentation (where critical evidence is scattered across slides, spreadsheets, and emails), weakening the traceability required for effective engineering (Romero et al., 2022).

1.3. Interaction Problems in Design Reviews

Alongside these process management problems, interaction problems are also documented. These include difficulties in how participants communicate, share viewpoints, and coordinate their attention during DRs, which often lead to misunderstandings and friction in the DR. For instance, in the context of remote and hybrid DRs that rely on shared CAD models, previous research introduced the concept of *friction situations*, which are events that interrupt and hinder progress, often leading to delays or disruptions in the workflow (Rivera et al., 2025). This concept was developed through the study of DRs that combined CAD and videoconferencing, though it need not be limited to these situations. The initially identified friction situations included challenges in verbal negotiation, expressing viewpoints and positions, articulating design ideas, referencing specific elements in the model, and evaluating ergonomics (Rivera et al., 2025). More broadly, studies of DRs indicate that conversational floor control (i.e., how speaking turns are managed and who naturally holds the implicit right to speak) can become unstable. Interruptions and monopolization of speaking turns often restrict balanced participation and contributions, and limit the exploration of design ideas (Hussein & Peña-Mora, 1999; Lee & Ostwald, 2022; Margariti et al., 2022). In situations where visual and physical interaction methods are limited, teams often rely excessively on verbal descriptions, which frequently lack the spatial detail found in gestures and sketches (Härkki et al., 2018; Rohrer et al., 2025; Stempfle & Badke-Schaub, 2002).

In response to these challenges, the literature describes a range of technology-supported approaches for conducting DRs. Notably, immersive environments, most commonly extended reality (XR), have been shown to improve spatial understanding, facilitate more immediate referencing of geometric elements, and broaden participation to non-CAD specialists by enabling in-scene cues and interactive manipulations (Eckertz et al., 2020; Horvat, Martinec, et al., 2022; Paes et al., 2021; Van Goethem et al., 2020). However, while current research demonstrates notable advances in the technical capabilities that support interaction, a persistent gap remains in integrating these developments into the formal DR processes required for reliable team coordination and enable downstream engineering tasks (Camba et al., 2017; Freeman et al., 2018; Howie et al., 2023; Umair et al., 2022).

1.4. Technology-supported Approaches for Design Reviews

In parallel with these research developments, several industries (e.g., automotive, aerospace, and architecture) have adopted a set of interaction technologies that may be referred to as *advanced collaboration tools*. These tools are grounded in ideas similar to those identified in sections 1.2 and 1.3, for enhancing collaboration in the PDP, including shared 3D context, in-scene cues, and persistent capture. Yet, different advanced collaboration tools typically target specific industry sectors, phases, and product development domains. This category of software spans multiple industries and software classes, including*: review-oriented multi-user cross-platform viewers* such as Campfire (Campfire, California, USA) and VRED (Autodesk, California, USA); *cloud authoring tools* including Arkio (Arkio, Reykjavík, Iceland), ShapesXR (ShapesXR, California, USA), and Gravity Sketch (London, UK); *cloud co-authoring CAD platforms* such as Onshape (PTC, Boston, USA); and *process-centric issue platforms* including CoLab (CoLab, Newfoundland, Canada).

These collaboration tools are strategically positioned to support different stages of the PDP, ranging from early ideation to detailed design and pre-production, as well as domain-specific software tailored to industries such as architecture and vehicle design and manufacturing. These tools are built around various features useful in DRs, including virtual multi-user spatial environments, and independent and shared viewpoints. For note taking, these tools provide both anchored annotations and mechanisms to capture decisions and discussions around specific model states. At the same time, most tools emphasize

a specific design domain or moment in the PDP. For example, Gravity Sketch focuses on concept sketching and form exploration in early design; VRED provides high-fidelity visualization and review for detailed assemblies; CoLab supports workflow and issue management for teams coordinating changes across disciplines; and Campfire serves as a cross-platform visualizer with an integrated issue-tracking system. This variety reflects the diverse technical needs of engineering DRs across industries and stages; however, the availability of such specialized software has not yet been matched by the development of clear procedural frameworks for their effective integration.

While several studies have advanced the understanding of technology-supported DRs, their contributions tend to concentrate on specific phases or aspects of the review process. For instance, studies examining immersive environments have focused primarily on interaction during the review session itself (Horvat et al., 2022; Paes et al., 2021; Van Goethem et al., 2020), with limited attention to preparation or post-review handoff. Where procedural models have been proposed, they tend to address specific domains such as architectural usability reviews or reporting strategies (Gebert et al., 2020; Liu et al., 2018) rather than offering a cross-domain framework. Notably, none of these studies provides validated procedural guidance that spans the full DR cycle, from preparation through execution to integration with product data systems, across multiple industrial cases. This is the gap the present study addresses

1.5. Gap in Integrating Advanced Collaboration Tools into Design Review Workflows

Despite these shared interactive capabilities, a clear gap remains between what advanced collaboration tools enable at the interface level and how DRs are actually managed within the PDP. While some studies show that such tools can improve interaction and spatial understanding (Eckertz et al., 2020; Horvat, Martinec, et al., 2022; Paes et al., 2021; Van Goethem et al., 2020), they rarely explain how these capabilities translate into clear procedures for preparation, role coordination, and outcome capture. Consequently, many organizations struggle to embed them into standard DR workflows or connect them to their product data system.

In response to this gap, this study proposes a set of evidence-informed guidelines for planning, conducting, and documenting DRs with advanced collaboration tools. The guidelines explicitly link interactive features, such as shared viewpoints and anchored annotations, to operational considerations, including scope definition, scheduling, and responsibility assignment. The goal is to provide a transferable framework that organizations can adapt to embed these tools within their established, auditable DR processes and practices.

2. What are Advanced Collaboration Tools?

In this study, we refer to advanced collaboration tools as digital systems, whose interactive capabilities (e.g., multi-user shared scenes, in-scene deictic tools, independent navigation with optional follow) offer specific affordances to users, shaped by their roles and skills, tasks, and context setting. Drawing on Gibson's work, we understand an affordance as a relational property: an action possibility offered by the environment to a specific actor (Gibson, 1979). In this context, a software feature is defined as the technical capability that enables (or constrains) that relationship (Norman, 2013).

In the context of DRs, we identify a set of core affordances that distinguish advanced collaboration tools. These include mutual visibility of focus (seeing others' viewpoints and actions), deictic reference to geometry (indicating "this part, here"), and coupled viewpoint control (synchronizing or decoupling perspectives on demand). These affordances, in turn, support the interactional phenomena that are especially important in DRs, workspace awareness, grounding, and joint attention, during the inspection of complex geometry (Gutwin & Greenberg, 2002; McGrenere & Ho, 2000; Tang, 1991). Ideally, advanced collaboration tools in DRs also support the persistence of records tied to model states. However, given the variability in product data systems and integration policies across organizations, implementing a dedicated digital records repository is not always possible. For this reason, we treat

record keeping tied to model states as a desirable extension rather than a core requirement, noting that many tools already offer it.

Beyond this core, advanced collaboration tools typically include extended capabilities including review-safe manipulation, such as reveal and hide, sectioning, and slicing, along with role management and role-specific permission controls. When advanced collaboration tools include mechanisms to capture and record outcomes, decisions, rationales, and actions, recorded data is typically bound to model-state. In some cases, data recorded during the DR can be connected to existing repositories, such as CAD and PLM systems or issue trackers. These capabilities directly address documented needs for traceable capture and process linkage in DRs (Camba et al., 2016, 2017; Huet et al., 2007; Romero et al., 2022).

Advanced collaboration tools are not limited to a single presentation modality and can include software systems that leverage both XR and traditional 2D monitor-based systems. Thanks to their immersive qualities, XR systems are particularly capable of enhancing specific affordances such as spatial perception, natural deictic references, and scale cues. However, essential capabilities such as independent viewpoint control and anchored cues do not strictly require XR to be effective (Khanna et al., 2025; Vieira et al., 2024). Therefore, a simple videoconference with screen sharing and no in-scene cues or independent navigation does not meet our criteria, even if CAD is visible, since it lacks the necessary affordances for effective joint inspection and grounded referencing (Paes et al., 2021; Vieira et al., 2024).

Finally, advanced collaboration tools are used in diverse ways across industries and stages, from early concepts to detailed design and pre-production DRs. What unifies them is the set of affordances they provide to enable collaborative reasoning about the design proposal, including its geometry, requirements, and functional constraints (Berg & Vance, 2016; Eckert et al., 2017; Huet et al., 2007; Liu et al., 2018). Table 1 shows a summary of the required capabilities of advanced collaboration tools as defined in this study.

Table 1. Capabilities of Advanced Collaboration Tools

Capabilities Category	Collaborative Function in DR Context	Typical Implementations and Examples
Core affordances (minimum to be recognized as an Advanced Collaboration Tool)		
Shared Spatial Environment	Allows multiple users to co-exist in the same 3D scene. This includes the ability to see other users in the scene; establishes a common spatial frame of reference.	Real-time shared scene graph, multi-user presence.
In-scene Deictic Referencing	Enables participants to indicate specific parts of the geometry directly, reducing reliance on ambiguous verbal cues.	Pointing, highlighting, anchored sketches, or annotations attached to model geometry.
Independent and Shared Navigation	Supports autonomous exploration while allowing temporary synchronization around a presenter's view; ensures balanced participation.	Free viewpoint navigation, presenter-follow mode, focus synchronization toggle.
Extended affordances (complementary or "nice-to-have" capabilities)		
Review-safe Manipulation	Facilitates quick visual tests without compromising model integrity.	Hide/reveal, sectioning, slicing, model state bookmarking.
Roles and Permissions Management	Organizes participation and supports traceable authorship and access control.	Host/contributor/observer modes, access rights, moderation tools.
Integrated Communication Utilities	Consolidates communication within the collaborative workspace to reduce context switching.	Embedded voice/video chat, avatar gaze or pointer cues.
Session Capture and Traceability	Ensures that decisions and rationales persist beyond the review session and link back to managed data.	Action-item logs, auto transcription, review recording, CAD/PLM issue linking.
Domain-specific Analysis Tools	Extends the platform's utility by integrating discipline-specific evaluations within the collaborative context.	Digital human models, ergonomic checks, collision detection, layout simulation, material visualization.

By adopting this functional definition, we prioritize the interactional logic of the tools over their specific feature sets, ensuring that the analysis remains relevant as software platforms evolve. Crucially, the distinction between core and extended affordances serves as the structural logic for the procedural guidelines developed later in this study. In the proposed framework, Extended affordances (such as PLM integration or automatic capture) enable optimized workflows, referred to as the "Ideal Path." However, to ensure the guidelines remain universally applicable, we also define "Fallback Paths" that rely solely on Core affordances to achieve the same process goals through manual coordination. This layered approach prevents the guidelines from becoming rigid instructions dependent on specific high-end features, instead offering a resilient workflow that adapts to the technological maturity of the tool in use.

3. Methods

This section describes how the study was designed and conducted to develop evidence-informed guidelines for conducting DRs using advanced collaboration tools. The method includes the industrial context, recruitment of participants, preparation of the DR cases, workshop-based evaluations, and the procedures for collecting and analyzing the data. The aim is to provide sufficient detail for understanding how the cases were constructed, how participants engaged with the tool, and how the observations and workshop outcomes informed the guideline development.

Following the Design Research Methodology (DRM) framework (Blessing & Chakrabarti, 2009), this study combines descriptive and prescriptive elements. The descriptive component identifies coordination breakdowns through empirical observation of DR practices, while the prescriptive component translates these findings into structured procedural guidelines.

The remainder of this section is organized as follows: Section 3.1 describes the industrial context; Section 3.2 presents the case selection; Section 3.3 covers recruitment; Section 3.4 details the software and apparatus; Section 3.5 describes the study procedure; Section 3.6 presents data collection; and Section 3.7 explains the data analysis strategy.

3.1. Industrial Context

This study was conducted as part of a broader collaboration between [anonymous university] and two automotive companies [referred to here as Company 1 and Company 2 for anonymity] within the scope of an ongoing research initiative [details removed for anonymity]. The selection of cases was shaped by the ongoing collaboration between the research team and both companies, where several of their DR practices had previously been observed and discussed within the broader research project. As part of this collaboration, and drawing on their familiarity with the project aims, the companies proposed concrete DR cases that could be shared without intellectual property risks yet remained representative of their operational practice.

Currently, both companies use established digital workflows in which DRs focus on CAD-based evaluations. These sessions typically involve a hybrid setup: on-site participants gather around a conference room display, while distributed team members join via videoconferencing. In this format, the group's spatial understanding is constrained by the presenter's control of the shared view. Motivated to improve this dynamic, the companies aim to reduce reliance on the single-presenter model and strengthen spatial comprehension for remote engineers. To this end, they identified representative DR cases to be examined using advanced collaboration tools, serving as a testbed to understand how virtual environments might enhance their ongoing process development.

3.2. Case Selection

At Company 1, the case concerned a fixture used to locate and secure a car's body panels during welding. This type of fixture is evaluated in manufacturing-oriented DRs to ensure that operators and robots have sufficient access for welding, that clamping operations are feasible, that tolerances and positioning requirements can be met, and that the fixture can be maintained and adjusted over time. Because these fixtures directly influence assembly ergonomics, welding quality, and manufacturability, they represent a typical topic for cross-functional DR discussions (Figure 1). At Company 2, the case focused on the lower pallet that supports a car's front subframe assembly and related components during the assembly process. In DR meetings, this pallet is evaluated to verify the feasibility of the planned assembly sequence, assess tool access and clearance, evaluate operator posture and reach, and identify potential collisions or constraints that could cause downstream production issues. The pallet, therefore, provides a realistic context for examining spatial access, ergonomics, and process planning decisions, all of which are central concerns in the company's DR practices (Figure 2).

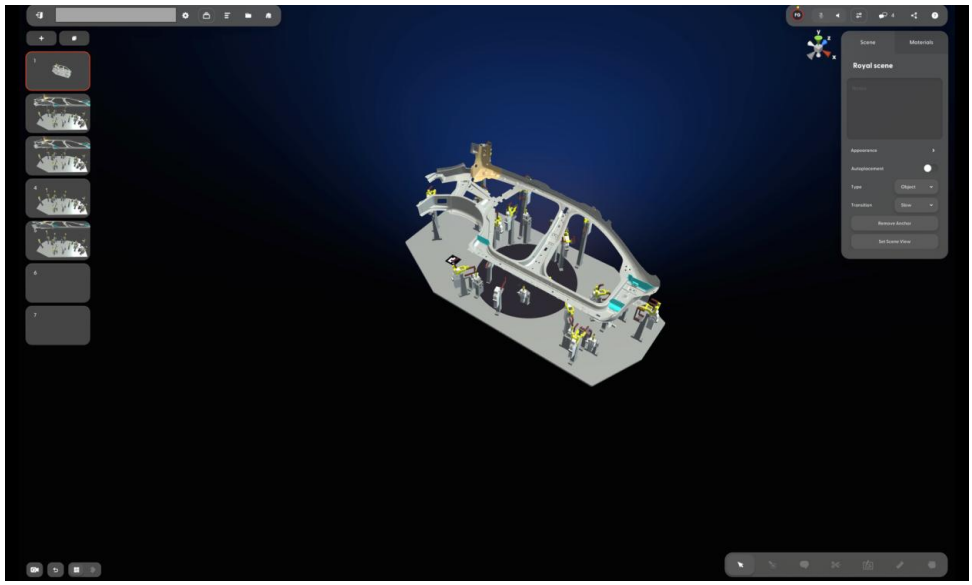


Figure 1. DR case 1 in Campfire

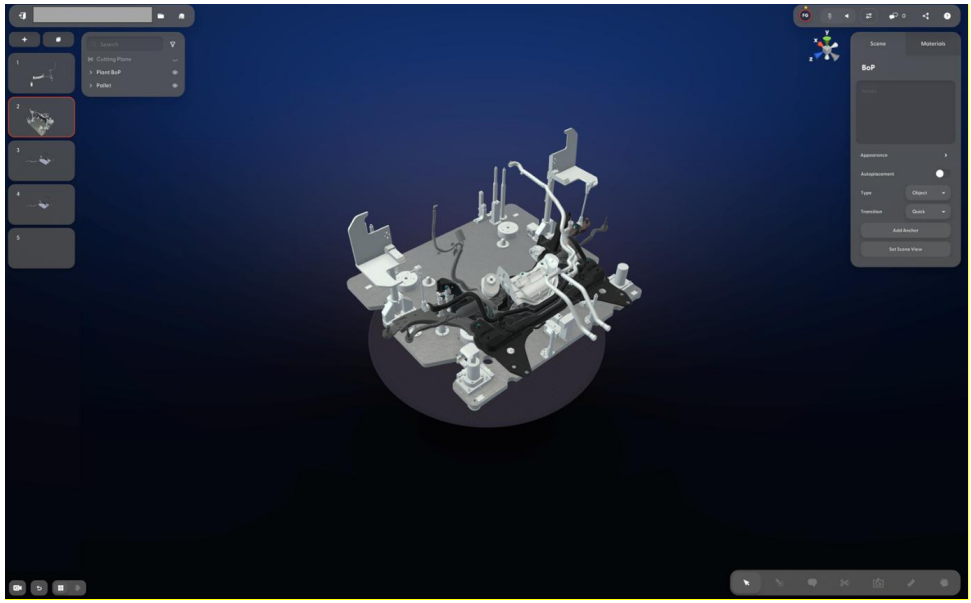


Figure 2. DR case 2 in Campfire

While both cases share a manufacturing-oriented DR focus and a similar hybrid baseline (conference room display with remote participants joining via videoconferencing), they differ in the specific product to be reviewed, and team composition. Table 2 summarizes these differences.

Table 2. Differences between the case at Company 1 and the case at Company 2.

Dimension	Company 1	Company 2
DR case	Welding fixture (body panels)	Assembly pallet (front subframe)
DR focus	Welding access, clamping, tolerances	Assembly sequence, tool clearance, ergonomics
Phase 2 workshop participants	6 (3 managers, 3 engineers)	4 (1 manager, 3 engineers)
Phase 3 workshop participants	5 (2 managers, 3 engineers)	6 (mix of 3 managers and 3 engineers)
Current DR practice	Hybrid: conference room display + Zoom for remote members	Hybrid: conference room display + Zoom for remote members
PLM system referenced	Siemens Teamcenter	Siemens Teamcenter

3.3. Recruitment

In both companies, a project manager and an engineer acted as the main internal contacts responsible for coordinating the study and preparing the DR cases. The project managers recruited participants from teams that regularly engage in DRs and were familiar with the selected components. Recruitment targeted participants who combined general DR experience with specific technical knowledge of the case components.

At Company 1, the project manager recruited six participants, including manufacturing engineers and a team leader responsible for process validation. At Company 2, the project manager recruited five participants, including manufacturing engineers, and a team leader overseeing the assembly zone. Participants had between five and twenty years of experience contributing to or leading DRs in their organizations.

The engineers acting as internal contacts supported the preparation of the CAD models and accompanying documents so that the cases would reflect the information normally reviewed in their DRs. They also ensured that the selected subassemblies were non-sensitive and could be openly discussed within the research setting. Participation was voluntary and took place during scheduled work hours with managerial approval. Confidentiality was secured through formal agreements covering the use of CAD data, workshop recordings, and the anonymization of all personal and organizational identifiers.

3.4. Software and Apparatus

The primary digital environment for the study was Campfire (Campfire, California, USA), a multi-user 3D collaboration platform. This specific software was selected because it provides the core affordances required for distributed spatial collaboration:

- **Shared Digital Environment**: The platform supports a real-time shared scene where multiple users can interact. It provides visibility into other participants' focus and actions (via avatars and gaze direction), establishing a common spatial frame of reference essential for joint activity.
- **Synchronized and Independent Navigation**: Users can dynamically toggle between "following" a presenter's view to maintain joint attention, or exploring the model geometry independently to verify details.
- **Deictic Tools**: The environment includes laser pointers and highlighting tools to support referencing during discussion.
- **Persistent Annotation**: Users can place anchored comments and text notes directly onto the 3D geometry.

To ensure ecological validity, the 3D scenes were reconstructed using each company's native CAD data. Assemblies were imported into Campfire, preserving the original product hierarchy, so that the digital artifacts matched the teams' existing vocabulary and operational context.

The study utilized a multi-device setup to simulate realistic deployment scenarios. The apparatus included:

- **Standard Workstations:** Two Asus XR-capable laptops (with a 3070ti graphics card, 32 GB of RAM, and Intel i9 processor) and one Macbook Pro laptop (M1 pro processor and 16 GB of RAM) were used for desktop participation, supporting keyboard/mouse interaction.
- **XR Devices:** Two Meta Quest 3 standalone headsets were used to enable immersive inspection and spatial manipulation.
- **Tablets:** An iPad Pro, 5th generation, with M1 processor was available to verify cross-platform accessibility.

The workshops were run in a large conference room with a large stand-up table placed in the middle. The devices were arranged as shown in Figure 3. While Campfire served as the active working environment, other tools such as Gravity Sketch (London, UK) and the DHM tool IPS IMMA (Industrial Path Solutions - Intelligenty Moving Manikins, Gothenburg, Sweden) were referenced during discussions to compare specific interaction capabilities. However, they were not used for the primary data collection tasks.

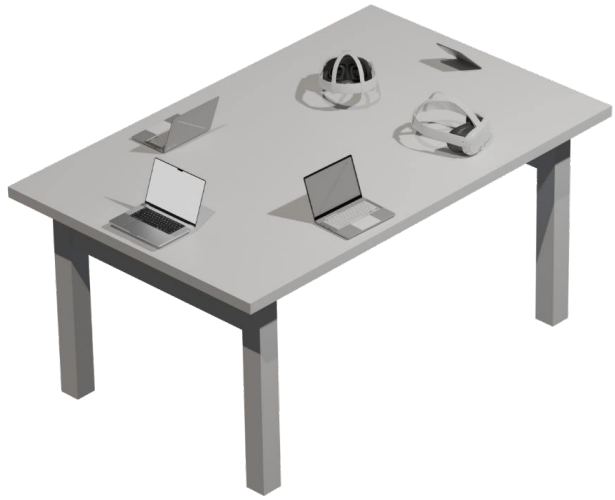


Figure 3. Illustration of the device set up at workshops

3.5. Study Procedure

The study followed an iterative research design process structured into three distinct phases. The process was designed to move from individual technical competence to collective procedural negotiation, ensuring that the final guidelines were based on group consensus and not on individual preference. As shown in Figure 4, the procedure progressed through three phases: Phase 1: Operational Fluency and Baseline; Phase 2: Procedural Mapping and Iteration; and Phase 3: Refinement and Validation. In the following sections, the objectives, participants, activities, and specific configurations of each phase are laid out.

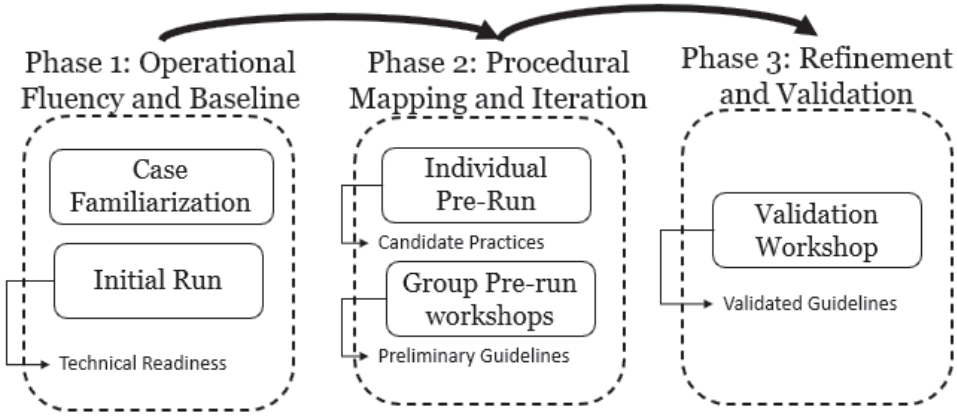


Figure 4. Study procedure and key activities per phase

Table 3 shows a compact overview of the study activities, participation, and data collection across all phases.

Table 3.Compact overview of the study activities

Phase	Activity	Format	Dur-ation	Company 1 Participants	Company 2 Participants	Devices Used	Data Captured	Analytic Purpose
1	Case Familiarization	Remote (Zoom)	~60 min	1 engineer	1 engineer	Desktop only	Screen recording, audio transcript	Validate scene reconstruction; establish shared vocabulary
1	Initial Run	Remote (Zoom)	~60 min	1 engineer	1 engineer	Desktop only	Screen recording, audio transcript	Build tool fluency; remove novelty effects
2, Step 1	Individual Pre-Run	Remote (Zoom)	~60 min	1 engineer	1 engineer	Desktop only	Screen recording, think-aloud transcript, in-scene artifacts (named views, annotations)	Generate candidate practices from individual expert reasoning
2, Step 2	Group Pre-Run Workshop	Co-located (conference room)	~90 min	6 (3 managers, 3 engineers)	4 (1 manager, 3 engineers)	2 XR-ready Windows laptops + 2 Meta Quest 3 + iPad+ Macbook	Audio recording, researcher field notes, in-scene artifacts	Collectively negotiate first set of candidate practices
3	Validation Workshop	Co-located (conference room)	~90 min	5 (2 managers, 3 engineers)	6 (3 managers and 3 engineers)	2 XR-ready Windows laptops + 2 Meta Quest 3 + iPad+ Macbook	Audio recording, researcher field notes, annotated physical guideline cards	Converge on final guideline wording via group consensus

Phase 1: Operational Fluency and Baseline

The objective of Phase 1 was to build technical readiness and remove novelty effects related to the software interface before introducing complex procedural tasks (Koch et al., 2018).

- **Participants:** The researcher and the contact engineer at each company.
- **Activity:** This phase consisted of two remote sessions:
    1. **Case Familiarization:** The researcher and the engineer jointly inspected the reconstructed 3D scene to ensure that hierarchy, naming, visibility layers, zones, and part organization matched the company's DR case. Minor mismatches were corrected on the spot. This session also surfaced initial DR routines (e.g., typical agenda, preparation habits), which later informed the sequencing of Phases 2 and 3.
    2. **Initial Run:** The researcher demonstrated the core interactions (shared/independent navigation, pointing, highlighting, anchored comments). Crucially, the researcher avoided describing how these features should be used for DRs, maintaining procedural neutrality and minimizing bias.
- **Specific Configuration:** Sessions were conducted remotely via video conference (Zoom) using desktop mode only. This constrained setup allowed the engineer to build confidence with the software interface.

Phase 2: Procedural Mapping and Iteration

The objective of Phase 2 was to generate the initial set of "candidate practices" by mapping the tool's capabilities to specific day-to-day DR tasks. Within this phase there were two steps:

Step 1: Individual Pre-Run (Remote)

- **Participants:** The researcher and the contact engineer at each company.
- **Activity:** The engineer walked through a hypothetical but realistic DR agenda using a think-aloud protocol, verbalizing how they would normally prepare, structure, and coordinate each step of the DR. While reasoning aloud, the engineer actively manipulated the scene by creating named viewpoints for expected discussion points, placing anchored comments where remarks

would commonly be documented, highlighting components relevant to particular agenda items, and adjusting the scene layout to match how they would typically prepare a DR in their own environment. This direct manipulation produced a set of concrete artifacts that mirrored the company's natural preparatory behavior. These artifacts were saved in the shared scene and later functioned as the initial baseline for the group pre-run workshops, allowing subsequent participants to critique, refine, and extend a workflow that originated from authentic company practice rather than abstract instructions.

- **Specific Configuration:** Remotely on desktop computers, same as both activities in Phase 1.

Step 2: Group Pre-Run workshops (In-Person)

- **Participants:** Company 1 involved six participants (three managers and three engineers), while Company 2 involved four participants (one manager and three engineers).
- **Activity:** This activity consisted of two on-site workshops (one at each company), lasting ninety minutes each:
    - **Protocol:** The sessions were conducted as moderated discussions in which participants systematically explored the tool's capabilities to identify necessary coordination routines. The same Campfire scene, already populated with named views and annotations from the preparatory run in Step 1, was used as a reference point to examine and debate the preliminary practices. Mediated by the researcher, the dialogue tracked the DR's workflow, examining the specific coordination required before the DR to ensure preparedness, during the session to manage attention and decisions, and after the DR to guarantee a valid handoff. Rather than performing a DR, participants explored specific affordances in response to targeted prompts, moving between desktop and XR to articulate how each capability would support or disrupt their established DR routines. As they tested actions such as following, bringing everyone, pointing, or anchoring comments, they described the coordination problems these actions might solve or create, including viewpoint drift, loss of shared focus, visual clutter, and weak traceability. Their comments frequently drew on concrete examples from their own processes, such as the need for one-click PLM launch to ensure latest-model trust or the importance of keeping comments tied to geometry to avoid losing context. Throughout the session, the researcher guided the conversation, ensured balanced participation, and asked clarifying questions about ownership, timing, and evidence to surface implicit assumptions about who performs each action and under what conditions. The interactions and on-model annotations produced during these discussions formed the first collectively negotiated set of candidate practices.
- **Specific Configuration:** Sessions were co-located in a conference room with a shared display. The hardware setup included two XR-ready laptops and two standalone Meta Quest 3 headsets. Device access was voluntary and rotational; participants were encouraged to switch between desktop and XR throughout the session to evaluate the workflow from different modalities.

Phase 3: Refinement and Validation Workshops

The objective of Phase 3 was to converge the draft practices into a stable, agreed-upon set of guidelines.

- **Participants:** Company 1 involved five participants (two managers and three engineers), while Company 2 involved six participants (a mix of managers and engineers).
- **Activity**: A second series of group workshops was held to refine the identified guidelines. Each session lasted approximately ninety minutes.
    - **Protocol:** The final phase consisted of in-person workshops where participants systematically refined the draft guideline set by working through each preliminary rule printed on physical cards and arranged across the five themes of pre-read, attention management, documentation and traceability, integration and handoff, and modality. The researcher guided the group through the cards one by one, prompting participants to read the wording carefully, question its completeness, and specify who performs the action, when it occurs, and what evidence or handoff it must produce in day-to-day DR work. Participants annotated the cards with sticky notes to add constraints, exceptions, or revised formulations, drawing directly on their operational experience; typical

additions included conditions tied to PLM integration, limits on concurrent pointers, expected ownership of named views, or clarification of when follow mode should be enforced versus optional. Throughout the discussions, participants frequently moved between desktop and XR to test whether a rule was technically feasible or whether an alternative wording better matched actual in-scene behavior, using quick demonstrations to check issues such as viewpoint re-centering, remark placement, or the practicality of switching from desktop to XR mid-discussion. The researcher maintained the flow of conversation, ensured balanced participation, and used targeted prompts to surface latent assumptions about roles and timing. The annotated cards, together with verified in-tool actions and participants' explanations, produced a consolidated and mutually agreed-upon guideline set that captured both the procedural intent and the operational boundaries identified across the two companies.

- **Specific Configuration:** The configuration was identical to the setup used in the Group Pre-Run (Phase 2, Step 2). Sessions were co-located in a conference room with a shared display. The hardware setup included two XR-ready laptops and two standalone Meta Quest 3 headsets. Device access was voluntary and rotational; participants were encouraged to switch between desktop and XR throughout the session to evaluate the workflow from different modalities.

3.6. Data Collection and Analysis

Data collection employed a multimodal approach to capture the full context interactions with the companies, ranging from interface mechanics to social negotiation. As summarized in Table 4, the dataset was aggregated across all three phases.

Table 4. Summary of data collection activities and sources

Phase and Activity	Observational Visual Data	Verbal Data Transcription	Generated Artifacts	Analytic Purpose
Phase 1: Operational Fluency and Baseline	✓	✓		To build technical readiness and remove novelty effects related to the software interface and tool novelty
Phase 2: Step 1: Individual Pre-Run	✓	✓	✓	To generate the initial set of "candidate practices" by mapping the tool's capabilities to specific day-to-day DR tasks
Phase 2: Step 2: Group Pre-Run Workshops		✓	✓	To identify necessary coordination routines and form the first collectively negotiated set of candidate practices
Phase 3: Refinement and Validation Workshops		✓	✓	To converge the draft practices into a stable, agreed-upon set of guidelines endorsed by the two companies

Observational and Visual Data

Visual data were collected using two different methods depending on the setting:

- **Screen Recordings (Remote Sessions):** During the Case Familiarization (Phase 1, Session 1), Initial Run (Phase 1, Session 2), and Individual Pre-Run (Phase 2, Step 2), the researcher captured the entire video conference feed (via Microsoft Teams) as shown in Table 4. This provided a record of the user's screen, cursor movements, and interface interactions, allowing for the precise identification of usability "friction points".
- **Observational Logs (Co-located Workshops):** During the in-person Group Pre-Run (Phase 2, Step 2) and Validation Workshops (Phase 3), room video recording was restricted for confidentiality, therefore, the sessions were captured primarily through audio recordings (Table 4). Additionally, as the researcher was actively moderating the sessions, continuous structured field notes would have disrupted facilitation. Instead, the researcher captured short, unstructured notes at specific conversational moments that signaled potential procedural

significance. These notes were taken whenever participants explicitly highlighted a coordination breakdown, proposed a rule they considered essential, questioned the feasibility of a practice, or demonstrated contrasting interpretations of roles, timing, or evidence. In practice, this meant writing brief markers such as “pointer clutter issue,” “decision read-back important,” or “VR vs desktop debate” immediately after they emerged in discussion. These notes did not aim to document full interaction sequences but served as anchors for later review during the systematic post-hoc review of the audio recordings.

Verbal Data and Transcription

Verbal exchanges were the primary source for understanding how participants negotiated roles and procedures.

- **Audio Capture:** For remote sessions, system audio was recorded directly. For co-located workshops, audio was recorded using mobile devices placed centrally on the conference table to capture multi-party dialogue clearly.
- **Think-Aloud Protocols:** During the Individual Pre-Run (Phase 2, Step 2), the contact engineer was asked to verbalize their thought process while manipulating the model (e.g., "I would add a note here that this hole is in the wrong place").
- **Transcription:** All audio files were processed using a local implementation of the Whisper automatic speech recognition system (Radford et al., 2022). To ensure validity, the raw transcripts were manually verified and corrected by the researcher, specifically to fix domain-specific engineering terminology (e.g., "subframe," "subassembly").

Generated Artifacts

Beyond recordings, the study collected the artifacts generated in the workshops as direct evidence of consensus.

- **Digital Artifacts (In-Scene):** The state of the scene in Campfire was saved after every session. This dataset includes the specific Named Views created to structure the agenda, 3D Annotations anchored to geometry, and Comments left by participants. These artifacts serve as evidence of how procedural rules (like "Set views before meeting") were technically implemented.
- **Physical Artifacts (Validation Cards):** During the final validation phase, data were collected via physical guideline cards. Participants used sticky notes to write wording edits, boundary conditions, or role clarifications directly onto the cards. These notes were transcribed, and linked to their corresponding guideline to verify that the final text reflected the group's specific language and consensus.

Data Analysis Strategy

The analysis aimed to synthesize the workshop discussions into a consistent, company-neutral framework. To achieve this, the strategy to analyze the data aimed to remove subjective participant preferences and focus on recurring procedural routines that addressed specific coordination issues. The process combined detailed linking of actions and talk with two coding cycles and matrix-based verification.

Crucially, this analysis was not a single retrospective event but a progressive filtering process. As summarized in Table 5, each phase of the study served a distinct analytic function, where findings from one stage established the constraints, vocabulary, and entry criteria for the next. This cumulative logic ensured that only practices that survived multiple rounds of testing and refinement advanced to the final validation.

Table 5. Analytic progression and filtering logic across study phases

Study Activity	Analytic Contribution	Influence on Next Stage
Case Familiarization (Phase 1, Session 1)	Established the validity of the scene reconstruction and common vocabulary (naming conventions, zones).	Constraint: Ensured researchers used the companies' native terms during coding and was familiar with the DR case.
Initial Run (Phase 1, Session 2)	Verified the technical feasibility of the tool features.	Filter: Separated software issues from actual "process issues."
Pre-Run (Individual) (Phase 2, Step 2)	Generated candidate practices based on individual expert reasoning.	Input: These individual proposals became the discussion topics for the group sessions.
Group Pre-Run (Phase 2, Step 2)	Provided multi-party critique and refinement of the candidates.	Refinement: Identified which individual proposals failed in a group setting (e.g., "Meeting in space is too chaotic").
Validation Workshop (Phase 3)	Final "Member Check" of wording and boundary conditions.	Lock: Confirmed the final text of the guidelines based on explicit group agreement.

Unit of Analysis: Action-Proposal Pairs

To explicitly link software usage to procedural intent, the primary unit of analysis was defined as the Action-Proposal Pair. An analytic episode was defined as starting when a visible action occurred in the scene (e.g., toggling “Follow mode,” saving a named view, placing an anchor) and ending when the corresponding procedural proposal was verbally expressed (e.g., “We should always do this when we start,” “Only the organizer should save views”).

In practice, these episodes were identified by reviewing the video recordings and transcripts at the same time. When an interactional event was observed in Campfire, the corresponding timeline segment in the transcript was located and extracted as a single Action-Proposal Pair. For example, in one workshop a participant activated follow mode to re-center the group and then stated: “At the beginning of the meeting, we should force everybody to follow the organizer.” This became one episode linking the capability (follow mode) to a candidate rule (enforced synchronization at meeting start). This segmentation was necessary because analyzing speech alone would miss the technical context, while analyzing software logs alone would miss the user’s intent. Pairing them ensured that every resulting guideline was grounded in a specific, observable tool interaction.

Coding Strategy

Data processing followed an iterative thematic approach using two coding cycles to move from raw episodes to abstract categories (as exemplified in Table 6). Coding was conducted by the first author, who has a background in mechanical engineering and product development.

- **First-Cycle Coding (In vivo and Descriptive):** In the first cycle, Action-Proposal Pairs were coded using participants’ own terminology (e.g., “bring everyone,” “point at it”) alongside short descriptive labels (e.g., “force follow at start,” “named views per agenda item”). This strategy prevented the researcher from making external theoretical labels too early and ensured the final guidelines reflected the vocabulary and distinctions used by engineers.
- **Second-Cycle Coding (Pattern Mapping):** In the second cycle, first-cycle codes were clustered to identify stable relationships between specific problems and proposed solutions. For instance, several in vivo codes (“not move too much if you know that people are following you,” “the product owner should stay stable when leading the session”) were grouped into a higher-level pattern such as separating exploration from decision capture. This step was required to filter out isolated comments and identify structural patterns that appeared across different users, roles, and sessions.

Table 6. Examples of coding progressions from raw episodes to thematic patters

Raw Episode (Excerpt)	First-Cycle Coding (Action-Proposal Pair)	Second-Cycle Coding (Thematic Pattern)
"Send this scene before the meeting so we can comment on the right model."	**Action:** Share scene before meeting **Problem:** Late model access	**Pre-read & Preparation**
"Everyone press follow so we start from the same place."	**Action:** Start in follow **Problem:** Viewpoint drift	**Attention Management**
"Keep the note with the picture so we know what changed."	**Action:** Capture remarks in-scene **Problem:** Weak traceability	**Documentation & Traceability**
"If it opens straight from Teamcenter, you know it's the latest."	**Action:** Launch from PLM **Problem:** Version uncertainty	**Integration & Handoff**
"You don't want thirty avatars blocking the model; better fade a few."	**Action:** Hide or fade avatars **Problem:** Visual clutter	**Modality**

Verification via Analytic Matrices

To verify the stability and relevance of the emerging patterns, two analytic displays were constructed.

- **Practice × Problem Matrix:** This matrix (see Figure 5) mapped every proposed guideline (practice) to the specific coordination breakdown it was claimed to solve (problem). Rows represent the candidate practices identified during the Pre-Run and Group Sessions (e.g., "starting in follow," "limiting concurrent pointers," "launching from PLM"), while columns capture the recurring coordination problems defined in the friction analysis (e.g., "viewpoint drift," "referential ambiguity," "visual clutter," "weak traceability"). The matrix was derived directly from the coded transcripts, where an analytic episode was generated whenever a participant explicitly linked a tool action to a specific problem. Consequently, the cell values indicate the recurrence of these specific practice-problem pairs across different sessions and

companies. Stable links were identified when a pair appeared repeatedly or was independently confirmed at both sites, distinguishing robust procedural reasoning from isolated suggestions.

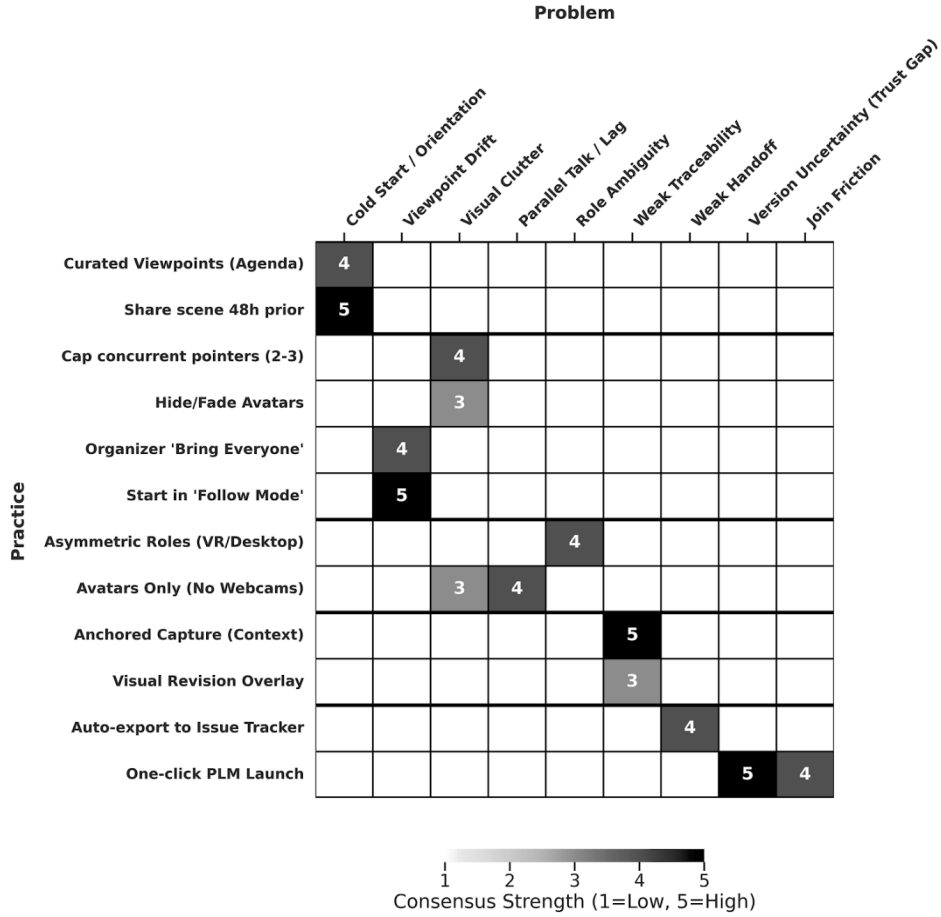


Figure 5. Extract from the Practice x Problem Matrix

- **Role × Move Matrix:** A second matrix (see Figure 6) summarized how coordination responsibilities were distributed among roles during group discussions. Rows represented the distinct functional roles identified in the workshops and in previous interactions:
    - **Organizer:** The facilitator responsible for meeting logistics, scene curation (pre-read), and managing flow (e.g., "bringing everyone").
    - **Presenter (Product Owner):** The technical lead responsible for explaining the design and driving the visual narrative (e.g., controlling "follow mode").
    - **Scribe (Integrator):** The role responsible for capturing evidence, recording rationale, and executing the handoff to the product data system.
    - **Reviewer (Domain Specialist):** The expert contributor responsible for inspecting specific geometry and providing feedback.

  Crucially, while the *Scribe* is analytically distinct as a documentation function, this responsibility often falls to the *Organizer* in practice, meaning one individual may perform these two roles simultaneously. Columns captured common coordination moves, such as starting in follow, bringing everyone in, granting permissions, creating named views, reading back decisions, and exporting action items. Role assignments were inferred from how

participants described responsibility in speech. Statements such as “the organizer should bring everyone back before we decide” created a link between the *Organizer* and *bring everyone*. At the same time, “the reviewer can ask for a view if they get lost” linked the *Reviewer* and *request named view*. Links that appeared repeatedly across different exchanges or were elaborated with reasoning or conditions were treated as stable. For example, the *Organizer-bring everyone* and *Presenter-read back decision* occurred in both companies. The display, therefore, provided a structured view of how participants themselves defined and negotiated the division of labor during DRs, forming the basis for the role definitions used in the guideline cards.

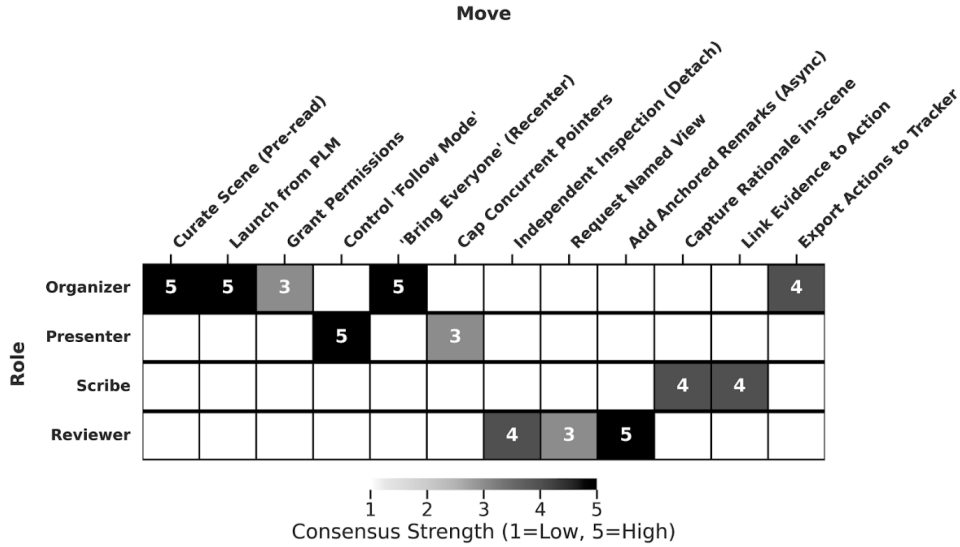


Figure 6. Extract from the Role x Move Matrix

Decision Rules for Inclusion in Guidelines

Consensus was reached through a structured convergence process. In the Phase 3 workshops (refinement and validation workshops), each candidate practice was printed on a physical card and presented to the group. Participants discussed the card’s wording, scope, and boundary conditions, annotating it with sticky notes. Consensus was considered reached when all participants present at the workshop agreed that the card’s final wording accurately captured the intended rule. Where disagreement arose (e.g., on the number of concurrent pointers or the timing of follow-mode enforcement), the researcher facilitated discussion until a shared formulation was reached or the point of contention was recorded as a contextual note rather than a core rule. This card-by-card procedure ensured that the final guidelines reflected explicit group agreement.

To ensure the final guidelines were robust enough for industrial application, a candidate practice was included in the framework only if it met three criteria:

1. **Cross-Case Confirmation:** The rule had to be confirmed in both companies independently, either through direct observation or explicit endorsement in workshop discussions. This approach was followed to reduce the risk that a guideline reflected a single company’s culture, tooling setup, or product type.
2. **Problem Relevance:** The rule addressed at least one documented friction point in the Practice × Problem Matrix. Practices that did not map to a recurring problem, or that only offered marginal convenience, were excluded or relegated to contextual notes. This criterion acted as a utility filter, preventing scope expansion by ensuring that every guideline targeted a

demonstrable improvement in DR reliability or efficiency rather than simply highlighting software features.

3. **Observability:** Compliance with the rule had to be visually or procedurally verifiable in the scene or associated artifacts (e.g., presence of named views, visible anchors, an exported issue list). This approach allowed the guidelines to be more auditable. A rule like "communicate better" is vague, whereas a rule like "leave an anchor with an owner and due date before closing a topic" can be checked directly.

Only practices satisfying all three criteria were included as core guidelines in the final framework. Practices that failed one or more conditions were documented as contextual recommendations or design implications for tool developers rather than procedural rules.

4. Results

The analysis highlighted a set of guidelines grouped in five thematic areas that structure the DR process from preparation to handoff. These themes emerged from the synthesis of Action-Proposal Pairs and were validated through cross-case consensus at both automotive companies.

The guidelines are summarized in Table 7. These rules are not software-specific features but procedural routines that rely on the core affordances of advanced collaboration tools (shared presence, deictic referencing, and persistent capture). The technological requirements presented in Table 7 should be interpreted as operationalized expressions of the core and extended affordances defined earlier in Table 1. While Table 1Table 1 establishes the theoretical capability classes that characterize advanced collaboration tools (e.g., Shared Spatial Environment, Independent and Shared Navigation), the enablers listed in Table 7 represent the specific functional manifestations needed to support each validated procedural guideline. For example, Table 1 defines Independent and Shared Navigation as a general platform capability, whereas the guideline on Attention Management requires a specific control mechanism, the Focus Toggle, to enforce the "bring everyone" routine operationally (Table 7). Thus, the terminology differs intentionally: Table 1 defines the tool category, whereas Table 7 specifies what the tool must do to support the workflow identified in this study. In the following sections, each thematic area is detailed by describing the context and problem definition, the validated procedural guideline, and the specific technological enabler required to support that practice.

Table 7. Summary of the DR guidelines

Theme	Validated Procedural Guideline	Required Technological Enabler
1. Pre-Read and Preparation	**Asynchronous Entry Routine:** • **Preparation window:** Organizer shares scene 24-48 hours prior to synchronous session. • **Curation Entry:** Presenter creates a "spatial agenda" using named views. • **Reviewer responsibility:** Reviewers leave anchored remarks before the meeting starts.	**Persistent Scene States and Named Views (Core Affordance):** System must support "spatial bookmarking" to guide non-experts to specific agenda items without getting lost.
2. Attention Management	**Guided Alternation:** • **Establish Context (Guided):** Presenter explicitly alternates between Guided (Follow mode) for context and Free for inspection. • **Visual Economy:** Limit active pointers to the speaker/decision-maker (max 2–3) to prevent visual chaos.	**Independent and Shared Controls (Core Affordance):** • **Focus Toggle:** Mechanism to "bring everyone to me" and release them. • **Clutter Control:** Ability to hide/fade avatars in crowded zones.
3. Documentation & Traceability	**In-Context Annotation:** • **The "Why" Requirement:** Scribes must record the rationale behind decisions, not just the action item. • **Visual Context:** Remarks must be visually anchored to geometry (Ideal) or embedded as screenshots (Fallback) to prevent context loss.	**Contextual Anchoring (Core Affordance):** The system must allow text/voice notes to attach to specific geometric coordinates. (Extended: Visual Versioning/Diffs).
4. Integration and Handoff	**PLM-First Discipline:** • **Version Verification:** Launch session directly from the system of record (PLM) to guarantee version trust. • **Closed-Loop Handoff:** Post-meeting transcription moves actions from the tool into the official tracker (closed loop).	**Frictionless Interoperability (Extended Affordance):** • **Direct Launch:** One-click initialization from PLM. • **Direct Sync:** Automated connection to issue trackers (e.g., Jira/SharePoint).
5. Modality and Role Distribution	**Asymmetric Participation:** • **Role-Based Hardware:** Key examiners use XR for spatial/ergonomic checks; Scribes/Managers use Desktop for documentation. • **Protocol:** Prioritize avatars over video feeds to reduce lag and distraction.	**Cross-Platform Synchronization (Core Affordance):** • **Parity:** Desktop users must see XR pointers/gestures without latency. • **Avatar Representation:** System must support non-intrusive presence indicators.

4.1. Theme 1: Pre-Read and Preparation

Context and Problem Definition

Participants from both companies identified a recurring inefficiency in traditional DRs, described as a "cold start". Company 1 noted that valuable meeting time is often wasted on basic orientation rather than substantive decision-making, stating, "We are not using the time on a meeting to... [look at a] blank screen". The baseline process often leaves stakeholders seeing the model for the first time during the session, leading to delays as they orient themselves. Ideally, participants should have "seen the fixture before", so the live meeting focuses strictly on "walking through the remarks".

Procedural Guideline

To mitigate orientation latency, we propose asynchronous entry. The study validates a strict Pre-Read Routine that shifts discovery to the asynchronous phase.

- **Preparation Window:** The meeting organizer is required to share the review scene 24-48 hours prior to the synchronous session.
- **Curated Entry (Ideal Path):** The organizer must not simply "dump" the model; they must curate the scene with premade comments, preliminary agenda, points of interest, and discussion points. Company 2 emphasized that "each user can make a section and explain," effectively creating a spatial agenda before the meeting starts.
- **Reviewer Responsibility:** Reviewers are expected to enter the environment asynchronously to leave preliminary "anchored remarks," allowing the live session to focus on resolving conflict rather than identifying obvious issues.

Technological Enabler

For this procedure to be viable, the software must support Persistent Scene States and Named Views.

- **Persistent Scene States:** Company 2 participants argued that simply sharing a raw 3D model is insufficient for non-experts. The tool should allow the organizer to create "camera viewpoints" that serve as "key positions" corresponding to agenda items.

- **Name Views:** These pre-defined views act as entry points, ensuring that asynchronous reviewers can navigate to the relevant geometry without getting lost in the 3D space.

4.2. Theme 2: Attention Management

Context and Problem Definition

In multi-user environments, "viewpoint drift" emerged as a critical coordination breakdown. Company 2 participants described the concept of "dancing," where users attempting to follow one another end up in a disorientation loop. Furthermore, visual clutter was identified as a significant distraction in larger groups. Company 2 noted that with larger teams, "you don't want fifteen [avatars] everywhere", blocking the view. Additionally, unrestricted use of deictic tools can lead to chaos; participants noted that if everyone points simultaneously, "it will be Star Wars," making it impossible to identify "who is watching" or focusing on the relevant detail.

Procedural Guideline

The validated protocol requires a deliberate alternation between "Guided" and "Free" modes to balance group cohesion with individual verification.

- **Establish Context (Guided):** The session leader must initiate the review of a new part by gathering the group. Company 1 noted that without a mechanism to "bring them to you somehow," the discussion becomes fragmented as users look at different areas.
- **Release for Inspection (Free):** Crucially, the presenter must explicitly release control. Company 1 emphasized that reviewers require the "freedom to... step out of the following mode" to inspect details relevant to their specific domain, but if it happens during a presentation, it should only be occasionally for “something really specific”.
- **Visual Economy:** To manage clutter, the group agreed on a protocol of "visual economy," limiting active pointers to the current speaker and key decision-makers. Company 2 suggested capping concurrent pointers (e.g., "three pointers or two pointers at once") to ensure the shared view remains readable.

Technological Enabler

To support this alternation without friction, the software must provide specific Independent and Shared Inspection Controls (Core Affordance).

- **Focus Synchronization Toggle:** The system must support a mechanism to "bring everyone to me". This ensures balanced participation by allowing the presenter to temporarily synchronize the group around a specific view or detail.
- **Free Viewpoint Navigation:** Equally important is the support for autonomous exploration. As noted by Company 1, the system must allow users to "step out of the following mode" seamlessly to verify details, without breaking the session for others.
- **Clutter Management:** To handle the visual noise inherent in shared inspections, the tool should offer specific visibility toggles, such as "hiding avatars". Participants noted that this capability is essential when many users inspect a small component, preventing social visualizations from occluding the engineering geometry.

4.3. Theme 3: Documentation and Traceability

Context and Problem Definition

A significant challenge observed in the current DRs involves the information discontinuity between the visual DRs and the formal documentation. Company 1 noted that feedback is often distributed across general-purpose tools such as OneNote or Excel, which can unintentionally separate comments from their spatial context. This separation requires engineers to correlate text-based feedback with specific geometric coordinates manually. Company 2 highlighted the long-term implications of this workflow, noting that while action items are captured, the underlying rationale is sometimes lost in brief notes, making it difficult to reconstruct the reasoning behind a decision "six months ago".

Procedural Guideline

To address this discontinuity, the study validates a strict guideline regarding rationale and visual evidence, regardless of the specific tool capabilities.

- **The "Why" Requirement:** The Scribe or Reviewer must explicitly record the rationale behind a decision, not just the action item. Company 2 emphasized that without capturing the reasoning at the moment of decision, the outcome cannot be "trusted" or traced effectively in later stages.
- **Visual Context (Ideal Path):** Remarks should be anchored directly to the geometry within the scene. Company 1 participants expressed a strong preference for workflows that allow them to "jot [remarks] down within the comments" to "keep everything together in the same place".
- **Visual Context (Fallback Path):** If the tool lacks native anchoring (e.g., it functions only as a viewer), the procedure mandates that the Scribe manually capture a screenshot of the specific view and embed it into the record immediately. Company 2 participants argued that "pictures, screenshots" are essential because text descriptions alone are often insufficient to describe complex geometric issues.

Technological Enabler

While the procedure can be executed manually, the workshops highlighted that Contextual Anchoring (Core Affordance) is a highly desirable capability that significantly reduces process friction.

- **Contextual Anchoring:** Participants favored tools that allow remarks to be "anchored" directly to the geometry, effectively creating a 3D version of the meeting minutes.

4.4. Theme 4: Integration and Handoff

Context and Problem Definition

While Theme 3 focuses on *capturing* information, Theme 4 addresses the critical friction of *moving* that captured information, specifically the uncertainty regarding data currency. Company 1 participants expressed significant concern about whether the CAD product model being reviewed was actually the correct version, stating, "You need to know that you are looking at the latest one... otherwise you can look at something [that] is a few minutes old, but it could be very different." The baseline practice of manual file preparation (exporting, saving to disk, re-uploading) was rejected as a barrier that is "super hard to scale." Company 2 described the current fragmented workflow as "stitching together a bunch of things," where information hops from Excel to Teams to PLM, creating a fragile chain of custody. This can be especially hard to navigate for individuals who are involved in several different teams, each with its own system in place to document the process in various tools.

Procedural Guideline

To ensure process integrity, the study suggests a strict discipline that establishes the organization's product data system (e.g., PLM or issue tracker) as the single source of truth.

- **Version Verification (Ideal Path):** Ideally, the session is launched via a "one-click" integration from the PLM to guarantee the version.
- **Version Verification (Fallback Path):** If the tool lacks integration (standalone app), the Organizer must explicitly validate the file version at the start of the meeting. Company 1 noted that users "shouldn't care about where the files are", implying that if manual loading is required, the Organizer must be the one ensuring the file matches the "latest one that you have in the [PLM]" before the DR session begins.
- **Closed-Loop Handoff:** The process is only complete when decisions are logged in the official repository. If the tool does not sync automatically, the procedure requires a post-meeting transcription step in which the Scribe moves action items from the tool/notes into the official tracker (e.g., SharePoint/Jira) and links it to the PLM system used to ensure they are not lost in the "silo" of the review tool.

Technological Enabler

To enable this collaboration, the tool must rely on Session Capture and Traceability (Extended Affordance) to prevent the DR from becoming an isolated data silo disconnected from the official product data system.

- **Direct Launch:** Company 1 noted that a critical "wish" for future implementation is direct launching from the PLM system to eliminate the "trouble that you do not have the latest one".
- **Direct Sync:** Company 2 highlighted the desire for the tool to connect directly to platforms like SharePoint or Jira, allowing action items to be populated automatically rather than requiring manual transcription. This extended affordance would transform the DR tool from a data silo

into a verified lens for the product data system, similar to the way "GitHub" manages code versions in software development.

4.5. Theme 5: Modality and Role Distribution

Context and Problem Definition

The workshops revealed that enforcing a single hardware standard (e.g., "VR for everyone") is impractical and often detrimental to the DR meeting's flow. Company 2 noted that attempting to replicate a face-to-face meeting by enabling webcams ("faces") inside the 3D view causes technical lag and acts as a visual distraction. Company 1 observed that while VR offers "richness" for inspection, requiring it for all attendees creates a barrier to entry; if a user is struggling with the hardware setup, they feel like they are "standing outside," effectively missing the meeting.

Procedural Guideline

To maximize attendance and comfort, the study validates a model of Asymmetric Participation, where hardware selection is determined by the user's role rather than a blanket mandate.

- **Key Examiners (XR):** "Key users" or those performing spatial manipulations are recommended to utilize XR. Company 1 noted that these users benefit from the "deeper reality" required to verify ergonomics or complex assemblies.
- **Observers and Scribes (Desktop):** Management or peripheral stakeholders are recommended to join via desktop or tablets to maintain a synchronized view with the presenter, allowing them to focus on decision-making and documentation without the cognitive load of navigating the 3D space. Company 1 concluded that the process must be flexible: if users "don't want to use [XR]... they keep their role, but they just shoot it from [desktop]".

Technological Enabler

To make asymmetry viable, the tool must incorporate Cross-Platform Synchronization (Extended Affordance) to prevent different hardware devices leading to a fragmented understanding of the design.

- **Shared Spatial Environment:** The system must ensure that the "common spatial frame of reference" is maintained across hardware devices (XR and desktop) without latency (Core Affordance). As noted by Company 1, even if a user is on a desktop, they must clearly see the actions of XR users, specifically the In-scene Deictic Referencing (pointing, measurements, and drawing notes), to maintain trust in the collaboration.
- **Integrated Communication Utilities:** To support the "visual economy" guideline (Theme 2), the platform should prioritize avatar-based cues over embedded video feeds (Extended Affordance). Company 2 explicitly preferred avatars to maintain performance and focus, noting that video feeds inside the 3D space are "disturbing" and introduce unnecessary lag. This aligns with the definition of consolidating communication within the workspace to reduce distraction.

5. Discussion

This study develops and validates a set of grounded guidelines for conducting DRs with advanced collaboration tools, summarized in Table 7. While individual components of these guidelines, such as preparing the scene or managing turn-taking, may appear to be standard practice, this research moves them from general assumptions to evidence-informed coordination routines. Earlier research effectively demonstrated the technical feasibility of shared 3D environments and XR-based visualization but rarely explained how such affordances should be integrated into repeatable industrial workflows. We extend this line of inquiry through cross-case analysis by emphasizing that effective collaboration depends less on tool novelty or immersion than on specific procedural conditions that align the system’s affordances with the team’s workflow. By explicitly linking interaction-level evidence, such as shared viewpoints and anchored annotations, to organizational routines, such as auditable documentation, we propose a transferable framework that transforms these tools from isolated visualizers into reliable and managed elements of the product development process.

The findings on Pre-Read and Preparation (Theme 1) show that participants value examining the model and record anchored remarks beforehand. By doing so, the DR shifts from an exploratory session to a

structured conversation that builds on already externalized insights. This behavior extends the notion of grounding (Clark & Brennan, 1991). It aligns with recent work on asynchronous collaboration, which demonstrates that preparatory artifacts facilitate communication among distributed contributors and streamline subsequent synchronous interactions (Wang et al., 2024). By enabling participants to externalize reasoning before the meeting, pre-read activities promote the early development of shared mental models (Cannon-Bowers et al., 1993; Casakin & Badke-Schaub, 2014, 2017; Van Den Berg et al., 2017), giving teams a standard perceptual and conceptual frame that reduces clarification and negotiation during the session. Anchored annotations thus serve as boundary objects that sustain mutual understanding over time, turning preparation into an active phase of collaboration. For practitioners, these findings highlight that DRs efficiency depends less on meeting duration than on the quality of pre-meeting alignment, while for advanced collaboration tool developers, they stress the importance of features that support verified model access, anchored commenting, and persistent annotations that carry pre-read insights into the main review record (Company et al., 2023; Wang et al., 2024).

Participants framed Attention Management (Theme 2) as a deliberately organized routine for shared work, rather than something that happens naturally. Across sessions, engineers reasoned that presentation-type moments should begin in follow mode to establish a common view, with deliberate, short detachments for individual inspection and quick re-alignment via named viewpoints or a "bring everyone" control before decisions. This logic echoes classic observations about how shared visual spaces, gestures, and viewpoints facilitate coordination, and why explicit cues are necessary to maintain joint focus (Gutwin & Greenberg, 2002; Tang, 1991). At the same time, participants emphasized the flexible use of this approach, following it to synchronize key transitions while preserving autonomy between them, which resonates with design-team communication work on structuring turns and maintaining mutual orientation, now translated to virtual contexts (Stempfle & Badke-Schaub, 2002). Recent reviews and industry-oriented studies of distributed/co-located VR collaboration similarly highlight the need for simple, built-in mechanisms to realign perspectives and maintain common ground when multiple people navigate complex geometry (Lapointe et al., 2021). Together, these discussions position attention management as a practical coordination habit that turns follow toggles, named views, and re-centering into conversational resources for sustaining a shared frame of reference.

The findings on Documentation and Traceability (Theme 3) indicate that the effectiveness of a DR depends on maintaining direct connections between remarks, rationales, and decisions and the geometry under review. When documentation is detached from the model and dispersed across slides or notes, it loses the spatial and contextual cues that give meaning to design reasoning. This observation aligns with earlier work on design rationale capture, which cautions that post-hoc documentation often fails to reflect how decisions emerge during DRs accurately (Huet et al., 2007). By recording remarks and changes as anchored snapshots within the shared scene, participants transform documentation into a live artifact that evolves with the model. This behavior also aligns with the idea of knowledge continuity in product development, where insights are progressively linked across iterations to maintain a consistent digital thread (Liu et al., 2018). For practice, these results suggest that documentation should not be treated as a separate administrative task but as an integral part of the review conversation and as data for continuous improvement in the PDP. For software developers focusing on advanced collaboration tools, the implication is that collaboration platforms should enable users to create and compare snapshots, overlay revisions, and link action items with evidence in the same environment so that traceability occurs naturally as the DR happens.

The theme of Integration and Handoff (Theme 4) shows that a lot of the value of collaborative DRs depends on whether outcomes flow into the company's product data system with traceable continuity. In both companies, engineers emphasized the need for "one-click" launches from PLM and automatic integration of remarks and screenshots to their product data system, warning that manual exports and copy-paste workarounds erode confidence in model validity and break traceability. This emphasis aligns with the configuration-management principles defined in ISO 10007 (*ISO 10007*, 2019), which emphasizes the need to maintain the identification and traceability of design information throughout the lifecycle, and with broader quality-management requirements outlined in ISO 9001 (*ISO 9001*, 2021). It also echoes findings in configuration-management research that stress the importance of a single authoritative source to ensure coherence across digital tools and teams (Liu et al., 2018). For practice,

this implies that every DR should begin from the verified model and conclude with actions and evidence already visible in the product data system. For tool developers, it points to the importance of open integration paths and low-effort transitions so that advanced collaboration tools become natural extensions of existing PLM ecosystems rather than isolated visual spaces.

The findings on Modality and Role Distribution (Theme 5) show that mixed-device participation is not a transitional stage toward full immersion but a mature collaborative configuration in its own right. Participants described how XR, desktop, and tablet users enact complementary cognitive roles within the same review: XR participants bring embodied awareness and the ability to make spatial relations visible through gesture and scale, while desktop and tablet participants provide symbolic reasoning through annotation, documentation, and external data reference. This asymmetry reflects what Hutchins (1995) conceptualized as a *distributed cognitive system*, where collective understanding emerges from the coordinated use of different perceptual resources rather than identical experiences. Similar insights have been reported in studies of asymmetric collaboration in XR, where immersed and non-immersed users occupy complementary roles that increase efficiency and communicative clarity (Burova et al., 2022; Rivera et al., 2025; Zheng et al., 2025).

Participants' suggestions to fade avatars, cap concurrent pointers, or avoid webcams reflect an intuitive recognition that every visual signal competes for limited attentional bandwidth. This aligns with cognitive-affective models of immersive learning (Makransky & Petersen, 2021) and recent XR ergonomics research showing that over-embodied environments fragment focus and reduce coordination efficiency (Souchet et al., 2022). By negotiating which cues remain visible, teams effectively self-regulate collective attention, a procedural analog to the "attention economy" Gutwin & Greenberg (2002) described for real-time groupware (collaborative software). The pattern observed here suggests that the strength of advanced collaboration tools lies not in achieving parity of experience but in enabling functional asymmetry: allowing each participant to contribute through the modality that best fits their cognitive and task-specific strengths. For practitioners, this means that successfully using XR depends less on how immersive it feels and more on deciding who experiences what from which device, when, and for what purpose. For tool developers, it implies that progress will come not from maximal fidelity and features but from interfaces that maintain legibility across heterogeneous devices, preserving the cognitive continuity of the DRs.

Although the current study offers grounded insights, its scope remains intentionally narrow. The evidence comes from two large automotive companies and primarily from participants in manufacturing and production engineering roles. As a result, the patterns reported here reflect the workflows, toolchains, and coordination logics typical of that sector and may differ in early design or service-oriented domains. Furthermore, the evaluation in this study is based on workshop-based assessment and practitioner consensus, which supports claims about the guidelines' understandability, relevance, and perceived feasibility. However, this evaluation does not measure downstream impact on decision quality, time or rework reduction, traceability metrics, or follow-up completion rates. Demonstrating such effects would require longitudinal deployment within live DR workflows, which remains a priority for future research. The findings should therefore be interpreted as contextually valid rather than universally generalizable. Future research could expand validation across other industries, disciplines, phases of the PDP, and team configurations, or combine qualitative analysis with performance metrics to examine how these practices influence review efficiency and decision quality.

Taken together, the five themes outline a connected workflow that mirrors how DRs are conducted in industrial practice, before, during, and after the meeting. We argue that effective collaboration depends less on immersion or tool novelty than on affordances that support grounding, shared focus, traceable documentation, and continuity between decision and implementation. In this sense, advanced collaboration can be understood not as a technology class but as a set of repeatable conditions that make DRs auditable, comprehensible, and manageable within existing engineering systems. By articulating how these conditions interact, the study connects experimental insights into multi-user 3D interaction with the procedural realities of industrial DRs.

6. Conclusion

This study developed empirically grounded guidelines for using advanced collaboration tools in DRs, connecting what teams do and say during collaborative sessions to structured routines for preparation, coordination, and documentation. It demonstrates how affordances such as shared viewpoints, anchored annotations, and follow controls can be embedded into the preparation, execution, and documentation of DRs, effectively operationalizing advanced collaboration tools through structured guidelines. The proposed framework illustrates how traceability, shared understanding, and data continuity can emerge when cognitive and organizational routines are aligned with a system's affordances, rather than when immersion alone is prioritized. For industry, the guidelines offer concrete procedures that can be immediately implemented within existing DR workflows; for software developers focused on advanced collaboration tools, they provide a functional blueprint for designing interaction mechanisms that prioritize workflow support over visual fidelity; and for researchers, they provide an analytic template for deriving grounded, transferable practices from situated evidence. Future studies should test these routines across diverse industries and product stages, quantifying their impact on coordination efficiency, knowledge retention, and the quality of design decisions. Future work will focus on examining how the proposed routines influence collaboration and decision-making when adopted at an organizational scale. This involves studying how teams incorporate these guidelines into their established workflows, how they adapt them over time, and what measurable effects they produce on coordination, knowledge retention, and implementation speed. Observing this process in a real-world organizational context will enable the evaluation of not only the feasibility of the guidelines but also their impact on the broader effectiveness of DRs as part of product development practices.

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FRANCISCO GARCIA RIVERA

Francisco Garcia Rivera holds a BSc in Mechanical Engineering from the University of Málaga in Spain and an MSc in Intelligent Automation from the University of Skövde. His research interests include Computer-Supported Cooperative Work (CSCW), Human-Computer Interaction (HCI), Extended Reality (XR), and collaboration in product development processes.

In this dissertation, Francisco investigates how collaborative technologies mediate interaction and coordination in engineering design reviews, and how advanced tools can resolve the friction associated with teamwork. His work identifies the structural limitations of traditional presenter-centred screen sharing, evaluates the specific affordances of immersive and non-immersive tools, such as independent navigation and in-scene spatial referencing, and introduces validated procedural guidelines for deploying these systems in industrial practice. Together, these contributions enable a more inclusive, sustainable, and structurally robust approach to cross-functional decision-making in modern product development.

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